

Advanced Manufacturing

Gas turbines — the crown jewel of advanced manufacturing

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- **Decade-long upcycle: AI power, energy transition, structural supply shortage**
- **Nested constraints at OEM and hot-section components**
- **Initiate coverage on Yingliu (Siemens' H-class supplier) and DEC (fully-indigenous exporter) with Outperform**

A US\$200b AI power market. AI power demand, energy transition and structural supply shortage are supporting a decade-long upcycle of Gas Turbine where we are still at early stage. Global DC electricity grows 14–20% CAGR, from 485 TWh (2025) to 945–1,200 TWh (2030F), ~40% of all incremental global demand. With gas supplying 35–45% of the DC increment, AI alone drives 56–96 GW of incremental GT capacity. Layered on 175–190 GW of non-AIDC demand (coal-to-gas, grid balancing, replacement), total GT demand reaches 231–285 GW over 2026–30F. We size the new-equipment TAM at US\$92–114b (base to bull), or ~US\$139–171b incl. aftermarket services.

Big Three on 85% share, 6–7 year lead time, up to 62GW shortfall. The Big Three (GEV, Siemens Energy, MHI) hold ~85% share and ~228 GW of backlog at end-2025; lead times have stretched from 3–5 to 6–7 years and ASP are up ~20%. Even with announced expansion lifting combined capacity from ~31 to ~51 GW/yr by 2030, our base case shows a cumulative 7 GW supply gap over 2026–30F, widening to ~62 GW in the high-demand case. We lean to the upper end: AI power forecasts have a one-directional miss history (IEA +70% in one revision), inference demand is barely modelled, and grid queues of 3–7 years make behind-the-meter gas a necessity, not a choice. That gap is the structural opening for Chinese and Korean OEMs and the global component base.

Value cascades from OEM to hot-section components. OEM is a textbook oligopoly — combustion IP, certified field hours, 30-yr service lock-in, US\$1b+ / 10–15-yr barriers to entry. Below it, the value cascades down a waterfall (300MW+ → mid-frame → aero-derivative → diesel gensets) as buyers trade efficiency for delivery. Chinese / Korean electric OEMs, aeroderivative OEM and industrial genset OEM are taking up the remaining 15% share and impatient AIDC. Behind it, the supply chain tells another nested supply shortage story. Chinese participation is differentiated: genuine global penetration in hot-section blades (Yingliu — GE/Siemens/BKR certified) and cold-section rings/casings (Himile — all Big Three as customers); credible EPC packaging in North America (Jereh); but superalloy materials still face a 3–5-yr qualification wall (Longda, 688231 CH, the furthest along).

Value is skewed to the hot section (~40–50% of turbine cost) — single-crystal blades and vanes, the named global bottleneck, made by only few firms worldwide.

Peak orders, not peak earnings. Order intake might peak in 2026F after two consecutive years of over 100GW new orders i.e. GEV 2025/1Q26 +47%/59% YoY; Siemens 2025/1H26 +43%/55%. But with 6–7-yr lead times, the revenue and FCF cycle runs to 2030F+. The sector re-rates from order-momentum to delivery-and-margin-expansion. We believe profit-taking at peak orders is essentially (1) forgoing the future decade-long earnings uptrend, (2) posing upside risks from AI inferencing power expansion, and (3) missing the most valuable part of the GT business model i.e. aftermarket services. Five things we monitor with respect to the GT market cycle: (1) Big Three lead times, (2) GEV Gas Power margin, (3) hyperscaler capex, (4) Ni/Co prices, and (5) DEC's North American G50 conversion.

Sector trading strategy: Components over OEM, multi-pronged over pure play. The positioning question isn't whether to own the cycle but where. We prefer components over OEM (wider TAM, deeper technical / qualification moat, no involution risk), and China for scaling.

- **Yingliu (603308 CH, Outperform, TP RMB90) — primary pick.** Sole Chinese H-class blade supplier to Siemens; certified across GEV, BKR, Ansaldo, Safran, RR — a global oligopoly with Howmet/PCC that qualification alone can't fast-follow. Overseas revenue already 54%. 61% EPS CAGR, 1.5x PEG.
- **Dongfang Electric (1072 HK / 600875 CH, Outperform, TP HK\$45 / RMB49) — China OEM pick.** China's only fully-indigenous F-class GT in commercial production and export; G50 export optionality essentially unpaid for at 13x 2027F H-share P/E. The Chinese power leader should be benchmarking against Doosan (034020 KS) of Korea.
- **Global OEM analysis: Siemens Energy > MHI > GEV.** GEV purest GT play but a consensus trade after +250%; ENR = the GEV thesis at a discount plus faster-growing Grid Tech; MHI cheapest with a defense second engine; Doosan is the wildcard — the global 5th H-class, supplier of xAI Colossus, SMR optionality and a non-Chinese option for Western buyers at 104x 2027F P/E.
- **Substitutes & complements down the cascade.** As buyers step down the size ladder, adjacent plays open up: CSIC Power (600482 CH) — naval-to-land conversion, cheap-and-fast-growing (19x 2027F P/E / 62% EPS CAGR), Guanghan asset injection as catalyst; Weichai (2338 HK/000338 CH) — complementary backup-power layer, an H-share blue chip at 21x P/E (12% premium to A) on 21% EPS CAGR; Jereh (002353 CH) — the only Chinese EPC assembling large GT packages for North American AIDC, deeply tied to Baker Hughes; and in hot-section, Howmet (HWM US) as the global champion and Wedge (000534 CH) as China's no.2 — supplement to our Yingliu call.

Key risks AI capex slowdown; aggressive Big Three capacity expansion; geopolitical / IP restrictions on core-component trade; raw-material cost inflation.

Key gas turbine manufacturer valuation metrics

			Mkt cap	2027F P/E	2026F PEG	3-Yr CAGR
Company	Stock code	CCBIS rating*	(US\$ b)**	(x)	(x)	(%)
Yingliu	603308 CH	O	7	48.3	1.5	61.1
Dongfang H	1072 HK	O	16	12.5	1.1	23.0
Dongfang A	600875 CH	O	16	18.8	1.3	23.0
GEV	GEV US	NR	295	47.4	2.6	57.5
Siemens E	ENR GR	NR	166	29.0	1.4	67.2
Baker Hughes	BKR US	NR	57	21.0	1.5	7.9
Jereh	002353 CH	NR	25	34.5	1.4	31.0
Weichai	2338 HK	NR	46	20.9	1.2	20.9
Howmet	HWM US	NR	109	47.7	3.4	21.8
Wedge	000534 CH	NR	2	35.4	1.7	43.7

* CCBIS ratings: O = Outperform, N = Neutral, U = Underperform, NR = Not Rated

** Price as at close on 22 Jun 2026

Source: Bloomberg, CCBIS estimates

高端制造

燃气轮机——制造业皇冠上的明珠

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- 十年一遇的上行周期,由 AI、能源转型、结构性供应短缺驱动
- 整机与热端零部件形成两级产能瓶颈
- 推荐应流股份(西门子 H 级中国独供)、东方电气(中国争气机出口故事)

2000 亿 AI 能源市场。燃气轮机正值一个——由 AI 用电需求、能源转型、以及结构性供应短缺三者共同驱动——似十年计的上行周期。我们预计全球数据中心(DC)用电量以 14-20% 的复合增速增长,从 2025 年的 485TWh 升至 2030F 的 945-1,200 TWh,占全球用电增量约 40%。假设燃气满足 DC 增量的 35-45%,仅 DC 一项就带动 56-96GW 的新增燃机需求;叠加 175-190GW 的非 DC 需求(煤改气、电网调峰、存量置换),预计 2026-30F 燃机总需求达 231-285GW。我们测算新增设备 TAM 为 920 亿至 1,140 亿美元(基准至乐观情形),计入后市场服务则约 1,390 亿至 1,710 亿美元。

CR3=85%、订价排至 2030 年、供给缺口达 62GW。三巨头(GEV、西门子能源、三菱重工)合计市占约 85%,截至 2025 年底在手订单约 228GW;交货周期已从 3-5 年拉长至 6-7 年,ASP 上涨约 20%。即便计入已公布的扩产、使合计产能从约 31GW/年提升至 2030 年约 51GW/年,我们的基准情形仍显示 2026-30F 累计 7GW 的供给缺口,在高需求情形下扩大至约 62GW。我们倾向看高一端:AI 用电预测历来单向低估(IEA 曾一次上调+70%)、推理(inference)需求几乎未被纳入测算、而 3-7 年的并网排队使表后(Behind The Meter, BTM)燃气发电成为刚需而非选项。该缺口正是中国及韩国 OEM、以及全球零部件供应链的结构性切入点。

从整机到热端部件的价值渗透。整机(OEM)是教科书式的寡头垄断——燃烧室 IP、认证运行时长、长达 30 年的服务绑定、十亿美元+及 10-15 年的进入门槛。在其之下,价值沿产业链逐级下沉(300MW+ → 中型机 → 航改型 → 柴油发电机组),客户不惜以较低效率换取较快的交货期。中国/韩国电气巨头、航改燃、大型柴油发动机等争夺剩余 15% 市场及急不及待的 AIDC 客户。在其背后,供应链则是另一个供应瓶颈故事。中国的参与在价值链的多处有所体现:在热端叶片(应流——已通过 GE/Siemens/BKR 认证)及冷端环件/机匣(豪迈——三巨头均为其客户)已具备真正的全球渗透;在北美 EPC 总承包亦具竞争力(杰瑞);但高温合金材料仍面临 3-5 年的认证壁垒(隆达,688231 CH,进度最快)。价值高度集中于热端(约占整机成本 40-50%)——单晶叶片及导向叶片,正是公认的全球瓶颈,全球仅约数家厂商能造。

当前是订单高峰——而盈利高峰仍在后头。新接订单或于 2026F 出现连续两年逾 100GW 的新订单后见顶——GEV 2025/1Q26 同比 +47%/+59%;西门子能源 2025/1H26+43%/+55%。但在 6-7 年交货周期下,收入与自由现金流(FCF)周期将延伸至 2030F 之后。板块估值逻辑将从“订单动能”切换至“交付兑现与利润率扩张”。我们认为在订单高峰就获利离场,等于(1)舍弃一个长达十年的盈利上行周期;(2)曝露于 AI 推理需求爆发的上行风险;及(3)丢失燃气轮机投资逻辑最有价值的售后服务。我们跟踪五项指标帮助我们定位燃气轮机周期:(1)三巨头交货周期、(2)GEV 燃气电力业务利润率、(3)超大规模云厂商资本开支、(4)镍/钴价格,以及(5)东方电气的北美 G50 转化进度。

如何布局:零部件优于整机,多线发展优于沽注一掷。问题不在于是否参与本轮周期,而在于从何处切入。我们偏好零部件优于 OEM (TAM 更大、即时的置换件经常性收入、更深的技术/认证护城河、无内卷风险),并以中国作为规模化扩张的主战场。

- **应流股份 (603308 CH, 优于大市,目标价: 人民币 90)——首选。**唯一向西门子能源供应 H 级叶片的中国厂商;已通过 GEV、贝克休斯(Baker Hughes)、安萨尔多(Ansaldo)、赛峰(Safran)、罗罗(Rolls Royce)认证——与豪梅特(Howmet)/PCC 共同构成仅凭认证难以快速追赶的全球寡头格局。海外收入占比已达 54%。EPS 复合增速 61%, PEG 1.5 倍。
- **东方电气 (1072 HK/600875 CH, 优于大市,目标价: 港币 45.00/ 人民币 49.00)——中国整机首选。**中国唯一实现商业化量产并出口的完全自主 F 级燃机; G50 出口期权价值在 15 倍 2027F H 股市盈率下基本未被定价。中国电气龙头应对对标韩国斗山能源(034020 KS)。
- **全球 OEM 分析: 西门子能源 > 三菱重工 > GEV。**GEV 为最纯粹的北美燃气轮机整机厂,但在 +250% 涨幅后已属市场共识交易。西门子能源 = 打折版的 GEV 逻辑,叠加增速更快的电网业务;三菱重工估值最低,且有国防业务作为第二增长引擎;斗山能源(034020 KS)是黑马——带领韩国成为第五个掌握 H 级技术的国家,已签约 12 台美国 AIDC 机组并具备 SMR 期权价值,是西方买家在地缘政治上可接受的非中国选项。
- **沿产业链逐级下沉的替代与互补标的。**随着买方沿机组尺寸阶梯逐级下移,相邻标的随之浮现:中国动力(600482 CH)——舰改燃,低估值、高增长标的, 2027F 19 倍市盈率、62% EPS 复合增速,以广汉资产注入为催化剂;潍柴动力(2338 HK/000338 CH)——与燃机互补的备用电源环节,H 股工业蓝筹,21 倍市盈率(较 A 股溢价约 12%)、~21% EPS 复合增速;杰瑞股份(002353 CH)——唯一为北美 AIDC 组装大型燃机 EPC 包的中国厂商,与贝克休斯深度绑定;热端方面,豪梅特(HWM US)为全球龙头,万泽股份(000534 CH)为中国第二——均为我们应流主线的替代标的。

主要风险: AI 资本开支放缓;三巨头激进扩产;核心零部件贸易的地缘政治/IP 限制;原材料成本上涨。

主要燃气轮机制造商估值比较

公司	股票代码	建银国际证券评级*	市值 (十亿美元)**	2027F 市盈率 (倍)	2026F PEG (倍)	三年复合增长率(%)
应流股份	603308 CH	O	7	48.3	1.5	61.1
东方电气 H	1072 HK	O	16	12.5	1.1	23.0
东方电气 A	600875 CH	O	16	18.8	1.3	23.0
GE 能源	GEV US	NR	295	47.4	2.6	57.5
西门子能源	ENR GR	NR	166	29.0	1.4	67.2
贝克休斯	BKR US	NR	57	21.0	1.5	7.9
杰瑞股份	002353 CH	NR	25	34.5	1.4	31.0
潍柴动力	2338 HK	NR	46	20.9	1.2	20.9
豪梅特	HWM US	NR	109	47.7	3.4	21.8
万泽股份	000534 CH	NR	2	35.4	1.7	43.7

* 建银国际证券评级: O = 优于大市, N = 中性, U = 弱于大市, NR = 未评级

** 2026 年 6 月 22 日收市价

数据来源: 彭博, 建银国际证券

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Gas turbine supply chain – peer comparison

Role	Company	公司	Stock code	Mkt cap (US\$ m)	Share price (LC)	P/E FY0 (x)	P/E FY1 (x)	P/E FY2 (x)	PEG (x)	Sales CAGR (%)	EPS CAGR (%)	GPM (%)	NPM (%)	Net gearing (%)	ROE (%)	Control
OEM	GE Vernova	GE 能源	GEV US	303,006	1,127.59	132.5	53.1	47.4	2.6	16	58	20	6	(42.6)	20.7	POE
OEM	Siemens Energy	西门子能源	ENR GR	170,469	170.22	106.4	40.4	29.0	1.4	13	67	17	4	(26.8)	14.1	POE
OEM	Mitsubishi HI	三菱重工	7011 JP	83,270	3,909.00	39.5	45.6	33.0	1.2	6	13	22	7	13.9	11.4	POE
OEM	Dongfeng Elec. H	东方电气 H	1072 HK	16,207	25.24	19.1	15.4	12.5	1.1	11	23	17	5	(14.0)	9.3	SOE
OEM	Dongfeng Elec. A	东方电气 A	600875 CH	16,207	33.03	28.7	23.2	18.8	1.3	11	23	17	5	(14.0)	9.0	SOE
OEM	Shanghai Elec. H	上海电气 H	2727 HK	15,296	3.81	42.5	37.3	29.2	1.1	3	26	18	1	25.0	1.6	SOE
OEM	Shanghai Elec. A	上海电气 A	601727 CH	15,197	7.44	95.4	83.8	65.6	3.2	3	26	18	1	25.0	1.6	SOE
OEM	Harbin Elec.	哈尔滨电气	1133 HK	5,517	16.80	12.3	9.3	7.9	0.9	8	20	15	6	(55.5)	15.7	SOE
OEM	Doosan Enerbility	斗山能源	034020 KS	39,873	94,200.00	713.6	191.1	104.2	1.9	9	114	16	0	35.3	1.1	POE
naval-to-land	CSIC Power	中国动力	600482 CH	11,591	34.80	60.0	26.1	19.0	0.8	18	62	17	2	(52.9)	5.0	SOE
Aeroderivative	AECC Avi Power	航发动力	600893 CH	16,283	41.60	173.3	141.8	111.4	26.9	10	12	9	1	67.1	1.4	SOE
Aeroderivative	Baker Hughes	贝克休斯	BKR US	58,681	59.15	22.8	24.9	21.0	1.5	3	8	24	9	43.6	13.6	POE
EPC	Jereh	杰瑞股份	002353 CH	26,109	173.66	65.8	47.3	34.5	1.4	24	31	32	17	(6.6)	13.4	POE
Generator	Weichai H	潍柴动力	2338 HK	47,415	41.00	32.3	25.1	20.9	1.2	9	21	21	5	(4.5)	13.4	SOE
Generator	Weichai A	潍柴动力	000338 CH	40,984	31.29	24.6	18.9	15.7	1.4	9	22	21	5	(4.5)	13.1	SOE
Hot-Blade	Howmet	豪梅特	HWM US	112,174	280.36	74.4	57.1	47.7	3.4	14	22	34	19	12.6	30.1	POE
Hot-Blade	Yingliu	应流股份	603308 CH	6,906	69.26	135.8	74.7	48.3	1.5	33	61	36	12	97.2	8.3	POE
Hot-Blade	Wedge	万泽股份	000534 CH	2,360	31.54	79.3	47.2	35.4	1.7	36	44	72	15	79.6	15.6	POE
Cold-Forging	IHI Corp	石川岛重工	7013 JP	19,256	2,816.50	18.5	19.7	18.8	3.8	7	2	23	10	108.3	23.1	POE
Cold-Forging	Bharat Forge	巴拉特锻造	BHFC IN	10,486	2,103.80	107.6	56.9	43.2	3.3	13	40	55	6	17.0	14.4	POE
Cold-Forging	Guizhou Aviation	航宇科技	688239 CH	1,640	58.49	59.7	35.7	26.6	1.3	28	44	28	9	53.3	10.5	POE
Cold-Blade	Hyatech	航亚科技	688510 CH	993	26.02	66.7	35.6	26.6	1.4	36	51	37	15	5.6	10.8	POE
Cold-Casing	Himile	豪迈科技	002595 CH	8,650	50.78	24.5	20.3	17.3	1.3	17	19	34	22	(16.6)	21.8	POE
HRSO	Xizi	西子节能	002534 CH	2,678	21.82	38.3	33.2	28.1	1.6	19	17	19	7	3.1	10.2	POE
Superalloy	CISRI Gaona	钢研高纳	300034 CH	2,172	18.56	168.0	58.5	48.5	2.5	18	69	24	2	31.3	5.0	SOE
Superalloy	Longda	隆达股份	688231 CH	1,709	47.15	152.1	70.0	46.5	1.3	34	70	14	4	12.3	3.8	POE
Superalloy	Toland Alloy	图南股份	300855 CH	1,985	34.18	87.6	34.2	25.9	1.3	36	63	28	14	(6.5)	11.5	POE
Superalloy	Fushun Special Steel	抚顺特钢	600399 CH	1,309	4.52	-11.0	48.4	9.5	1.8	11	27	1	(10)	47.5	(10.1)	SOE
Control system	Woodward	伍德沃德	WWD US	25,431	426.82	61.9	45.4	39.9	2.7	13	22	27	12	24.6	17.8	POE

Note: Not Rated data sourced from Bloomberg consensus

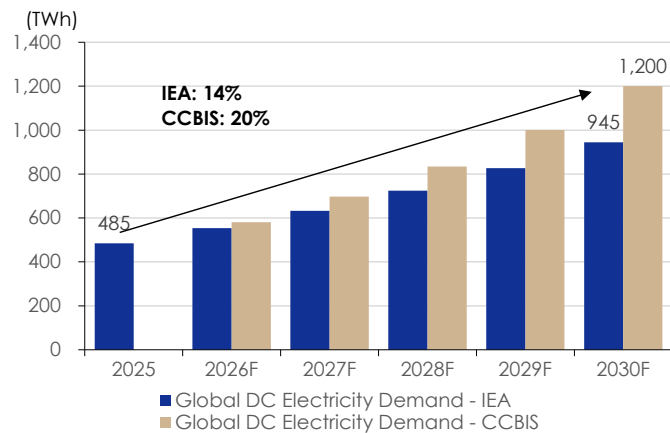
† Price as at close on 22 Jun 2026 (local currency)

‡ CCBIS ratings: O = Outperform, N = Neutral, U = Underperform, Not Rated = Not Rated

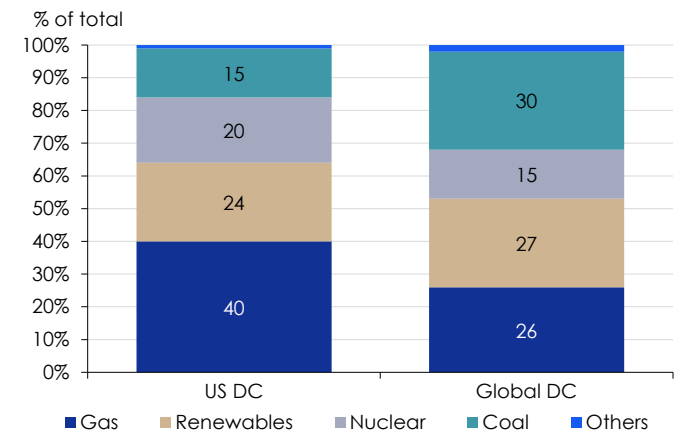
Source: Bloomberg, CCBIS estimates

FOCUS CHARTS

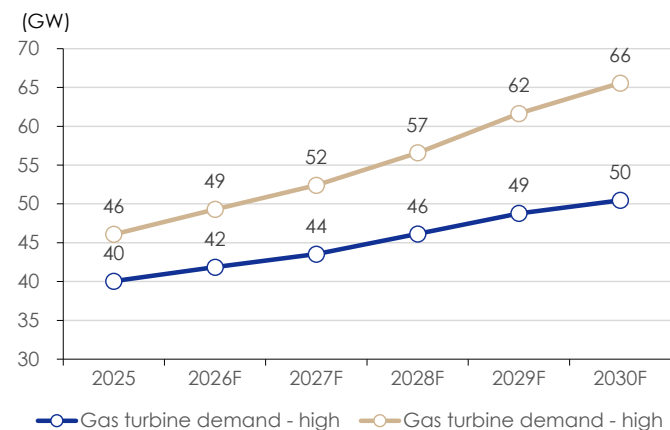
Global DC Electricity demand to grow at 14-20% CAGR in 2025-2030F



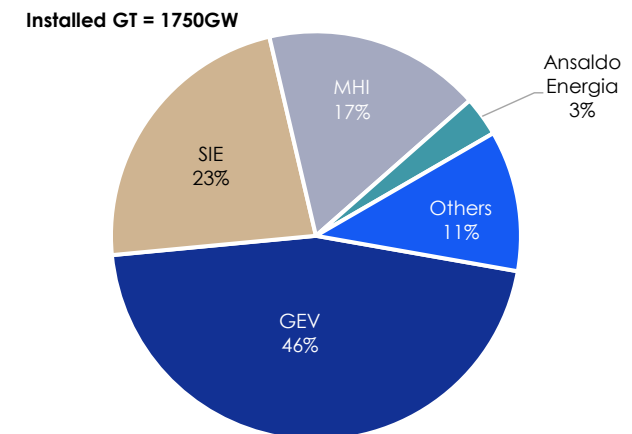
Gas powers 40% DC in US in 2024



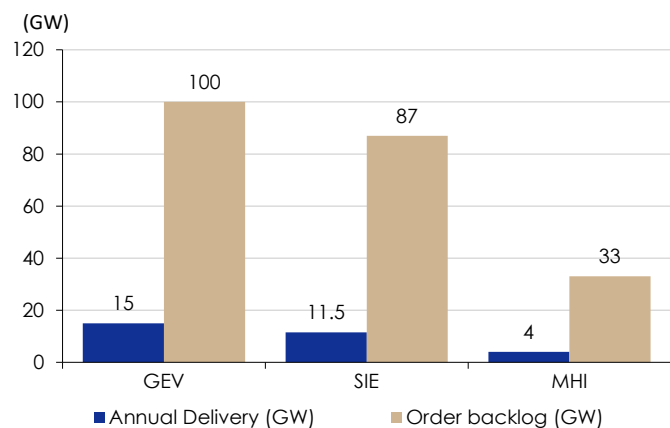
Overall gas turbine demand to be 231-286 GW in 2026-30F



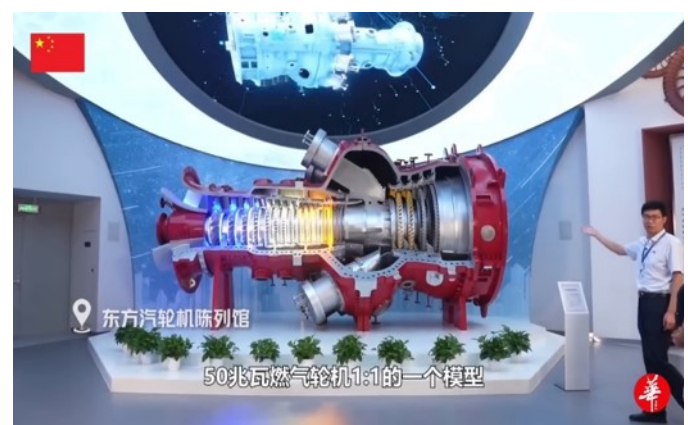
An oligopolistic market: CR3=85%



Global Big Three's order backlog implies 6-7 year lead time for delivery

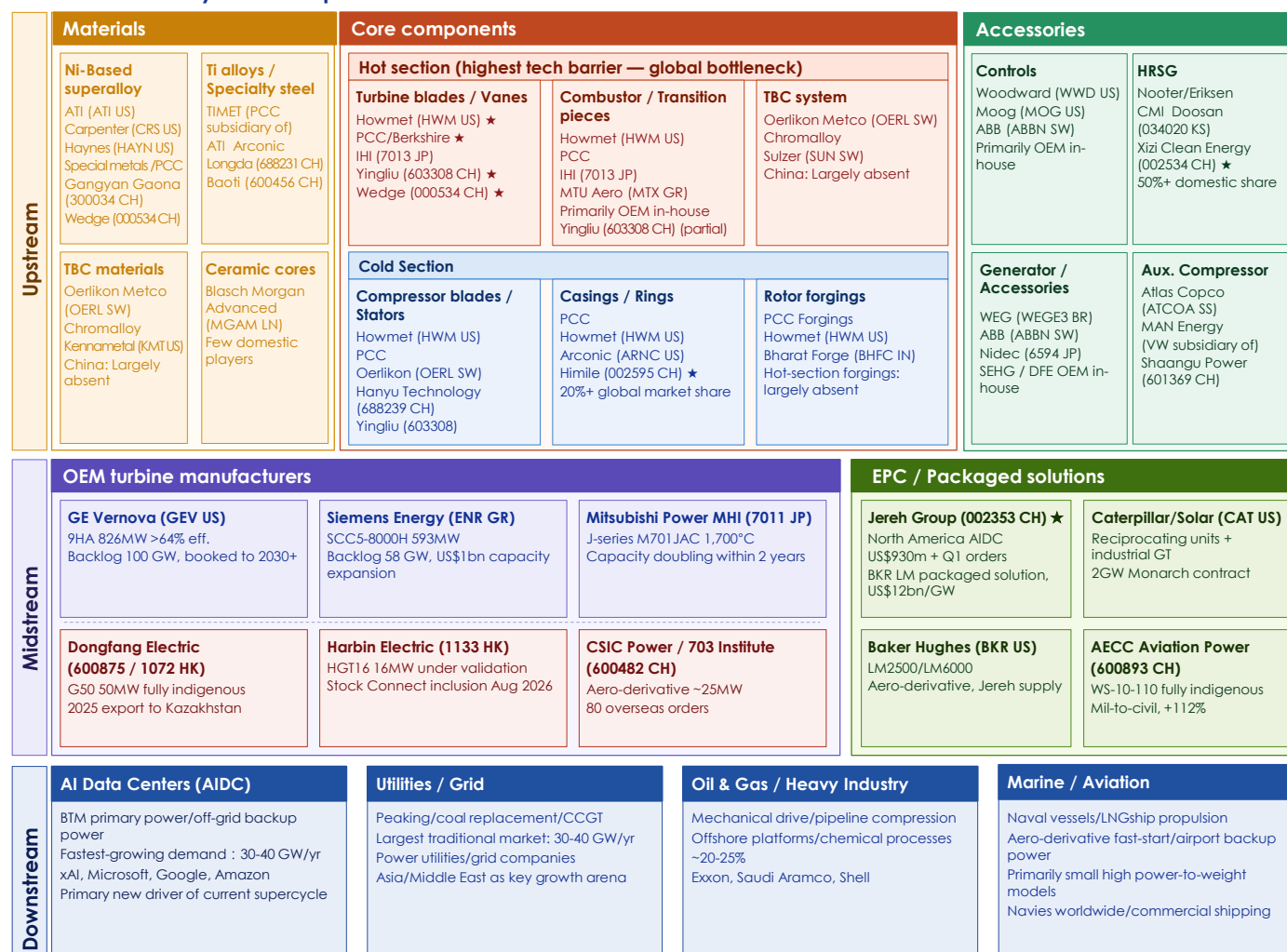


G50 China's first fully indigenous F-class heavy gas turbine



Source: Company, CCBIS

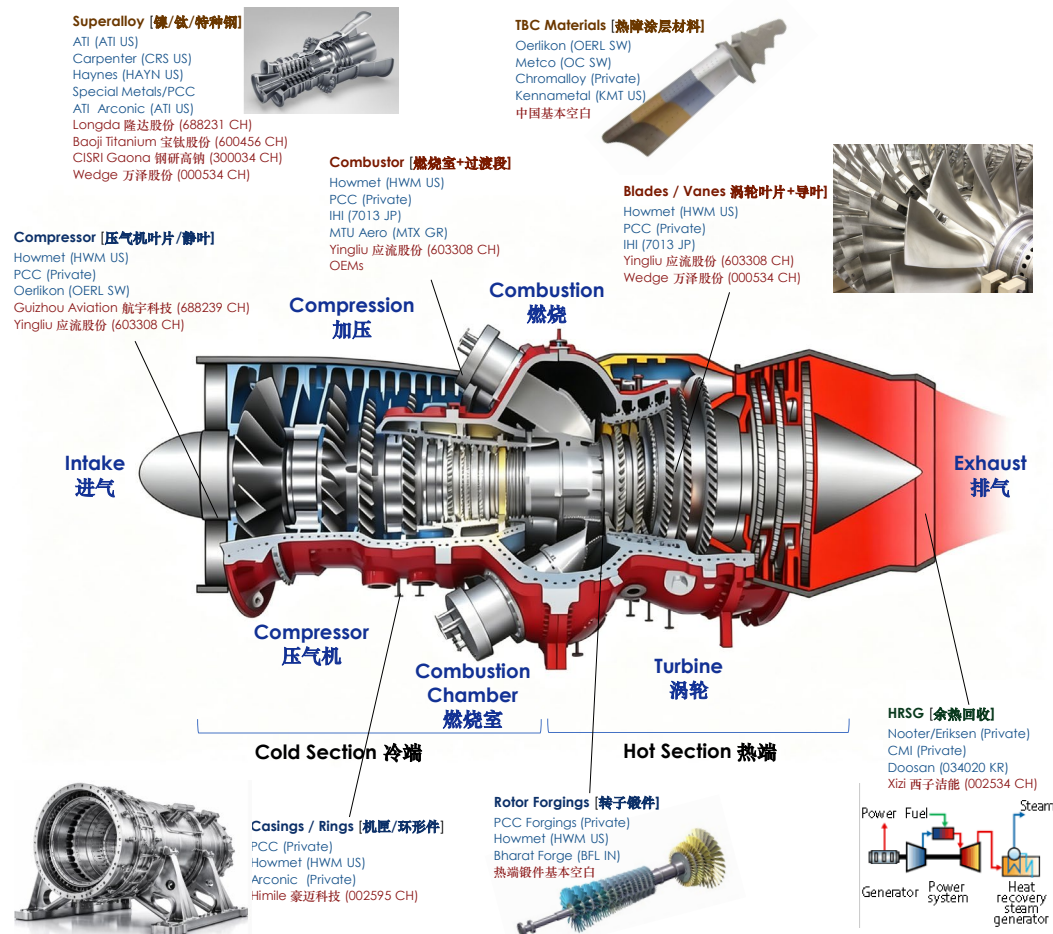
Gas turbine industry chain map



■ Blue = Global suppliers (Listed/Private) ■ Red = China suppliers (A/H share) ★ = Key stock picks

Source: Company, CCBIS

Gas turbine upstream opportunities

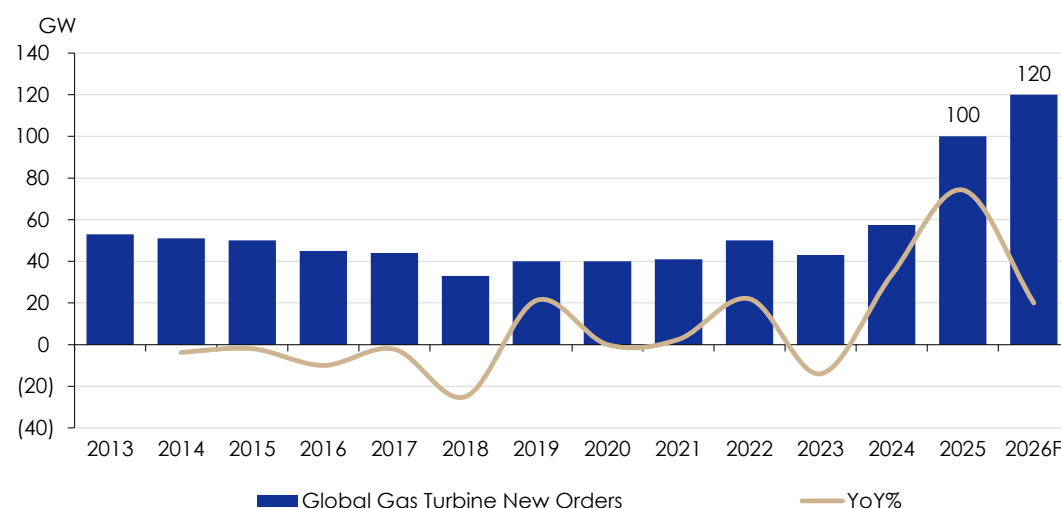


Source: Company, Ansys, Jungyi, Pengky, CCBIS estimate

GAS TURBINES: THE CROWN JEWEL OF ADVANCED MANUFACTURING

Gas turbines have spent a decade in the shadows of renewable energy hype. That is changing fast. The confluence of (1) AI-driven power demand, (2) energy transition bridging needs, and (3) a structurally constrained global supply chain has produced what we believe is the industry's most compelling setup in 30 years. The global big three OEM, namely GE Vernova (GEV US, NR), Siemens Energy (ENR GR, NR), Mitsubishi Heavy Industries (7011 JP, NR) are booked out to 2030–2031. Chinese OEMs – including Dongfang Electric (600875 CH/1072 HK, Outperform), Shanghai Electric (601727 CH/2727 HK, NR) and Harbin Electric (1133 HK, NR), long locked out of international markets by IP licensing restrictions, are now finding the door kicked open by desperate overseas buyers. Component suppliers are turning away orders. Prices are rising. This report maps the investment landscape across the full chain — from the global leaders to niche Chinese blade makers — and identifies where the risk/reward looks most compelling.

Global Gas Turbine New Orders enter into a decade-long upcycle



Source: McCoy, MHI, CCBIS estimates

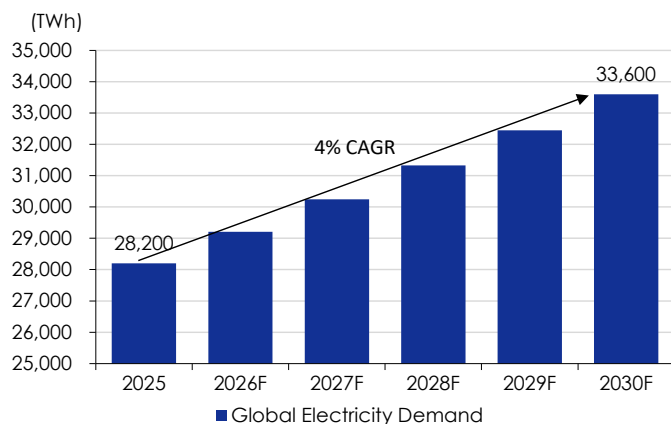
Three forces: AI Power Gap, Energy Transition & Supply Shortage

(1) The AI Power Gap — 20% CAGR

The most important development of the past 18 months is straightforward: AI datacenters need enormous, reliable, always-on power, and grid infrastructure cannot keep up.

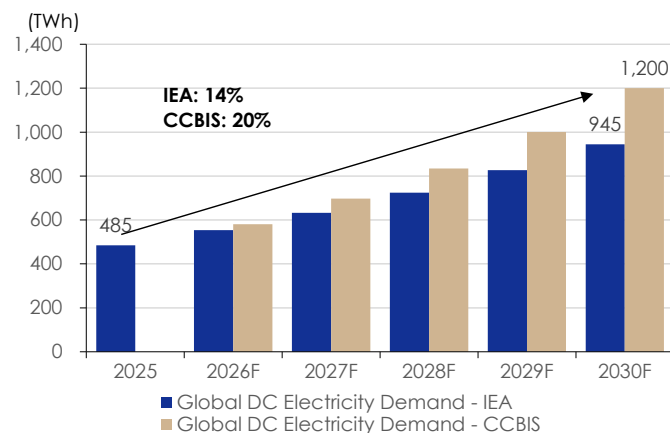
Global electricity demand is expected to grow at a 3.6% CAGR from 2025 to 2030, with US demand tracking roughly in-line with the global average at a similar rate. Within that broader growth, however, data centers stand out as a disproportionate driver: US datacenter power consumption alone is expected to account for c.29% of total incremental US electricity demand over this period — a remarkable share for a single end-use category.

Global Electricity demand to grow at 4% CAGR in 2025-2030F



Source: IEA, CCBIS estimates

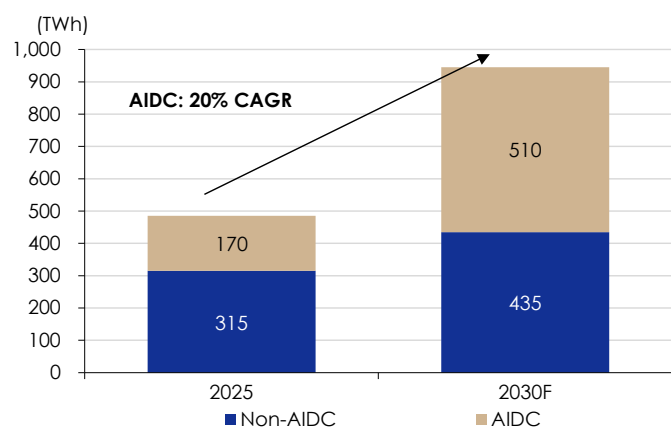
Global DC Electricity demand to grow at 14-20% CAGR in 2025-2030F



Source: IEA, CCBIS estimates

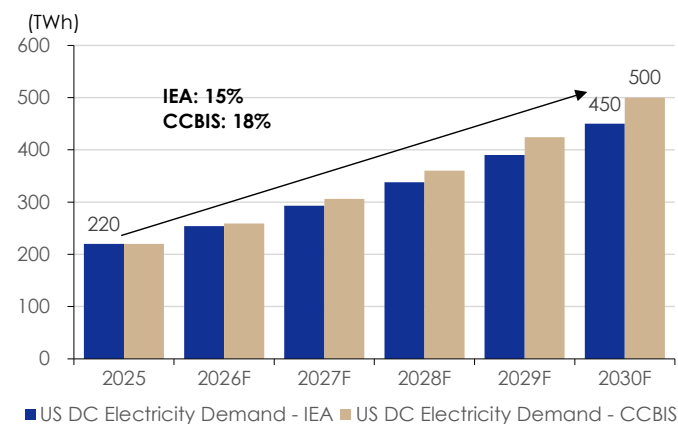
At the global level, datacenter electricity demand is forecast to grow at a 14–20% CAGR, rising from c.485 TWh in 2025 to 945–1,200 TWh by 2030. Translating this electricity growth into installed generation capacity terms, the datacenter sector will require an estimated 70–106 GW of new power generation capacity to be built over 2026–2030 to service incremental demand.

Global AIDC Electricity demand to grow at 20% CAGR in 2025-2030F



Source: IEA, CCBIS estimates

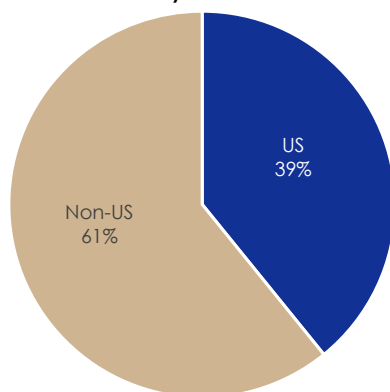
US DC Electricity demand to grow at 15-18% CAGR in 2025-2030F



Source: IEA, CCBIS estimates

39% of new DC are located in US

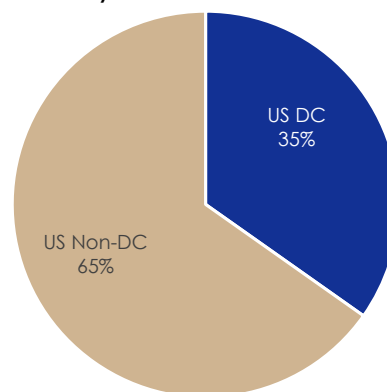
Global DC incremental electricity demand = 715 TWh



Source: IEA, CCBIS estimates

35% of US incremental electricity are consumed in DC

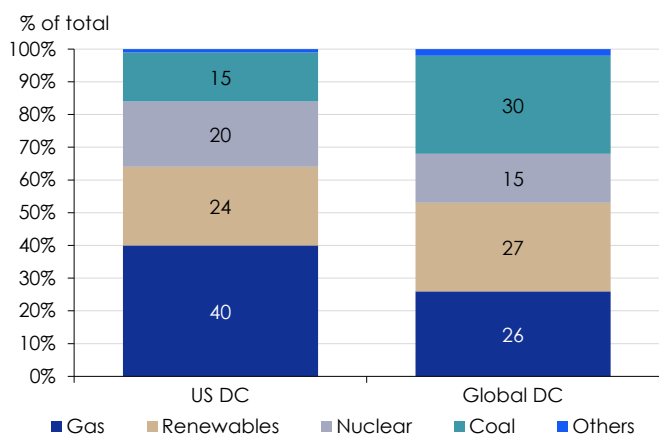
US incremental electricity demand = 805 TWh



Source: IEA, CCBIS estimates

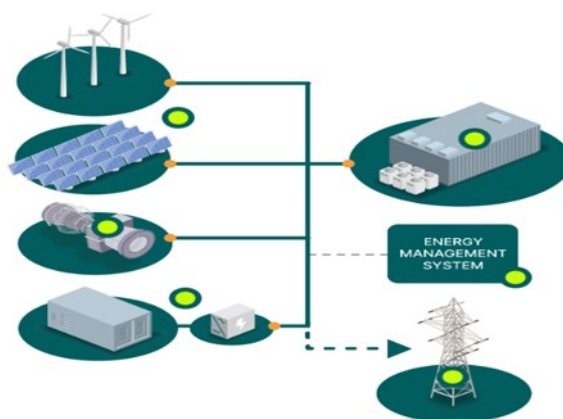
Gas Turbines supply 35-45% of DC electricity: Gas-fired generation — spanning both grid-connected peaking turbines and behind-the-meter (BTM) on-site generation — is expected to supply 35–45% of this new datacenter electricity, implying a direct power delivery requirement of c.31-48 GW attributable to gas. However, because gas turbines are not dispatched at full output continuously — grid-side peaking machines typically run at 40–50% capacity factor while BTM base-load backup units run at 70–80%, giving a blended average of c.50% — the actual installed gas turbine capacity required to reliably deliver this power is roughly double the energy delivery figure. On this basis, the pure AIDC-driven incremental gas turbine installed capacity requirement over 2026–2030 is estimated at **56–96 GW** — a demand increment that sits entirely on top of the 30–40 GW per year of pre-existing non-AIDC gas turbine demand from utilities, industrial users, and oil and gas applications.

Gas powers 40% DC in US in 2024



Source: IEA, CCBIS estimates

The setting of a renewable thermal hybrid microgrid



Source: GEV

The practical consequence is that a market which was already sold out — the Big Three OEMs were booked to 2029–2030 before the AI power surge — is now facing an additional structural demand layer with no near-term supply response. This is the core of the investment thesis.

Why gas turbines specifically for datacenters? Three reasons:

- Reliability: 24/7/365 uptime with rapid black-start capability (minutes, not hours)
- Emissions: ~40% lower CO₂ per kWh vs. coal; easier to get permits
- Economics: Technology is well understood; fuel (natural gas) is abundant and cheap in North America

Diesel has historically served as backup, but above 50MW generation output, facilities need industrial permits — and gas turbines produce far less emissions per kWh. Amazon required 105 backup diesel gensets at one facility to produce 674 MW; gas turbines achieve the same at a fraction of the footprint and regulatory risk. CCGT (Combined Cycle Gas Turbine) plants are too slow to start for backup; SCGT (Simple Cycle Gas Turbine) turbines are the sweet spot.

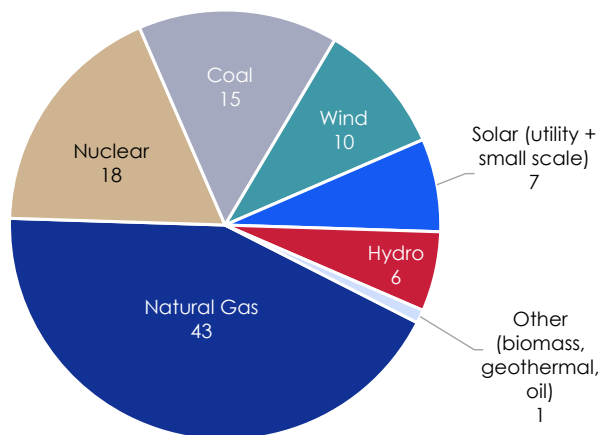
(2) Energy transition bridging — The medium-term anchor

Beyond datacenters, gas turbines occupy a structurally important niche in the broader energy transition. As coal retires and grids add more intermittent wind/solar capacity, flexible, fast-ramping gas turbines have become essential for grid stability.

- US natural gas accounts for 43% of electricity generation (2023); CCGT dominates the gas fleet
- Asia (particularly China and India) is adding gas power as coal transition bridges capacity
- Frequency regulation (调频), spinning reserve (旋转备用), and black-start services (黑启动服务) all favor gas turbines
- Global capacity under construction: 2/3 is in Asia (primarily China, India, Saudi Arabia)

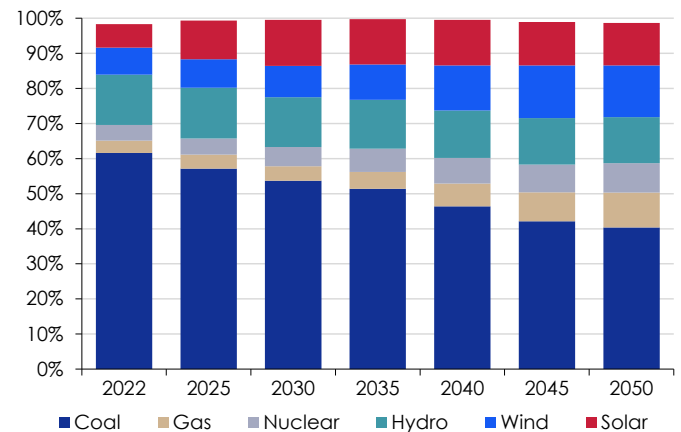
Long-term technology: the industry is also investing in hydrogen co-firing capability (掺氢燃烧技术, Mitsubishi's J-series already supports H₂/natural gas blends), extending the asset life in a decarbonizing world.

Gas plays a central role in US electricity generation



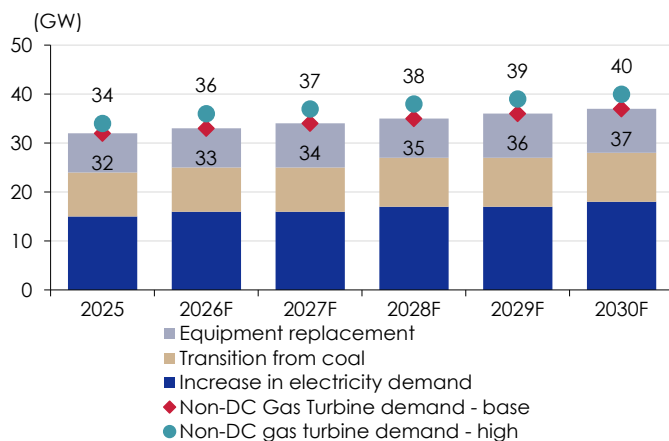
Source: EIA, CCBIS

Gas isn't abundant in China, yet it is fast-growing



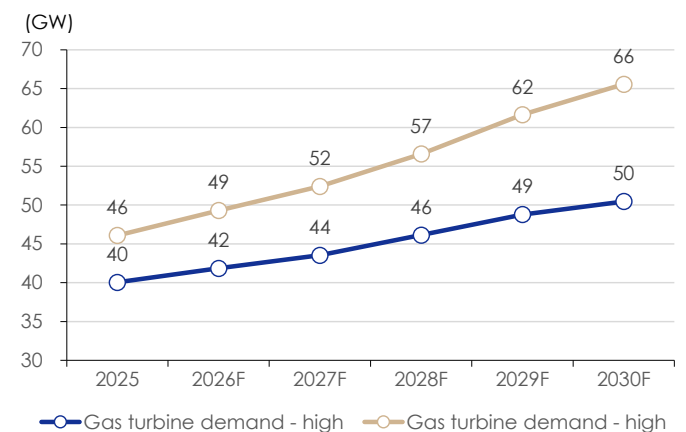
Source: State Grid, CCBIS

Non-DC gas turbine demand to be 175-190 GW in 2026-30F



Source: CCBIS estimates

Overall gas turbine demand to be 231-286 GW in 2026-30F



Source: CCBIS estimates

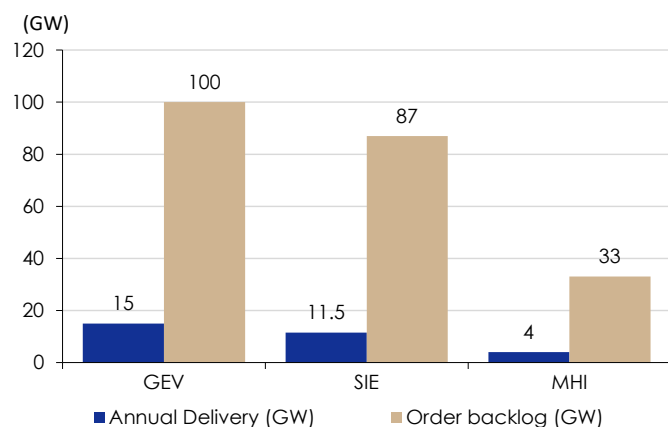
(3) Supply scarcity — The pricing power story

Heavy-gas-turbines are not scalable: The third pillar is simply that the global heavy-gas-turbine supply chain cannot respond quickly to demand. This is a bespoke, capital-intensive, highly engineered product — and lead times have extended dramatically as demand has surged:

- GE Vernova: backlog ~83 GW; booked to 2029–2031; investing US\$160m to expand capacity to ~24 GW/yr
- Siemens Energy: backlog ~78 GW (€146b+); investing US\$1b on US expansion; delivery to 2030
- Mitsubishi Power: +67 units (+84% YoY); planning to double capacity within 2 years; delivery to 2029
- Current delivery lead times: 6–7 years for the global majors (up from 3–5 years pre-2024)
- GEV turbine pricing: ~US\$2,000/kW → US\$3,000/kW (+50%); overall industry equipment prices +35% vs 2024; new plant total cost +49%. Jerih (China packager) repriced by ~20% in recent months alone.
- Global effective annual capacity: only ~50 – 60 GW/yr against 84 GW demand in 2025 → structural shortage of 30–35 GW/yr

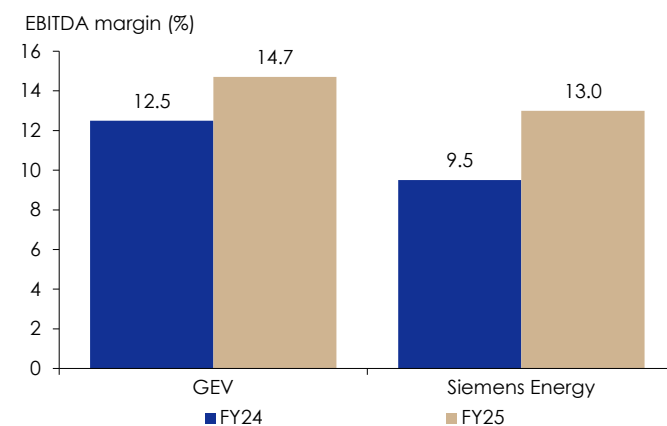
Value cascades down the waterfall: When customers cannot get heavy turbines, they work down the size ladder: 300MW+ → 50 – 100MW mid-size → 30MW aero-derivatives → 10 – 25MW small industrial → diesel gensets as last resort. This "waterfall" creates opportunity for Chinese OEMs that could not previously compete on the global stage.

Global Big Three's order backlog are stretching until 2029-2030F



Source: Company, CCBIS estimates

Margin is ramping up fast



Source: Company, CCBIS

The hot-end component bottleneck is a second-order constraint

Even if OEMs want to expand, turbine blades and vanes (single-crystal superalloy castings) take 6→18–24 months to procure — up from 6 months pre-cycle. Globally, Howmet (HWM US) and Precision Castparts (privatized by Berkshire Hathaway) command 70%+ of precision cast turbine airfoils; new entrants need 3–5 years to build capacity and 2–3 years to get certified. This makes component suppliers like Yingliu (603308 CH, Outperform) structurally scarce assets, not just cyclical beneficiaries.

The upshot: For the first time in decades, Chinese gas turbine OEMs are receiving direct inquiries from North American datacenter customers — not because they are preferred, but because the global majors have no near-term capacity.

TAM estimation for gas turbines: DC Power 14-20% CAGR

Our TAM estimation for gas turbines starts from the global electricity demand forecast. We based our assumptions on International Energy Agency (IEA)'s World Energy Outlook report. IEA expects global electricity demand to grow from 28,200 TWh in 2025 to 33,600 TWh in 2030F, implying 4% CAGR. The electricity demand from datacenter (DC) is estimated from 485 TWh in 2025 to 945 TWh in 2030F, implying 20% CAGR. The demand from DC will account for 9-13% of the incremental global electricity demand. We believe this 945 TWh is largely underestimated as a consistent track record of being wrong in one direction. Other forecasters like ABI Research and Deloitte are estimating 1,065-1,479 TWh electricity demand from DC by 2030F. Hence, we set our base-case and high-end estimates to be 945 TWh and 1,200 TWh respectively.

A. Global electricity demand to grow at 4%; DC electricity demand is expected to grow at 14-20%

Metric	Unit	2025	2026	2027	2028	2029	2030	CAGR 25-30	Note / Source
Global Electricity Demand (TWh)	TWh	28,200	29,206	30,247	31,326	32,443	33,600	4%	IEA
of which: DC Electricity Demand (TWh)	TWh	485	554	633	724	827	945	14%	McKinsey, IEA
of which: DC Electricity Demand (TWh, high)	TWh	485	581	697	835	1,001	1,200	20%	CCBIS estimates
US Electricity Demand (TWh)	TWh	4,195	4,345	4,500	4,661	4,828	5,000	4%	IEA
US DC Electricity Demand (TWh, base)	TWh	220	254	293	338	390	450	15%	50% of global
US DC Electricity Demand (TWh, high)	TWh	220	259	306	360	424	500	18%	CCBIS estimates

Source: CCBIS estimates

B. Data center capacity build-out & demand derivation

Metric	Unit	2025	2026	2027	2028	2029	2030	Cumulative	Note / source
Assumption: DC utilization rate	%	85	83	82	81	80	80		
Assumption: Gas share (Base)	%	40	40	40	40	40	40		35-45% range; grid-side + BTM combined
Assumption: Gas share (High)	%	45	45	45	45	45	45		Higher BTM deployment scenario
Assumption: Capacity factor	%	50	50	50	50	50	50		Grid-side 40%; BTM 70-80%; blended 50%
DC IT load (GW, @100% load)	GW	55.4	63.2	72.3	82.6	94.4	107.9		Global DC TWh ÷ 8,760h
Effective DC power demand (GW, base)	GW	65.1	76.2	88.1	102.0	118.0	134.8		= IT load ÷ utilization rate
Effective DC power demand (GW, high)	GW	65.1	79.9	97.0	117.7	142.8	171.2		High compute scenario (1,200 TWh by 2030)
Annual incremental DC power demand (GW, base)	GW/yr		11.1	11.9	13.9	16.0	16.8	69.7	Incremental capacity demand (annual additions)
Annual incremental DC power demand (GW, high)	GW/yr		14.8	17.1	20.6	25.2	28.4	106.1	High scenario annual additions
DC-driven GT capacity demand (GW, base) GW/yr			8.8	9.5	11.1	12.8	13.5	55.8	= Annual incremental DC demand × 40% ÷ 50%
DC-driven GT capacity demand (GW, high) GW/yr			13.3	15.4	18.6	22.6	25.6	95.5	= Annual incremental DC demand × 45% ÷ 50%

Source: CCBIS estimates

C. Non-DC Gas Turbine Demand is composed of (1) organic growth, (2) energy transition, and (3) replacement demand

Metric	Unit	2025	2026	2027	2028	2029	2030	Cumulative	Note / Source
Driver: Power Demand Growth (GW)	GW/yr	15.0	16.0	16.0	17.0	17.0	18.0		Non-DC annual electricity growth ~950 TWh × 20% gas share ÷ 45% CF
Driver: Coal-to-Gas / Energy Transition (GW)	GW/yr	9.0	9.0	9.0	10.0	10.0	10.0		US/EU coal retirement, replacing aging coal units
Driver: Aging Unit Replacement (GW/yr)	GW/yr	8.0	8.0	9.0	8.0	9.0	9.0		Installed base ~2,200 GW × 1/30-yr replacement rate
Non-DC GT Annual Demand Base (GW/yr)	GW/yr	32.0	33.0	34.0	35.0	36.0	37.0	175.0	= Power growth + coal-to-gas + replacement; +1 GW/yr from 2025
Non-DC GT Annual Demand High (GW/yr)	GW/yr	34.0	36.0	37.0	38.0	39.0	40.0	190.0	More aggressive scenario (accelerated electrification)

Source: CCBIS estimates

D. Global GT Demand is forecast to be 231-286 GW in 2026-2030F

Metric	Unit	2025	2026	2027	2028	2029	2030	Cumulative	Note / Source
DC GT Demand (GW/yr, Base)	GW/yr	8.0	8.8	9.5	11.1	12.8	13.5	55.8	IEA 945 TWh forecast
DC GT Demand (GW/yr, High)	GW/yr	12.1	13.3	15.4	18.6	22.6	25.6	95.5	CCBIS 1,200 TWh forecast
Non-DC GT Demand (GW/yr, Base)	GW/yr	32.0	33.0	34.0	35.0	36.0	37.0	175.0	Non-DC = 75% of total
Non-DC GT Demand (GW/yr, High)	GW/yr	34.0	36.0	37.0	38.0	39.0	40.0	190.0	Non-DC = 67% of total
Total GT Annual Demand (GW/yr, Base)	GW/yr	40.0	41.8	43.5	46.1	48.8	50.5	230.8	
Total GT Annual Demand (GW/yr, High)	GW/yr	46.1	49.3	52.4	56.6	61.6	65.6	285.5	

Source: CCBIS estimates

E. ASP of GT is forecast to be USD300-500m per GW; Price hike and Long-term Service Agreement provide further upside

Metric	ASP	Note
ASP assumption (Low) – Equipment average	US\$300M/GW	New unit equipment only; large H-class ~\$200-250/kW
ASP assumption (Base) – Equipment average	US\$400M/GW	Equipment + near-term delivery-related services; +10-30% price increases in recent years
ASP Assumption (High) – Equipment average	US\$500M/GW	Includes LTSA premium; mega H-class + BTM package
Services Market Multiplier (Full Market)	1.5×	Includes LTSA, spare parts, major overhauls; unit life ≥ 25 years

Source: CCBIS estimates

F. Equipment TAM is forecast to be USD92-114b in 2026-2030F; Full-market TAM to be USD139-171b; CAGR 5-7%

Metric	Unit	2025	2026	2027	2028	2029	2030	Cumulative	Note
Annual equipment TAM – Base	US\$ b	16.0	16.7	17.4	18.5	19.5	20.2	92.3	Base Demand × Base ASP
Annual equipment TAM – High	US\$ b	18.4	19.7	21.0	22.6	24.7	26.2	114.2	High Demand × High ASP
Full market TAM (Equipment + services, base)	US\$ b	24.0	25.1	26.1	27.7	29.3	30.3	138.5	Incl. LTSA
Full market TAM (Equipment + Services, High)	US\$ b	27.7	29.6	31.4	33.9	37.0	39.3	171.3	Incl. LTSA

Source: CCBIS estimates

The Bull Case: AI Power Demand Systematically Underestimated

The base case suggests Big Three capacity may be sufficient through the late 2020s — but only if the IEA forecasts are right. We hold a different view.

1. Forecasters have a consistent track record of being wrong, in one direction. The IEA raised its 2030 AI data center power estimate by 70% in a single revision cycle. Across the analyst community, forecasts produced in early 2023 have repeatedly come in 2–3x below subsequent actuals. If 2030 AI power consumption reaches 1,500–2,000 TWh rather than our base case 945 TWh, incremental gas turbine demand rises by 20–35 GW per year.

2. Inference is the driver nobody fully modelled. Training compute is lumpy and legible; inference scales continuously with user adoption and has no end date. Most forecasts written 18 months ago did not meaningfully embed inference demand — meaning a large share of future data center capacity additions is simply not in any model yet.

3. Behind-the-meter deployment is becoming the default configuration. Hyperscalers are co-locating gas turbines directly with their facilities in the 400–600 MW range, bypassing the grid entirely. BTM units run at 70–80% capacity factors versus ~40% for grid-peaking units, improving both the volume and quality of demand.

4. Grid bottlenecks have made on-site gas generation structurally necessary. US grid interconnection queues now run 3–7 years. Any data center operator facing a near-term capacity deadline has no practical alternative to BTM gas. This makes gas turbines a necessity, not a preference, establishing a demand floor that is largely insensitive to economics.

5. OEM management is telling you the demand runway is long. GEV's CEO expects the order book to be sold out through 2030 by end-2026. Siemens Energy describes demand visibility through the end of the decade. MHI is doubling manufacturing capacity. These are capital allocation decisions made by the people with the best information — and none of them are behaving as if a peak is near.

6. Every new geography re-runs the US playbook. AI data center demand is now replicating across Europe, the Middle East, Southeast Asia, and Japan. Each market follows the same path: hyperscale investment, grid constraints, BTM gas adoption. Geographic diffusion alone is sufficient to absorb any moderation in US demand growth.

The 2026 backlog peak thesis assumes AI demand unfolds in line with pre-inference-era forecasts. If even a few of the above materialize, the demand ceiling moves higher, backlog peaks push to 2028–2029, and the TAM expands accordingly. The base case in this model is a floor, not a midpoint.

AN OLIGOPOLISTIC MARKET: 85% BY GLOBAL BIG 3

The industrial gas turbine market is a textbook oligopoly at the top, graduating into a fragmented mid-tier as power output requirements fall. GE Vernova, Siemens Energy, and Mitsubishi Power collectively account for ~85% of the global market, their dominance resting on proprietary combustion chemistry, decades of certified field operating history, and aftermarket lock-in across 30-year asset lives. Below them, national champions (Ansaldo, Doosan, China's three power groups) have achieved F-class or emerging H-class capability; further down, aeroderivative turbines, small industrial machines, and large-bore reciprocating gensets serve distributed and backup applications where deployment speed and cost per kW matter more than thermodynamic efficiency.

The current supply environment has inverted the normal selection logic. The Big Three's combined production capacity of ~30 GW per year stands against 2025 global demand of 84 GW, stretching lead times to 6–7 years. Buyers who need power before 2030 are stepping down the product ladder — accepting lower efficiency and smaller unit output in exchange for availability. Each step down opens a wider field of suppliers, shorter timelines, and lower capital costs. The exhibit below maps this demand cascade across five tiers; the sections that follow describe each participant group in detail.

Classification of Gas Turbine

Tier 1: Heavy >100MW (G class)

GEV 9HA (826MW)



Siemens SCC6-8000H



MHI M701JAC



Ansaldo GT36



Doosan DGT6-300H



Tier 2: Mid-size 50-100 MW (F Class)

Doosan DGT6-H.S2
(in research, Hydrogen-compatible)

Dongfang Electric G50



Siemens SGT-800



GEV LM9000

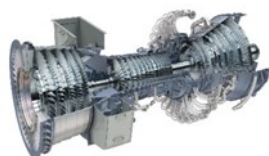


Mitsubishi M701F



Tier 3: 20-50MW Aeroderivative

Baker Hughes GE LM6000



Siemens SGT-A65



AEEC Taihang 25



Rolls-Royce Trent 60



Jereh Power to go



Tier 4: Small 10-25 MW Industrial / Marine

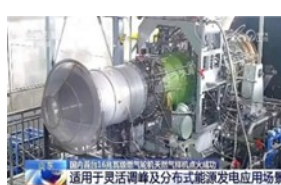
CAT / Solar Turbine
Titan 250



CSIC Power /
Guanghang CGT25-EA



Harbin Electric
16MW GT



Jereh
SGT-A05



MAN
MG16200



Tier 5: Large-bore reciprocating gensets 2-10 MW

Weichai 20M61 genset



CAT C175-20 diesel genset



Cummins QSK95 diesel genset



MTU Series 4000 genset



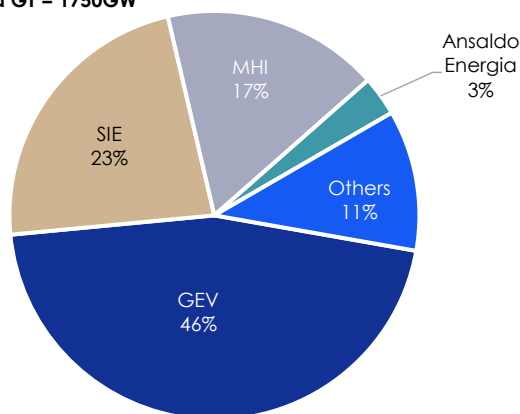
Source: Company, CCBIS

The Global Big Three = 85% market share

The oligopoly is stable and probably getting more so. CR3 = ~85% of global market. By installed/under-construction capacity, GE Vernova leads (~50+ GW), followed by Mitsubishi Power and Siemens Energy. In 2023 new order book by capacity, Mitsubishi Power briefly overtook both, capturing 36% market share and 56% of the advanced J/H-class segment.

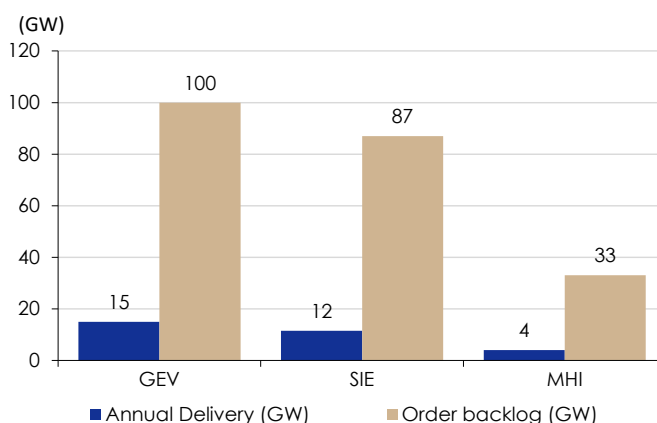
GT is a textbook oligopoly market with CR3 = 85%

Installed GT = 1750GW



Source: Company, CCBIS estimates

Global Big Three's order backlog implies 6-7 year lead time for delivery



Source: Company, CCBIS estimates

- Decades of proprietary combustion chemistry — thermodynamic IP accumulated over 50+ years, non-replicable without comparable R&D duration
- Certified field operating histories measured in millions of hours — the only bankable reliability reference for power project financing
- Digital monitoring platforms (GE APM, Siemens SPPA-T3000) creating aftermarket lock-in across 30-year asset lives
- Capital and time barriers: \$1b+ and 10–15 year development cycles required to produce a credible new heavy-duty turbine platform

Industry landscape of GT

Company	Stock code	Mkt share	Backlog	Capacity plan	Delivery	Key products
GE Vernova	GEV US	~46%	~100 GW (sold to 2029–30)	+US\$160m → 24 GW/yr	2030	9HA (826MW, >64% eff.), LM aero series
Siemens Energy	ENR GR	~23%	~33 GW (€146b+)	US\$1b US expansion	2030	SCC5-8000H (593MW max), SGT-800
Mitsubishi Power	MHI 7011 JP	~17%	87 units (+84% YoY)	2x capacity in 2 yrs	2029	J-series M701JAC (1,700°C), 1,600+ delivered
Ansaldo Energia	Private (12% owned by Shanghai Electric)	~3%	—	—	2028	GT36; mid-size specialist
Others	various	~11%	—	—	—	—

Source: Company, CCBIS estimates

GE Vernova (GEV US, NR) is arguably the global leader in technology breadth and product diversity (9HA for baseload, 7HA for mid-range, LMS100 for fast-start, aeroderivative LM series for offshore/marine). The 9HA.02 at 826MW CCGT with >64% efficiency is the industry's most thermally efficient machine. GEV is also the most "pure play" listed expression of the upcycle.

Siemens Energy (ENR GR, NR) rebuilt confidence post-Gamesa offshore wind losses. Gas turbine and grid infrastructure are the recovery thesis. Its gas portfolio spans heavy-duty, industrial, and aeroderivative, with max output 593MW. Valuation is more nuanced as the business is healing.

Mitsubishi Power (listed under MHI conglomerate) (7011 JP, NR) holds #1 market share by 2023 orders but is embedded in a diversified shipbuilding/defense/industrial group — making it a less pure play. The J-series reliability (99.5%) and 2.3M cumulative operating hours are industry-leading.

A. Big 3 Annual Manufacturing Capacity Plan (GW/yr)

OEM	2025	2026	2027	2028	2029	2030	Cumulative	Notes
GE Vernova (GEV)	15.0	20.0	22.0	24.0	25.0	26.0	117.0	2025: 15GW; 2026: 20 GW (US\$600m capex); 2028: 24GW; 2030F: 30 GW CEO: sold out through 2030
Siemens Energy (SIE)	11.5	12.5	13.0	13.5	14.0	14.5	67.5	FY25: 194 units; 2028F +20% = 13.2-14.4GW; \$1b US capacity expansion
MHI (Mitsubishi Heavy Industries)	4.0	5.5	6.8	8.0	9.0	10.0	39.3	2026: 2-year capacity doubling plan; 2028F: 10-15GW; Savannah plant expansion
Big 3 Total (GW/yr)	30.5	38.0	41.8	45.5	48.0	50.5	223.8	= GEV+SIE+MHI; 2025: 32.5GW → 2030: 63GW

B. Global Demand vs. Big 3 Supply → Gap Analysis

Metric	2025	2026	2027	2028	2029	2030	Cumulative	Notes
Global GT Demand Base (GW/yr)	40.0	41.8	43.5	46.1	48.8	50.5	230.8	From Leg 1 Base Scenario
Global GT Demand High (GW/yr)	46.1	49.3	52.4	56.6	61.6	65.6	285.5	From Leg 1 High Scenario
Big 3 Total Capacity (GW/yr)	30.5	38.0	41.8	45.5	48.0	50.5	223.8	= Big 3 capacity from table above
Gap (Base Demand – Big 3, GW)	9.5	3.8	1.7	0.6	0.8	(0.0)	7.0	+ve = Demand > Supply (China/Others opportunity); -ve = Supply > Demand
Gap (High Demand – Big 3, GW)	15.6	11.3	10.6	11.1	13.6	15.1	61.7	Bull case gap is larger and more persistent

C. China Gas Turbine Manufacturers TAM Estimate

Metric	2025	2026	2027	2028	2029	2030	Cumulative	Notes
Domestic OEM penetration assumption Base (%)	5.0	7.0	9.0	11.0	13.0	15.0		
Domestic OEM penetration assumption High (%)	5.0	8.0	10.0	14.0	17.0	20.0		
China OEM addressable market Base (GW)	2.0	2.9	3.9	5.1	6.3	7.6	25.8	= Total global demand × penetration rate
China OEM addressable market High (GW)	2.3	3.9	5.2	7.9	10.5	13.1	40.7	= Total global demand × penetration rate
Assumption: Domestic ASP Base (US\$/GW)	\$250M	\$250M	\$250M	\$250M	\$250M	\$250M		~35% discount to Big 3; primarily F-class
Assumption: Domestic ASP High (US\$/GW)	\$300M	\$300M	\$300M	\$300M	\$300M	\$300M		Narrower discount; H-class penetrates
China OEM equipment TAM Base (\$B)	0.5	0.7	1.0	1.3	1.6	1.9	6.5	= GW demand × \$250M/GW
China OEM equipment TAM High (\$B)	0.7	1.2	1.6	2.4	3.1	3.9	12.2	= GW demand high × \$300M/GW

Source: IEA, CCBIS estimates

The China Big Three: filling up the gap

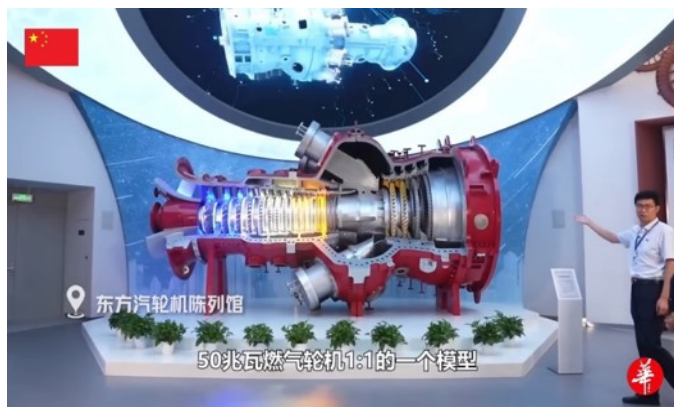
The three are state-owned enterprises that historically licensed technology from the global majors — Mitsubishi, Siemens, and GE — for domestic use, with IP restrictions preventing meaningful exports. That paradigm is cracking, driven by the global supply shortage that has forced AIDC customers to consider Chinese OEMs for the first time.

- **Rapidly closing technology gap:** F-class capability achieved domestically; H-class in development — second-tier but accelerating
- **Proven domestic track record:** thousands of operating hours in China's grid provide a growing commercial reference base
- **High CP value and fast market responsiveness:** not always best-in-class, but quick to adapt to emerging demand (AIDC, distributed power) and competitively priced
- **SOE backing:** national capital support enables aggressive R&D investment, pricing flexibility, and policy-driven market access

Dongfang Electric 东方电气 (600875 CH / 1072 HK, Outperform) — the domestic champion going global

- **China's largest power equipment conglomerate:** full energy spectrum (coal, hydro, nuclear, wind, solar, gas) with dominant domestic positions across all segments
- **Gas turbine R&D since the 1950s:** licensed MHI F-class technology in 2003, progressing over two decades from localization to autonomous development — culminating in the G50
- **The national pride G50:** China's first fully indigenous F-class heavy gas turbine (争气机) — 100% domestic hot-end manufacturing, 12,000+ operating hours, ~36.5% simple cycle efficiency; decisive edge is 13-month delivery vs. Big Three's 6–7 years
- **Export breakthrough:** 2025 orders in Kazakhstan and Iraq; May 2026 — 10-unit North American datacenter contract at US\$2b marks first developed-market penetration; further orders in discussion

G50 China's first fully indigenous F-class heavy gas turbine



Source: Company, CCBIS

Harbin Electric's flagship HGT16 is a 16MW GT



Source: Company, CCBIS

Harbin Electric 哈尔滨电气 (1133 HK, NR) — Value play on the power equipment upcycle: gas turbine as option

- **Traditional power equipment company:** core businesses in coal, hydro, and nuclear generation, themselves in a multi-year upcycle driven by China's baseload expansion
- **Gas turbine development 1–2 years behind Dongfang Electric:** HGT16 (16MW) rolled off production line in 2025 but still in validation phase
- **HGT16 sub-scale, fast deployment:** HGT16 targets distributed generation, industrial applications, and AIDC backup power — smaller form factor is a differentiator in faster-deployment, sub-scale use cases
- **Power equipment + Stock Connect:** Investment case anchored on conventional power equipment earnings (+40% profit CAGR consensus 2024–2027); near-term catalyst: anticipated Stock Connect inclusion (Aug 2026)

Shanghai Electric 上海电气 (601727 CH / 2727 HK, NR) — Most peripheral of the three

- **Gas turbine licensed from Siemens:** Gas turbine exposure primarily via longstanding Siemens JV — manufacturing licensed F-class machines for the domestic market; new-generation F-class passed 168-hour full-load trial in June 2025
- **Least focused on indigenous gas turbine** among the China Big Three; trailing DEC by 1–2 years in commercial operating hours, no export track record
- **Highly diversified businesses:** nuclear components, offshore wind, electrical and industrial equipment, integrated services — gas turbines are a small, non-disclosed segment of a complex group

Business exposure of Global Big 3 and China Big 3

Business		GEV	SIE ENR	MHI	Dongfang E	Harbin E	Shanghai E
Traditional	Coal	5%		6%	26%	41%	17%
	Gas	43%	31%	19%	8%	5%	
	Hydro	2%			5%	12%	
	Nuclear	2%		7%	7%	12%	5%
Renewable	Wind	24%	25%		21%		10%
	Solar				1%	1%	10%
Power Infrastructure		25%	28%		9%	10%	18%
Other Infrastructure				17%			
Services				22%	11%	6%	8%
Industrial	Automation		15%				4%
	Others		3%	3%	13%	13%	28%
Aerospace				27%			

Source: Company, CCBIS estimates

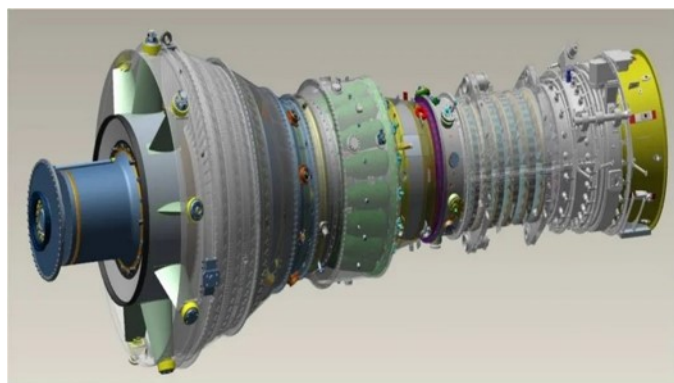
Cross-over & Adjacent Players = Lower Efficiency for Shorter Delivery Time

These companies are not classical heavy gas turbine OEMs. They enter the AIDC power theme via different technology routes — naval-to-land conversion, aero-derivative turbines, Korea's emerging H-class platform, and large-bore reciprocating engines. Each addresses a different segment of the power demand waterfall described in the previous section.

CSIC Power 中国动力 (600482 CH, NR) — Naval-to-land: from warships to power plants

- **CSSC's listed power equipment arm** — covers gas, steam, diesel, and integrated electric propulsion across defense and civilian markets
- **Primary supplier of naval gas turbines (6MW and 25MW-class)** to the Chinese navy; 100% market share in large naval steam turbine systems — sole state-designated navy supplier
- **Guanghan Power subsidiary:** 25MW industrial turbine deployed across pipelines, offshore platforms, distributed energy, and barge power; stackable to 100–200MW for AIDC — key upside, but assets not yet injected into the listed entity
- **Brazilian oil project** speculation triggered +108% stock run (Dec 2025) before company clarification; export story directionally valid but early-stage

CSIC Power subsidiary Guanghan CGT25-EB



Source: Company, CCBIS

AECC Aviation Power Taihan AGT-110



Source: Company, CCBIS

AECC Aviation Power 航发动力 (600893 CH, NR) — Aero-derivative: Aviation technology meets the grid

- **Listed manufacturing arm of AECC** — China's only developer of a full-spectrum military aviation engine program; leveraging the aero-derivative pathway into industrial power (mirrors GE's LM series model)
- **Taihang 25 (~25MW):** commercially deployed in distributed generation; achieved first grid synchronization on NEA gas turbine innovation demonstration project
- **Taihang 110 (110MW, fully indigenous IP):** passed national acceptance review — strategically significant, but commercial entity (AECC Gas Turbine Co.) sits outside the listed vehicle for now; injection timing unknown
- **Export progress:** April 2026 — five Taihang 7 turbines contracted for PETRONAS FPSO project in Malaysia; first domestically-made Chinese turbine in the high-end offshore engineering market

Doosan Enerbility 斗山能源 (034020 KS, NR) — The fifth country to master large gas turbine technology

- **Fifth country possessing gas turbine:** Only non-Big-Three OEM to independently develop world-class heavy-duty gas turbines from scratch — “fifth country” after US (GE), Germany (Siemens), Japan (MHI), and Italy (Ansaldo)
- **DGT6-300H S2 (380MW H-class):** world's most comprehensive full-scale, full-speed, full-load test facility at Changwon; 15,000+ commercial operating hours; fully Korean-made CCGT offering (turbine, steam turbine, HRSG, controls) — no Chinese OEM has matched this validation infrastructure
- **xAI Colossus:** 12 cumulative units as of March 2026 (including xAI Colossus datacenter) — from zero international references 18 months ago; compelling proof of export breakthrough
- **Dimension-reduction strike to mid-size GT for AIDC:** Expanding to 90MW mid-size segment targeting AIDC hyperscalers — filling Big Three capacity cascade; fully Korean-made CCGT carries geopolitical appeal for Western buyers wary of Chinese OEMs

Doosan's DGT6-300H S2 makes Korea the fifth country to own large-scale GT



Source: Company, CCBIS

Weichai's 5MW genset 20M61 is a large-bore diesel engine



Source: Company, CCBIS

Weichai Power 潍柴动力 (2338 HK/ 000338 CH, NR) — Large-bore diesel engines: The same theme, a different layer

- **Diesel engines for AIDC:** Different technology (large-bore reciprocating diesel engines), same AIDC macro catalyst — dominant supplier of core diesel backup genset components for data centers (70–80% of genset cost)
- **Disrupting the Western oligopoly (Cummins, Caterpillar, MTU):** foreign/JV brands held ~80% of China's datacenter engine market in 2024; Weichai datacenter engine sales up 491% YoY in H1 2025; full product range complete, max 5MW per unit
- **Delivery lead time under 2 months** vs. foreign brands' 10+ months — same speed-vs-technology dynamic driving Chinese gas turbine OEM export gains
- **SOFC optionality:** 20% stake in Ceres Power + manufacturing license signed Nov 2025; SOFC (Solid Oxide Fuel Cell) efficiency 55–65%, deployable in 90 days vs. 1–2 years for gas turbines

AIDC Power Equipment Comparison: large-scale GT (>100 GW) are most preferable in terms of efficiency; yet delivery time (supply shortage) pushes AIDC to adopt mid-scale GT

Maker	AIDC-relevant model	MW (SC)	SC eff. (%)	CC eff. (%)	Start-up (mins)	Fuel	Lead time	Units/ GW
Overseas Big 3 — AIDC on-site direct (modular mid-size, self-generation behind the meter)								
GE Vernova	LM6000PF+ (aeroderivative)	53.8	42	56	< 5	Nat. gas	3–5 yr	19
GE Vernova	LM2500+G5 (aeroderivative)	38.3	41	56	< 5	Nat. gas	3–5 yr	26
Siemens Energy	SGT-800 (industrial frame)	57–62.5	40–41	58.5	~13	Nat. gas	3–5 yr	16–18
Siemens Energy	SGT-A35 (aeroderivative)	32–39	41.9	~56	10	Nat. gas	3–5 yr	26–31
Mitsubishi Power	H-25 (industrial frame)	32	~35	~52	~15	Nat. gas	3–4 yr	31

Overseas Big 3 — Utility-scale grid machines (bought by utilities to serve large AIDC campuses via grid)

GE Vernova	7HA.02 / .03	384 / 430	42–43	> 61	~30	Nat. gas	2029–30	2–3
Siemens Energy	SGT5-4000F	329–385	41–41.5	> 60	~30	Nat. gas	2030+	3
Mitsubishi Power	M501JAC (J-class)	~470	42	> 62	~30	Nat. gas	2029+	2

Doosan Enerbility — Korea's 5th indigenous GT maker; 23 units ordered, 12 destined for US data centers

Doosan Enerbility	DGT6-H.S2 (H-class, 380 MW)	380	43	~60	~30	Nat. gas	2029+ (7 units to US DC from May 2029)	3
Doosan Enerbility	~90 MW mid-size (in dev.)	~90	~40*	~58*	~15	Nat. gas	TBD — targeting AIDC/ hyperscalers	11

Baker Hughes / Jery — Aeroderivative (same mid-size class; sold as integrated AIDC packages)

BH via Jery	LM6000PF+ (packaged system)	53.8	42	56	< 5	Nat. gas	2–3 yr	19
BH via Jery	LM2500+G5 (packaged system)	38.3	41	56	< 5	Nat. gas	2–3 yr	26

China OEMs — Self-developed, freely exportable machines

Dongfang Electric	G50 (F-class, fully indigenous)	50	36.5	~53	~20	Nat. gas	12–14 mo	20
China Power (CSSC)	Longjiang Guanghan ~25 MW	25	~34*	~50*	~20	Nat. gas	< 12 mo	40
Harbin Electric	HGT16 (prototype validation)	16	~33*	~48*	~20	Nat. gas	2027+ est.	63

AECC Aviation Power — Technically superior but civilian supply near-zero

AECC Avn. Power	Taihang 110 (aeroderivative)	110	~38*	~55*	< 5	Nat. gas	Military priority — booked to 2030	9
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Weichai Power — Diesel genset (emergency standby; not continuous primary power)

Weichai Baudouin	20M55 diesel genset	4.25	~43*	—	< 30 sec	Diesel	< 6 mo	235
Weichai Baudouin	20M61 diesel genset (2025)	5.0	~43*	—	< 30 sec	Diesel	< 6 mo	200

Source: Company, CCBIS estimates

Turbomachinery family and estimated market size

Category	Description · key OEMs	Est. annual market (new equip., 2025)
Turbomachinery (涡轮机械)	Any machine that exchanges energy between a rotor and a fluid following Brayton cycle—includes gas turbines, steam turbines, and hydraulic turbines	~US\$500b+ all segments combined (equip. + services)
Gas turbines (燃气轮机)	Air-breathing: compress atmospheric air as oxidiser → combust fuel → expand through turbine.	~US\$80-100b/yr all industrial GT incl. services; excl. aviation MRO
Industrial gas turbines (工业燃气轮机) ★ THIS REPORT	Stationary power generation & mechanical drive. Sub-divided by size and design origin	~US\$25-30b/yr new equipment only; +US\$41b services (2024)
Heavy-duty large frame (重型框架) ≥100 MW	Slow start, massive output. H/J-class: TIT 1,600–1,700° C, CCGT eff. >62% GEV 9HA · Siemens SGT5-9000HL · MHI M701JAC · Dongfang G50 · Doosan DGT6-H.S2	~US\$15-20b/yr equip. only; our report 5-yr TAM USD100-140b incl. services + AIDC surge
Aero-derivative (航改燃) 20–75 MW	Aircraft jet engine redesigned for power. Light, fast-start (mins), higher USD/MW vs heavy-duty Baker Hughes LM9000 (73 MW) · LM6000 (54 MW) · LM2500 (32 MW) · RR Trent 60	~US\$4-6b/yr new equipment; ~220 GW installed globally
Small industrial frame (小型工业) 5–50 MW	Purpose-built for small-scale industrial use — not derived from aircraft. Modular, cost-effective for CHP and O&G drive CAT Solar Turbines (Titan, Mars, Centaur) · MAN Energy MGT · Siemens SGT-800 (62 MW)	~US\$3-5b/yr 10,000+ units globally
Aviation jet engines (航空发动机)	Generates THRUST, not shaft power. Turbofan, turbojet, turboprop, turboshaft. Not in this report CFM56 / LEAP (GE+Safran JV) · GE90/GE9X · P&W GTF · Rolls-Royce Trent · AECC WS-15	~US\$60-65b/yr new equip. only; ~US\$100b+ incl. MRO; ~60,000 active engines
Marine / naval propulsion (舰舰推进)	Mostly aero-derivative gas turbines adapted for naval vessels. Commercial shipping mainly uses diesel; GT used in warships, LNG carriers BKR LM2500 (32 MW, US Navy destroyers) · RR MT30 · 703 QD-70 (PLAN)	~US\$2-4b/yr GT-specific portion; BKR LM2500: 2,300+ naval deliveries
Rocket propulsion (火箭推进)	Carries own liquid oxidiser (LOX). Different operating principle from gas turbines. Turbopumps inside are technically gas turbines SpaceX Raptor (full-flow staged combustion) · ULA Vulcan · ISRO · CASC LM series	~US\$25-30b/yr commercial + govt launch market; ~250 launches in 2024

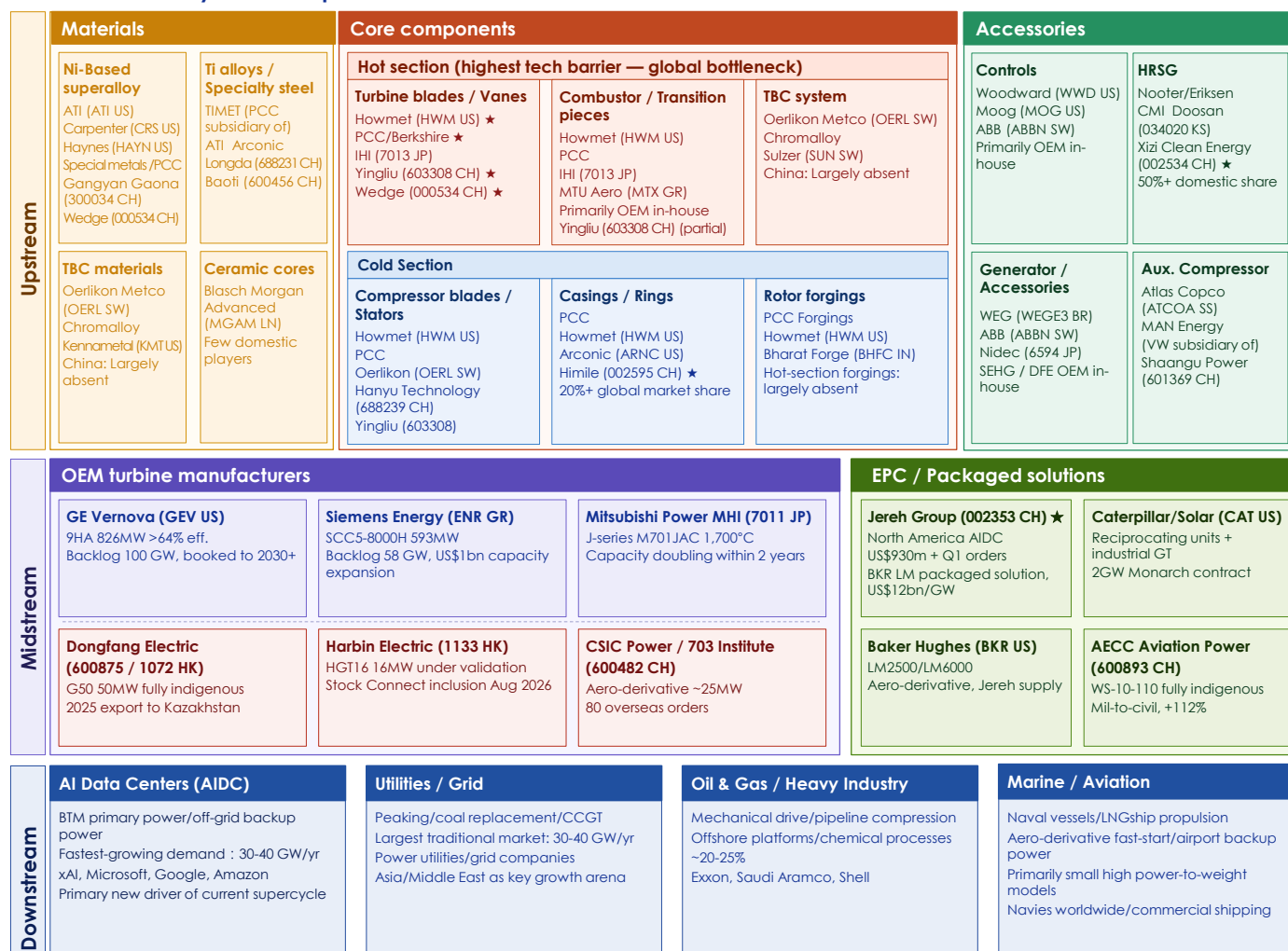
■ In scope (this report) ■ Out of scope ■ Different principle (rocket) ★ Primary focus

Source: GMIinsights, Precedence Research, Mordor Intelligence, CCBIS

THE VALUE CHAIN OPPORTUNITIES LIE IN THE HOT SECTION

Gas turbines are among the most technically complex machines ever built. The supply chain creates investment opportunities at every tier.

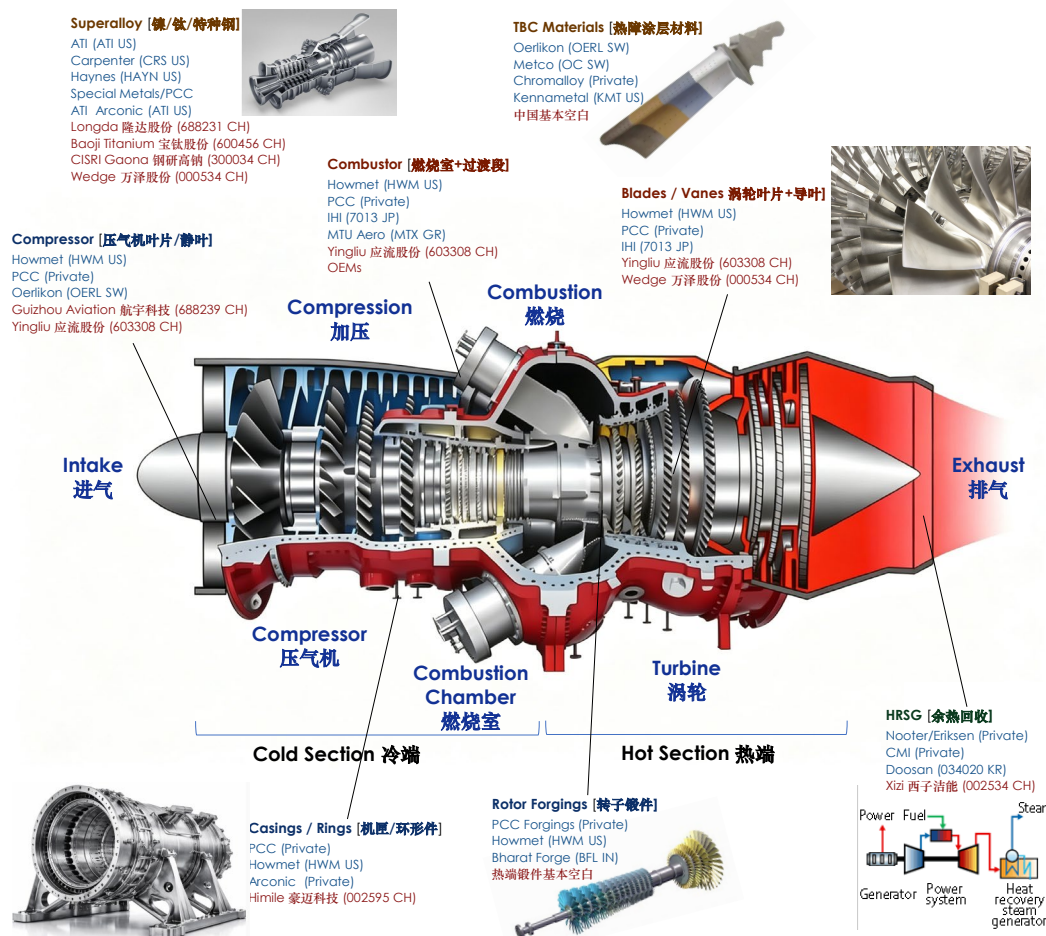
Gas turbine industry chain map



■ Blue = Global suppliers (Listed/Private) ■ Red = China suppliers (A/H share) ★ = Key stock picks

Source: Company, CCBIS

Gas turbine upstream opportunities



Source: Company, Ansys, Jungyi, Pengky, CCBIS estimate

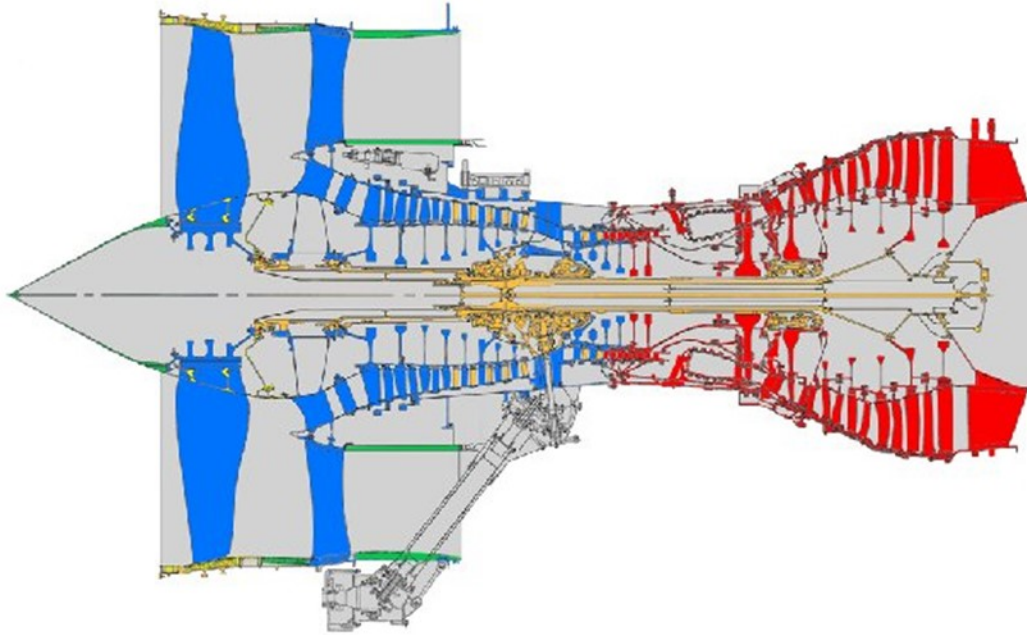
Supply Chain Map

A gas turbine is arguably the most technically complex machine in commercial power generation. The supply chain reflects this: each tier has distinct technical barriers, concentrated competitive dynamics, and different implications for investors. Below we map the chain using the same structure as our supply chain diagram — six tiers from raw materials to packaged solutions.

Upstream — Materials: the secret recipe is not replicable and is getting privatized

What it is: The foundational inputs for turbine manufacture — primarily nickel-based superalloys (for hot-section components), titanium alloys and specialty steels (for cold-section structural parts), thermal barrier coating materials (ceramic powders applied to hot-section surfaces), and ceramic investment casting cores (used to create internal cooling channels in turbine blades).

The different materials used in a Rolls-Royce jet engine. Titanium in blue, ideal for strength and density, but not high temperatures; Nickel-based superalloys in red for high temperature; Steel in orange for the static parts of the compressor



Source: University of Cambridge

Value share: Materials account for roughly 15–20% of total turbine cost, but their impact on performance is disproportionate. The choice of superalloy and the precision of its chemistry and microstructure directly determines how hot the turbine can run — and thermal efficiency scales with operating temperature.

Technical barriers: Nickel superalloy formulation is one of the most tightly held bodies of knowledge in manufacturing. Single-crystal directionally-solidified alloys — required for the most demanding hot-section blades — require precise control of composition, grain growth, and heat treatment. Qualification cycles with OEMs typically take 3–5 years. Thermal barrier coatings (TBC) are a separate sub-discipline: yttria-stabilised zirconia applied via plasma spray or electron-beam physical vapour deposition (EB-PVD), requiring ceramic chemistry expertise and proprietary deposition processes.

Nickel-based superalloys are used in hot section



Source: Fushun Special Steel

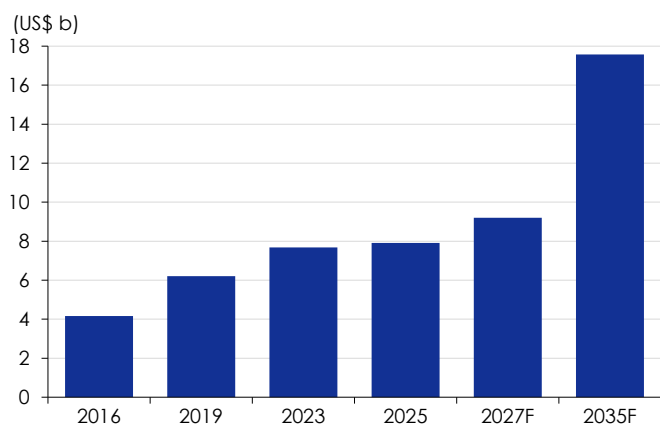
Titanium alloys are used in cold section



Source: MIT

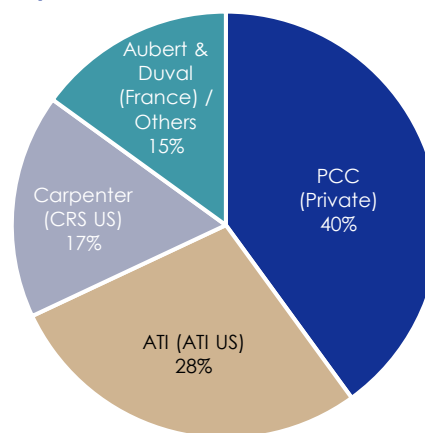
Competitive landscape: The global **superalloy** market is dominated by US-listed specialists — ATI (ATI US, NR), Carpenter Technology (CRS US, NR), and Haynes International (HAYN US, NR) — alongside Special Metals (a PCC subsidiary) and Aubert & Duval (France). These companies supply all three global OEMs and have decades of OEM qualification histories that are extremely difficult to replicate. For **thermal barrier coatings**, Oerlikon Metco (OERL SW, NR) is the global technology leader, with Chromalloy (private) and Kennametal (private) as secondary players. In China, CISRI Gaona (300034 CH, NR) holds ~35% domestic superalloy market share with ~60% gross margins, and Wedge (000534 CH, NR) and Longda (688231 CH, NR) address single-crystal and titanium alloy needs respectively. However, Chinese superalloy producers remain largely domestic-focused; achieving OEM qualification for export supply is a multi-year undertaking and none have broken through materially at the global level.

Nickel-based superalloys are used in hot section



Source: Allied Market Research, Academic Industrial Review Forecast, Commercial Market Intelligence, CCBIS estimates

Titanium alloys are used in cold section



Source: MIT

Upstream core components — Hot-section vanes and blades are the highest value-adding

What it is: The turbine blades, nozzle guide vanes (导叶), combustion liners, and transition pieces that operate immediately downstream of the combustion chamber. These are the highest-temperature, highest-stress components in any gas turbine, operating at 1,400–1,700°C with turbine inlet temperatures that exceed the melting point of the base alloy — only possible through a combination of advanced alloy chemistry, single-crystal casting, and intricate internal convective and film-cooling passages.

Value share: Hot-section components represent c.35–40% of total turbine cost, and a significantly higher share of lifecycle revenue — hot-end parts typically require replacement every 24,000–30,000 operating hours (c.3–4 years at data center utilization rates), generating recurring aftermarket revenue that OEMs price at a significant premium to the original part.

Technical barriers: This is the single most technically demanding segment in the entire supply chain, and the one Elon Musk specifically identified as the global bottleneck in a podcast interview. Single-crystal casting requires directional solidification furnaces, proprietary ceramic core geometries (to create cooling channels as narrow as 0.5mm), and post-casting processes including electrochemical drilling, coating deposition, and dimensional inspection at tolerances of ± 0.01 mm. A new supplier typically requires 3–5 years of process development and 2–3 years of OEM qualification testing before receiving production orders. Lead times for hot-section components have extended from 6 months pre-2023 to 18–24 months today, making this the physical bottleneck constraining OEM output growth — not the turbine assembly itself.

Competitive landscape: Globally, Howmet Aerospace (HWM US, NR) and Precision Castparts (PCC, Berkshire Hathaway subsidiary) together account for an estimated 75–80% of the world's single-crystal turbine airfoil capacity. IHI (7013 JP, NR) serves Mitsubishi Power's needs. MTU Aero Engines (MTX GR, NR) has limited industrial turbine exposure. Chromalloy (private) focuses more on MRO repair than new part production. In China, Yingliu (603308 CH, Outperform) is the standout: it holds 38% domestic market share for turbine blades, is the only Chinese company certified to supply Siemens F-class and H-class hot-end blades, and has

achieved c.50% global market share in Siemens' 4000F model specifically. Wedge (000534 CH, NR) addresses the single-crystal alloy and blade space at smaller scale but with high gross margins (~73%). The honest assessment is that China's domestic hot-end capability remains concentrated at the F-class level; H-class (>1,600°C inlet, the current frontier) remains largely a Western OEM domain.

Hot-section components represent the highest technological moat of GT manufacturing



Source: Company

Hot-section components represent 35-40% of total turbine cost

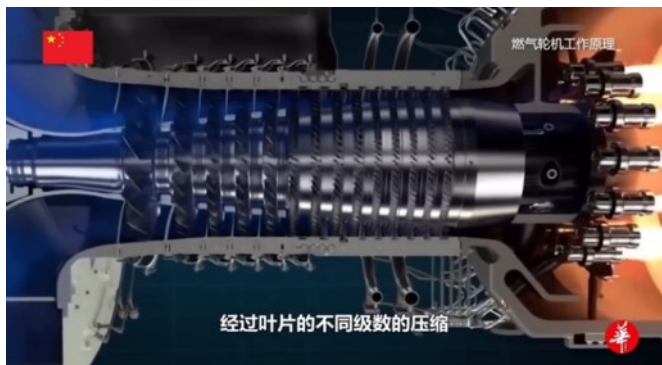


Source: Company

Upstream core components — Cold-section: lower moat but higher volume

What it is: Components in the compressor and structural sections of the turbine that operate at lower temperatures but under high mechanical loads — compressor blades and stator vanes, casings and ring components, and rotor disc forgings (the rotating discs onto which turbine and compressor blades are mounted).

Cold-section components include compressor and structural sections



Source: Company

Cold-section components represent 20-25% of total turbine cost



Source: Company

Value share: Cold-section components collectively account for c.20–25% of turbine cost. While individually less exotic than hot-section parts, the cumulative precision machining and forging requirements are substantial, and rotor forgings in particular are an emerging bottleneck — EPRI (the US Electric Power Research Institute) has flagged rotor forgings alongside turbine blades as the two primary supply chain constraints limiting OEM production output.

Technical barriers: Compressor blades require precision aerofoil profiling (± 0.02 – 0.05 mm tolerances) and fatigue-resistant titanium or steel alloys; the technical barrier is lower than hot-section but OEM qualification remains a 2–3 year process. Casing components (ring-shaped structural housings for compressor and turbine stages) require precision boring, grinding, and welding of large-diameter ring forgings — tight circularity tolerances prevent vibration and tip-clearance issues. Rotor forgings are a more concentrated bottleneck:

requiring large-press isothermal forging capability, only a handful of global companies (PCC Forgings, Howmet Forgings, Bharat Forge of India, and IHI) have the scale and qualification to produce critical rotating disc forgings for heavy-duty turbines.

Competitive landscape: PCC and Howmet again dominate precision casting of cold-section structural parts. For ring components and casings specifically, Himile (002595 CH, NR) is the most globally competitive Chinese company in this entire supply chain: it holds an estimated 20%+ global market share in ring/casing components, with all three Big Three OEMs as customers, and net profit growing 30%+ annually. This is not domestic substitution — this is genuine global market penetration achieved over two decades of investment in precision machining. Guizhou Aviation (688239 CH, NR) supplies casing ring forgings to GE and Siemens and is a second credible name in this tier.

Upstream components — Auxiliary & other: a few niche and oligopolistic markets

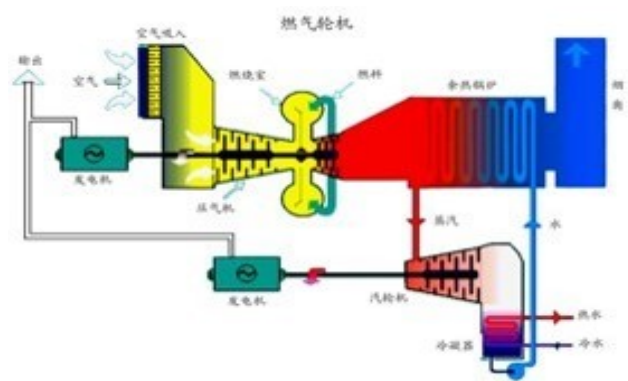
What it is: The systems that wrap around the core turbine to make it a functional power generation asset: control and protection systems (monitoring thousands of parameters and executing sub-second protective trips), heat recovery steam generators (HRSG, which capture exhaust heat in combined cycle configurations to drive a steam turbine for additional power output), generators, auxiliary compressors, air/fuel treatment systems, and emissions control equipment.

Vacuum Induction Melting and Casting Furnace is essential for superalloys, yet technical knowhow remains the irreplaceable moat



Source: ALD

HRSG can utilize the exhaust heat from the gas turbine



Source: Xizi Clean Energy

Value share: Auxiliary systems collectively account for 15–25% of turbine island cost, rising to 40–50% when the full combined-cycle power block is considered (adding steam turbine, condenser, and electrical systems). For simple-cycle configurations used at data centers, the proportion is lower.

Technical barriers: Control systems require deep integration with the specific turbine OEM's proprietary architecture; Woodward (WWD US) and Moog (MOG US) are the key independent control systems specialists, though OEMs increasingly develop in-house. HRSG design is complex — achieving maximum heat recovery without unacceptable back pressure or material creep failure — but the technology is more mature and the market less concentrated than hot-section components. The competitive intensity in auxiliary systems is higher, with more independent suppliers.

Competitive landscape: Woodward and Moog dominate turbine control systems internationally. For HRSG, the market is fragmented globally: Nooter/Eriksen (private, US), CMI Group (Belgium), Doosan (steam turbine + HRSG integration), and Mitsubishi Power (in-house). In China, Xizi (002534 CH, NR) holds over 50% domestic HRSG market share and has recently achieved the ability to export to US data center projects bundled with Chinese OEM turbine packages — a meaningful new revenue opportunity. For auxiliary compressors and fans, Shaangu Power (601369 CH, NR) is the domestic leader. The auxiliary segment is generally less attractive to investors seeking the highest-conviction supply chain names, as competitive intensity is higher and differentiation lower than hot-section parts.

Midstream — OEM: Two ends of the smiling curve: IP & aftermarket services are lucrative

What it is: The system integrators who design, assemble, test, and warranty the complete turbine — procuring components from across the supply chain and delivering a finished machine. OEM value is not primarily in component manufacturing (much is outsourced) but in proprietary aerodynamic design, thermodynamic optimization, combustion technology (particularly low-NOx dry combustors), digital monitoring software, and the installed base of service contracts that generate recurring revenue for 30+ years.

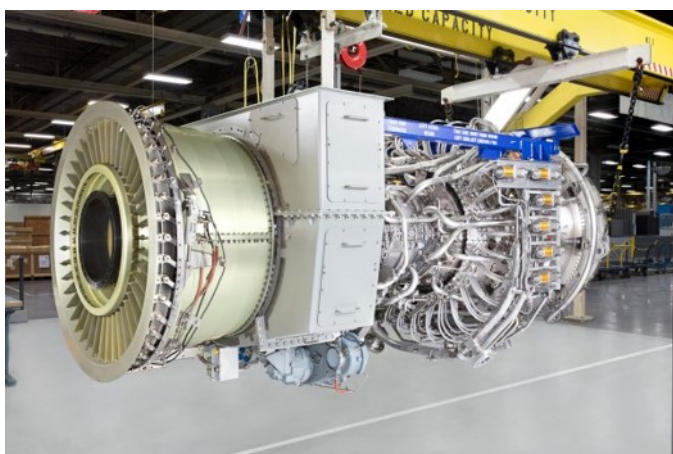
Value share: OEM integration and IP capture roughly 30–40% of the total turbine system value at point of sale, rising to 50%+ when aftermarket services over the full asset life are included. Service gross margins (40–50%) typically run 2–3x higher than new equipment margins (10–20%), and this aftermarket flywheel becomes more valuable with every unit commissioned.

Technical barriers and competitive dynamics: The Big Three's moat is multi-layered — decades of combustion technology development (particularly H-class turbines at 1,600–1,700°C), proprietary digital twin and remote monitoring platforms, and installed base lock-in (operators are highly reluctant to switch service providers mid-asset-life). Market share has been stable at CR3 ~85% for over a decade. New entrants face 10–15 year development timescales and billions in capital expenditure before achieving credible commercial references. China's three power groups (Dongfang Electric, Harbin Electric, Shanghai Electric) have only recently achieved genuine indigenous F-class capability with Dongfang's G50; H-class remains a future target. The key near-term dynamic is that Big Three capacity constraints have opened an unusual window for Chinese and Korean OEMs to win orders on the basis of delivery speed rather than technology differentiation.

Midstream — EPC packaging & alternative solutions: Most direct turnkey solutions to AIDC customers

What it is: This tier covers two distinct but related categories. First, **EPC and power packagers** — companies that procure turbine cores (often aero-derivative units) from OEMs and assemble complete turnkey power generation systems for customers who need power quickly and do not want to manage the engineering themselves. Second, technology adjacencies — **naval-to-land conversion turbines, aero-derivative units, and large-bore diesel gensets** that serve the lower end of the AIDC power stack where heavy turbine delivery timelines are prohibitive.

GT produced by Baker Hughes, packaged by Jereh



Source: Baker Hughes

EPC and power packagers deliver complete turnkey generation systems for customers i.e. AIDC



Source: Jereh

Value share: A packager like JerehJereh (002353 CH) typically captures 30–40% margin on the value-added engineering, integration, commissioning, and site services — the turbine core itself is pass-through at near-zero margin. The value add is speed, logistics expertise, and relationships with end customers.

Technical barriers: The technical barrier in packaging is lower than OEM, but execution capability is underestimated by markets. Delivering a multi-hundred-megawatt power system to a data center construction site in North America within 12–18 months requires regulatory navigation, site engineering, logistics, and an ability to source and stockpile turbine cores in advance. Jerih's competitive moat is its pre-existing US physical presence from oilfield services, its relationship with Baker Hughes for LM-series aero-derivative cores, and its willingness to hold inventory risk that most competitors avoid. For alternative solutions (naval-to-land, aero-derivative), the barriers are more technology-driven — decades of accumulated operating data and military qualification that cannot be replicated quickly.

Competitive landscape: In traditional packaging and EPC, Caterpillar (via Solar Turbines for industrial gas turbines, and its own large reciprocating gensets) and Baker Hughes (LM-series aero-derivative packages) are the major Western players. In China, Jerih has carved out a unique position as the only Chinese company with demonstrated ability to win and deliver large power contracts in North America. For naval-to-land conversion, CSIC Power 中国动力 (600482 CH, NR) via Guanghan Power (25MW) is the primary listed vehicle. For aero-derivative industrial turbines, AECC Aviation Power 航发动力 (600893 CH, NR) via the Taihang series (25–110MW) is developing capability, with the 110MW unit still in early commercial deployment. For large-bore diesel gensets — the backup power layer rather than primary — Weichai 潍柴动力 (2338 HK / 000338 CH, NR) is the domestic national champion, having taken share from the Cummins/CAT/MTU oligopoly on the strength of 2-month delivery vs. their 10+ months, with H1 2025 data center engine sales +491% YoY.

Downstream applications: Utility, oil & gas, industrial, marine/ defense, AIDC

Gas turbines serve five distinct downstream end-markets, and the demand mix is shifting meaningfully.

Downstream applications of GT include utility, oil & gas, industrial, marine, defense, data center etc.

End-market	Share*	Turbine type	Key driver / data point
Utility power gen	55–65%	Heavy-duty CCGT (>225MW = ~80% of 2024 orders)	Coal retirement — US planned gas additions doubled from <25 GW to >45 GW through 2030 (2021→end-2024)
Oil & gas	20–25%	Aero-derivative	LNG build-out — US export capacity +80% by Dec-2028; power + O&G packaged-GT value ~US\$163b over next decade
Industrial / cogen	10–15%	CHP & mechanical drive	Light industrial fastest-growing at ~7.5% CAGR
Marine / defense	5–7%†	Aero-derivative	Fleet renewal, all-electric warships, IMO-2030 LNG carriers; ~6.6% CAGR to 2030
AI data centers	New-flow leader	Heavy-duty + aero	65% of Siemens' 2025 GT orders; NA order share 14%→26%→40 – 50% (2023 / 2024 / 2025 – 30E)

Source: Company, CCBIS estimates

Takeaway: installed-base percentages in most research reports describe yesterday's demand. The new-order flow says hyperscalers — not utilities — are now the swing customer.

WHERE WE ARE IN THE CYCLE — AND WHAT TO WATCH

Peak Orders ≠ Peak Earnings

Wood Mackenzie forecasts global gas turbine order intake will peak in 2026, and the data broadly agrees: GEV booked 7 GW in Q1 2026 (tripling YoY) while Siemens Energy recorded its highest-ever quarterly intake at €8.87b in Q2 FY2026. These numbers are approaching the ceiling of what the market can physically absorb — when buyers can no longer pull forward delivery by ordering earlier, booking urgency fades. We believe 2026 might be the order peak.

The critical insight: **peak orders is not peak earnings**. With 6–7 year lead times, the revenue and FCF cycle from today's bookings runs to 2031–2033. The sector re-rates from order-momentum to delivery-and-margin-expansion — a transition that compresses growth multiples but does not impair the earnings trajectory. Investors who sell at peak orders likely sell two to three years too early.

Who Can Say AI Compute Demand is Enough?

Alongside the peak-order thesis sits an equally plausible counter-case: that **2026 might not be the ceiling**. Every AI compute demand forecast since 2022 has been revised upward within 18 months of publication. Three demand vectors are still at the beginning of their curves — (1) **reasoning models** that multiply per-query compute by 10–100x, (2) **always-on agentic workloads**, and (3) **physical AI** — none of which are meaningfully reflected in current power demand estimates. If AI has taught investors one lesson over the past five years, it is that the direction of revision is upward. Under that lens, the 60–100 GW AIDC-driven GT demand base case looks conservative; gas turbines, as the only dispatchable solution deployable at GW scale within two years, would see order momentum extend rather than taper.

Five indicators to monitor the industry cycle

Indicator	What signals trouble	Current reading
(1) Big Three delivery lead times	Compression 6–7yr → 4–5yr = supply catching up	No change. GEV targeting 24 GW/yr by 2028 vs 100 GW backlog = 4+ yr queue. Earliest compression: late 2027.
(2) GEV Gas Power EBITDA margin	Guidance cut = cost overruns or supply delays	On track. Q1 2026: 9.6%; full-year target 17–19%. Q1 FCF \$4.8b alone exceeded all of FY2025.
(3) Hyperscaler capex	Sustained reduction = AIDC demand softening	Accelerating. Q1 2026 earnings confirmed no slowdown across Microsoft, Meta, Google, Amazon. Risk: gradual normalization in 2027–2028 as initial build phase completes.
(4) Nickel and cobalt (LME)	Price reversal = immediate margin compression at component level before OEM earnings	Benign. Indonesian supply expansion keeping costs contained; Yingliu gross margin expanding toward 38%. Watch for supply discipline signals.
(5) Dongfang G50 North America	Cancellation or delay = re-rating reversal on China GT export thesis	Positive. Canada datacenter: 10 units (~RMB2b) contracted for end-2026 delivery. Second batch of 10 in discussion. Additional North American enquiries in pipeline.

Source: Bloomberg, Company, CCBIS estimates

The Transition

The next 12–18 months will test whether investors can hold through the narrative shift. Order headlines will slow — that is normal and expected. What replaces them is margin expansion (GEV's Gas Power moving from 10% toward 18%), FCF inflection (buybacks, de-leveraging), and the first evidence of the services compounding story. For China names, the transition is from "will Dongfang get any export orders" to "how many, how fast, and what margins" — a higher-quality problem to have.

SECTOR CALLS – WHERE TO OWN THE CYCLE

The positioning question is not whether to own the gas turbine upcycle but where in the value chain the risk-adjusted return is best. Our answer: own components over OEM, own China for growth, and within each tier concentrate in the highest-moat name.

Components Over OEM — Four Structural Reasons

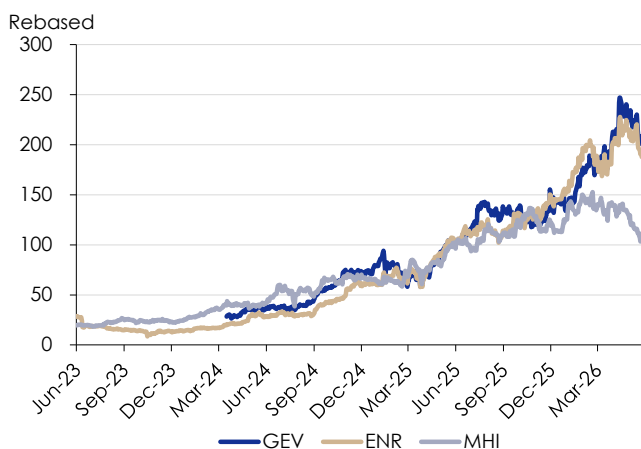
First, the TAM is wider. Hot-section blade manufacturers serve every OEM globally — GEV, Siemens, MHI, Dongfang, Doosan, Ansaldo. Chinese OEMs access only the residual market the Big Three cannot serve, a gap that shrinks structurally as Big Three capacity expands post-2028. Yingliu's growth does not depend on whether Dongfang or Siemens wins the order; it participates in both.

Second, the demand cycle is superior. Hot-section blades are replaced every 24,000–30,000 hours — roughly every 3–4 years at AIDC utilization. Every turbine commissioned today creates a compounding replacement annuity that starts immediately, not in 15 years when OEM service flywheels mature.

Third, the competitive moat is deeper. Single-crystal casting qualification takes 5–7 years per OEM program and designs the supplier in for the program's life. There is no involution risk — the barrier is accumulated process knowledge, not capital. OEM competition will intensify as Doosan, AECC, and CSIC Power join Dongfang. Blade casting will not.

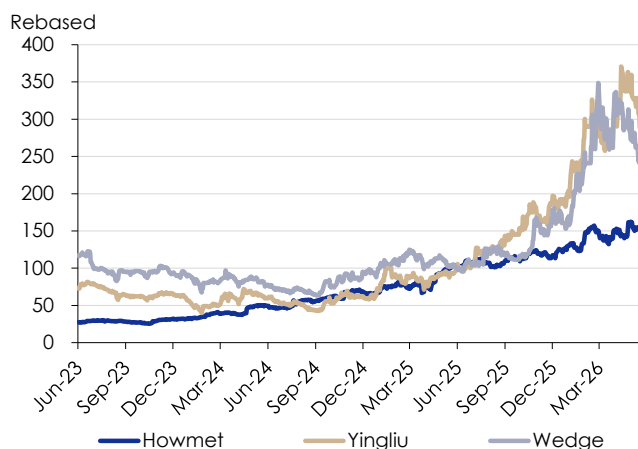
Fourth, the growth-adjusted valuation is more compelling. Yingliu at 1.5x PEG versus Dongfang at ~1.9x reflects superior earnings growth at comparable or lower relative cost.

Share prices of OEM are surging on the GT upcycle



Source: Company, CCBIS estimates

Share prices of component makers ride on the OEM



Source: Company, CCBIS estimates

Our Initiated Picks

Yingliu (603308 CH, Outperform, TP RMB90) — primary pick. The innermost bottleneck: sole Chinese H-class blade supplier to Siemens Energy, certified across GE Vernova, Baker Hughes, Ansaldo, Safran, and Rolls-Royce — a global oligopoly alongside Howmet and PCC that qualifications alone cannot fast-follow. 62% EPS CAGR, 1.5x PEG, overseas revenue already 54% of total.

Dongfang Electric (1072 HK / 600875 CH, Outperform, TP HK\$45 / RMB49) — China OEM pick. China's only fully-indigenous F-class gas turbine in commercial production and international export, at 13x 2026F H-share P/E with G50 export optionality essentially unpaid for. The export window will not stay open indefinitely — the Big Three are expanding capacity. This is the name to own the opportunity through.

Global OEM: ENR > MHI > GEV

We analyze the fundamentals of the Global Big Three OEM and rank them on a relative preference for investors sizing global exposure.

Siemens Energy (ENR GR) on grid + global exposure at discounted valuation

The gas turbine thesis is identical to GEV – backlogs, pricing power, margin expansion – but ENR adds two dimensions GEV cannot match: Grid Technologies (25-27% revenue growth, 18-20% margins), the fastest-growing division in global industrials, and a more globally distributed customer and supply chain base spanning Europe, Middle East, Asia, and North America. The 29x P/E vs GEV's 47x reflects a non-US listing discount.

Mitsubishi Power / MHI (7011 JP) diversified conglomerate

The once-cheapest of the Big Three at 40x trailing P/E with a diversified industrial base. JPY at historically weak levels provides additional upside for non-JPY investors. The conglomerate structure makes gas turbines harder to disaggregate — but the stability premium is worth something in the late stages of an order-momentum cycle.

GE Vernova (GEV US) remains the highest-quality franchise in the group

GEV remains the highest-quality franchise in the group with 9HA technology, deepest North American AIDC exposure, FCF inflecting strongly with a US\$10b buyback behind it. But at 47x P/E after 200%+ over 12 months is a consensus trade. GEV is suitable for the phase of fast-growing order momentum.

The Aftermarket Inflection — The Story Still Being Priced for Later

The most underappreciated element of this cycle is the services compounding that follows the equipment delivery wave. GEV's installed base services backlog (US\$87b) has crossed above its equipment backlog (US\$76b) for the first time — that inversion marks the transition from equipment growth to services compounding, which structurally commands a higher multiple.

The same logic applies to Dongfang at an earlier stage. Station services are 8% of revenue but 40%+ GPM and growing with every unit commissioned — the market still prices DEC as a cyclical equipment OEM. Each G50 sold internationally adds 20 years of LTSA income with no imported-component leakage. The investors who hold through the first equipment delivery peak in 2028–2030 will capture that re-rating. The key difference from GEV: at 13x H-share P/E, this is yet to be factored into the price.

MAJOR PLAYERS ON THE WATCHLIST

Core investment thesis of full value chain

Company	Stock code	Mkt cap (US\$ b)	Rating	Core thesis	2027F P/E (x)	EPS growth (%)
Yingliu	603308 CH	6.9	Outperform	Sole Chinese H-class blade supplier to Siemens Energy; nested bottleneck; 52% EPS CAGR	48	61
Dongfang Electric	1072 HK/600875 CH	16.2	Outperform	Only indigenous F-class OEM in China; G50 export breakthrough; six-power platform at 13x H-share P/E	12/19	23

OEM

GE Vernova	GEV US	303.0	NR	Purest global GT expression; Gas Power EBITDA 9.6%→17 - 19%; \$10b buyback; backlog \$150b	47	58
Siemens Energy	ENR GR	170.5	NR	GEV at a discount + Grid Technologies growing faster; Gamesa break-even as free option	29	67
MHI	7011 JP	83.3	NR	Cheapest Big Three (34x); GT + Japan energy/defence diversification; net cash	33	13
Doosan Enerbility	034020 KS	39.9	NR	Fifth country to crack H-class GT; 12 US AIDC units contracted; SMR as second growth leg	104	114
Harbin Electric	1133 HK	5.5	NR	Cheapest China power equipment (12-14x); Stock Connect Aug 2026 catalyst; GT as free option	9	20
Shanghai Electric	2727 HK/601727 CH	15.3	NR	Nuclear #1 + offshore wind; GT peripheral; core recurring profit only RMB2b	29/66	26
CSIC Power	600482 CH	11.6	NR	Guanghan 25MW naval-to-land GT; asset injection into listco is the thesis	19	62
AECC Aviation	600893 CH	16.3	NR	China military aero-engine monopoly; Taihang 25/110MW civilian GT emerging	111	12
Baker Hughes	BKR US	58.7	NR	LM-series aero-derivative platform; machine-head supplier to Jereh; oil & gas + LNG mix	21	8
Jereh	002353 CH	26.1	NR	Only Chinese company executing large-scale GT packages in North America; \$930m+ Q1 AIDC orders	35	31
Weichai Power	2338 HK/000338 CH	47.4	NR	AIDC backup-power layer; datacenter genset engines +491% H1 2025; 2-month delivery moat	21/16	21

Components

Howmet Aerospace	HWM US	112.2	NR	Global GT blade oligopolist; dual cycle aviation + IGT; Engine Products EBITDA margin 33%	48	22
Wedge	000534 CH	2.4	NR	China #2 hot-section; Siemens SGT5-4000F combustor; pharma cash cow; Saudi MRO contract	35	44
Himile	002595 CH	8.6	NR	Global 20%+ share in GT casings/rings; Big Three all as customers; cheapest valuation (20x P/E)	17	19
Xizi Clean Energy	002534 CH	2.7	NR	China HRSG monopoly (50%+ share); export optionality via G50 bundle	28	17
Longda	688231 CH	1.7	NR	Only Chinese superalloy supplier certified by GE, Siemens, BKR, RR, Safran; Malaysia plant	47	70
IHI	7013 P	19.3	NR	Japanese GT blade caster serving MHI; aero-engine MRO; Japan-focused	19	2

Source: Bloomberg, CCBIS estimates

Initiate Coverage On A Leading OEM And A Niche Supplier In China

Yingliu 应流股份 (603308 CH, Outperform TP: RMB90.00) – The hottest section

The innermost bottleneck — sole Chinese supplier of Siemens H-class blades, sitting at the tightest intersection of the AI power supply chain and the gas turbine component shortage.

- **Financials** 2025 revenue RMB2.92b (+16%), net profit RMB349m (+22%); 1Q2026 accelerating to +34% / +31% YoY; 61% EPS CAGR 2025–2030F.
- **Order book** New orders RMB800m+ in 1Q2026 (run-rate towards RMB3b full-year target); backlog RMB2.1b at end-1Q2026, extending into 2027. Three ALD directional-solidification/single-crystal furnaces due 2H2026 — critical capacity step.
- **Nested bottleneck** AI datacentres → gas turbines → hot-section blades. Yingliu is the innermost point: qualified by GE Vernova, Siemens Energy, Baker Hughes, Ansaldo, Safran, Rolls-Royce. Overseas revenue 54% of total in 2025.
- **Multiple drivers** Gas turbine (~40% of revenue, 38%+ GPM) is the primary driver; nuclear (sole Hualong-1 main coolant-pump casing supplier), aerospace (GE LEAP / Safran / Rolls-Royce), and commercial space (LandSpace rocket castings) draw on the same single-crystal platform — each a potential re-rating angle.
- **Valuation** 75x 2026F P/E narrows to 48x 2027F; PEG 1.5x vs. Howmet at 3.4x. TP RMB90 = 30x 2030F EPS discounted back to 2027F at 10%, consistent with Howmet and Yingliu historical averages. Implies ~30% upside from RMB69.
- **Key risks** Demand slowdown in gas turbine or aerospace; capacity/qualification slippage; post-2030 order cliff; OEM price pressure; balance-sheet leverage ahead of CB conversion.

Dongfang Electric (600875 CH / 1072 HK, Outperform TP: RMB49.00 [A] / HK\$45.00 [H]) – The big brother of China gas turbines

China's only fully-indigenous F-class gas turbine OEM — G50 export breakthrough as the re-rating catalyst for a cheaply-valued six-power platform.

- **Financials** Revenue RMB78.6b (+13%) in 2025; net profit RMB3.83b (+31%); 23% NP CAGR 2025–28F; net cash balance sheet throughout.
- **Order book** Total backlog RMB160b at 1Q 2026 (clean energy ~70%); gas turbine backlog covers 2.7x annual gas revenue; visibility to 2028.
- **The catalyst** G50 exports: Kazakhstan (3 units ✓), Iraq (2 units ✓), Canada (10-unit intent, deposit paid). Each unit sold internationally carries a 20-year LTSA service annuity — recurring high-margin income not yet in the market's valuation.
- **Why re-rate** Market still prices DEC as a cyclical OEM. Station services — 8% of revenue but 40%+ GPM — provide a GEV-style services annuity that deserves a higher multiple. H-share at further A/H discount widens the entry point; 13x 2027F H-share P/E vs. Siemens Energy at 29x and GEV at 47x.
- **Our SOTP** Core business (coal/nuclear/hydro/wind/services) valued at 15–30x 2027F by segment; gas turbine export option (25 units × RMB250m ASP × 40–50% GPM) at 30x adds ~RMB27b; total implies target market cap ~RMB140b.
- **Key risks** Gas turbine export conversion pace; wind margin recovery; slow recognition of large projects (Yarlung); non-recurring gains creating noise.

Global OEMs – Oligopoly With Distinct Investment Stories

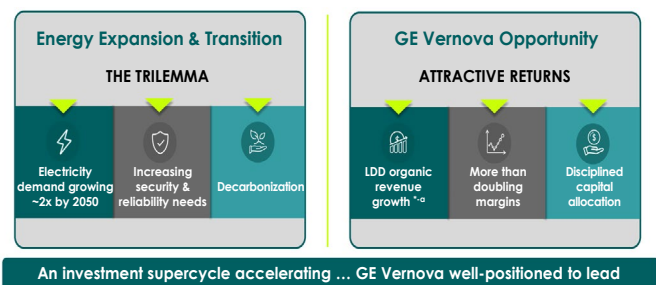
GE Vernova (GEV US, NR) – the global bellwether

- The global bellwether of gas turbine; the purest large-cap play; the domestic player in US market.
- Gas Power EBITDA margin tracking 9.6% (Q1 2026) toward full-year guidance 17–19%; Q1 2026 FCF US\$4.8b (exceeding FY25); US\$10b buyback; total backlog \$150b heading toward US\$200b (2028 target); Electrification backlog US\$42b is a second secular growth engine.
- Debate: 47x 2027F consensus P/E prices optimistic scenarios; pure beta play; Wind drag ~US\$400m EBIT persists. Not to miss: quarterly Gas Power margin progression; slot reservation → firm order conversion.

Siemens Energy (ENR GR, NR) – more balanced and more globalized

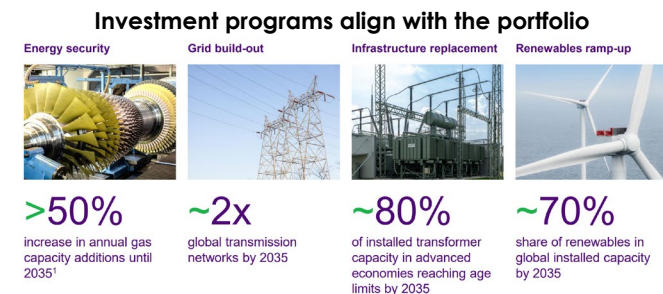
- Same GT upcycle as GEV at a discount, with a faster-growing Grid Technologies business and Gamesa break-even as a free option.
- FY2025: 194 turbines sold (nearly doubled YoY); 65% of orders from data centers; FY2026 FCF guidance ~€8b pre-tax; Grid Technologies guiding 25–27% revenue growth and 18–20% margin; 29x PE vs GEV's 47x. Q2 FY2026 Gas Services order intake €8.87b — record quarter.
- Debate: Gamesa wind remains tail risk (improved sharply; Q2 FY2026 loss €44m, but any re-escalation reopens old wounds); USD weakness creates EUR-reported earnings headwind. Not to miss: Gamesa quarterly updates; supply chain in China.

GEV: pure play in US power, bellwether of GT



Source: Company, CCBIS

Siemens Energy: strong global footprint and grid exposure



Source: Company, CCBIS

Mitsubishi Heavy Industries (7011 JP, NR) – defensive conglomerate

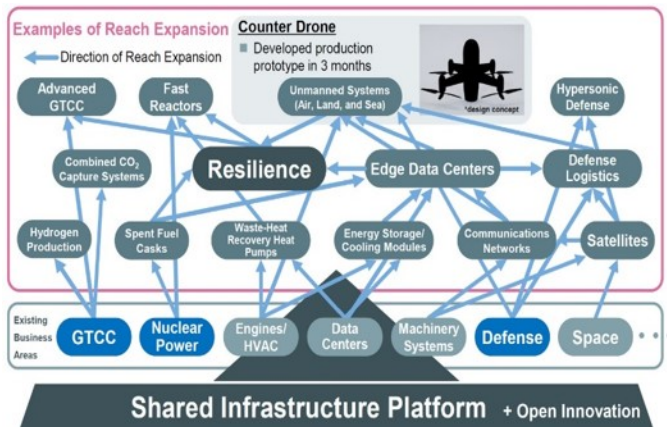
- Cheapest Big Three on trailing but caught up already; Japan defense budget doubling as a second independent super-cycle — a dual-engine structure GEV and Siemens Energy cannot offer.
- GTCC orders at all-time highs (35 large-frame units FY2025); ADS defence at back-to-back record revenues (next-gen fighter, missiles, vessels); cheapest multiple and only net-cash balance sheet in the Big Three; JAC series co-equals the 9HA as the world's most efficient CC turbine.
- Debate: conglomerate discount; defense exposure is controversial; JPY fluctuations pose uncertainties to business and financials; Logisnext forklift divestiture in progress. Not to miss: order updates on various businesses, capacity expansion progress.

Doosan Enerbility (034020 KS, NR) – aggressive fourth runner up

- The fifth country to independently develop H-class gas turbines — 12 US AIDC units already contracted, with SMR (small modular reactor) as a second secular growth option none of the Big Three can match.

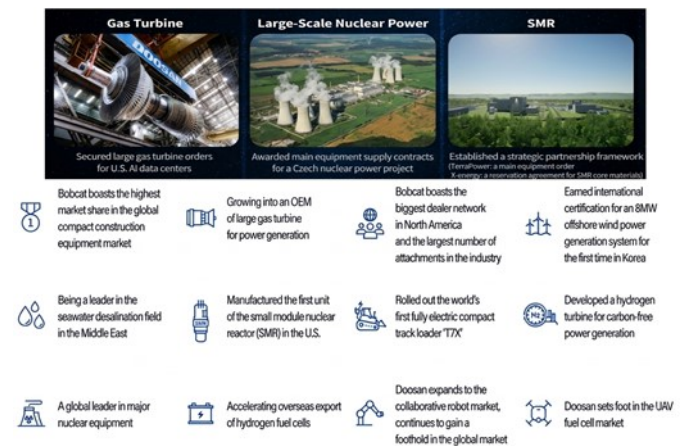
- Supplier to Colossus of SpaceXAI: 12 US AIDC units of 380MW DGT6-H.S2 contracted — first sustained international H-class export from a non-Big-Three OEM; 2025 revenue +74% YoY; 1Q2026 gross margin 18.3% (record); SMR: world's only end-to-end SMR manufacturer (Xe-100); 16 high-temp gas-cooled reactor core forgings contracted for Dow Chemical US project; 114% EPS CAGR the fastest-growing OEM at the most expensive 104x 2027F P/E.
- Debate: leverage still 68% D/E (down from 74%); Korea-listed (FX risks, market risks); single-digit GW annual production capacity delivery track record unproven. Not to miss: first two US datacenter deliveries (end-2026); second-wave US contracts; SMR Xe-100 groundbreaking.

MHI: GT + Nuclear + Defense + Industrials; Conglomerate Discount



Source: Company, CCBIS

Doosan: GT used in xAI's Colossus project; SMR venture weighs on valuation at the moment



Source: Company, CCBIS

China OEM – Diverse Fundamentals

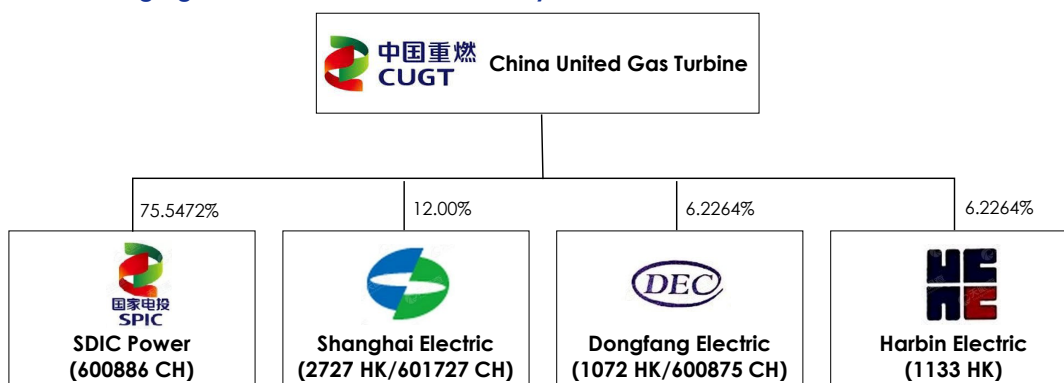
Harbin Electric (1133 HK, NR) – attractive valuation

- Cheapest China power equipment stock (9x 2027F consensus P/E) — coal/hydro/nuclear upcycle, gas turbines as a free option, and Stock Connect inclusion in August 2026.
- Stock Connect inclusion expected August 2026 is a mechanical re-rating catalyst; 2025 net profit +58% YoY, consensus 14% CAGR 2025–2028; operating cash flow turned positive in 2025 (RMB4.6b vs –RMB0.24b). HGT16 (16MW) rolled off production in 2025 but still in validation — any commercial milestone re-rates the gas option.
- Debate: coal still 62% of revenue (controversial); 14.3% gross margin is lowest among China Three; 16MW form factor limits addressable market vs Dongfang G50's 50MW. Not to miss: Stock Connect confirmation; HGT16 first commercial operation announcement.

Shanghai Electric (2727 HK / 601727 CH, NR) – storyline at nuclear, GT as option

- Nuclear equipment #1 and offshore wind leader — gas turbines are a peripheral option, not a reason to own it; least preferred of the China Three for this theme.
- 2025 net profit +60% YoY to RMB1.2b but core profit only RMB201m — most gains from asset disposals and subsidies; D/E ratio 75% is highest among the Three; gas turbine still 1–2 years behind Dongfang with no export track record. Nuclear (华龙一号 65% China share) and 26MW offshore wind are the real investment cases.

Chinese large gas turbine is under research by China United Gas Turbine



Source: Company, CCBIS estimates

Adjacent Players – Aeroderivatives and Others

CSIC Power (600482 CH, NR) – pending Guanghan injection

- The thesis is the asset injection — Guanghan Power's 25MW naval-to-land gas turbines are not yet in the listed entity; timing and terms are the only variables that matter.
- Guanghan Power's turbine business: 80 overseas orders, capacity 30→60 units/year planned; Brazil order over speculated under negotiation.
- Handle with care: CSSC is China's primary naval shipbuilder — elevated geopolitical sensitivity. Gas turbine thesis intact directionally but execution visibility and timeline uncertain.

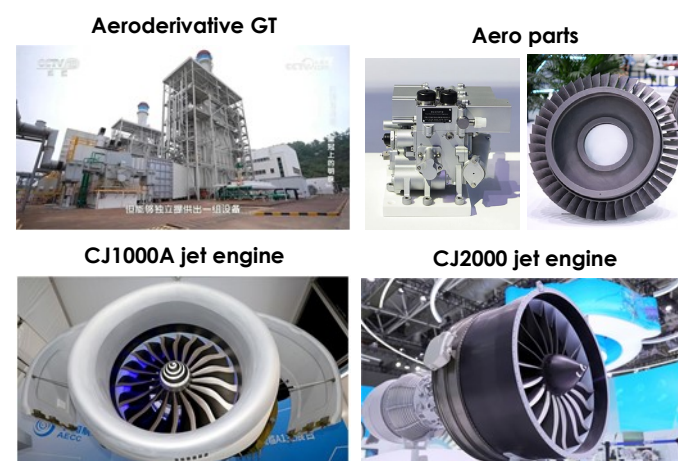
AECC Aviation Power (600893 CH, NR) – skyhigh valuation

- China's military aviation engine monopoly as the primary thesis, with Taihang 25–110MW civilian gas turbines as an emerging but structurally complex revenue stream.
- AECC is China's only full-spectrum military aero-engine developer; R&D +79.8% YoY to RMB995m in 2025. Taihang 110 (110MW) passed national acceptance review; first offshore export signed April 2026 (5x Taihang 7 for PETRONAS FPSO, Malaysia). Military aviation upside is the floor; civilian GT is the growth option.
- Structural issue: the Taihang 110 commercial entity (AECC Gas Turbine Co.) may sit outside the listed vehicle; stretched valuation at 111x 2027F consensus P/E.

CSIC Power: pending for Guanghan injection for naval-to-land GT business



AECC: Aviation, Defense, Aeroderivative GT



Source: Company, CCBIS

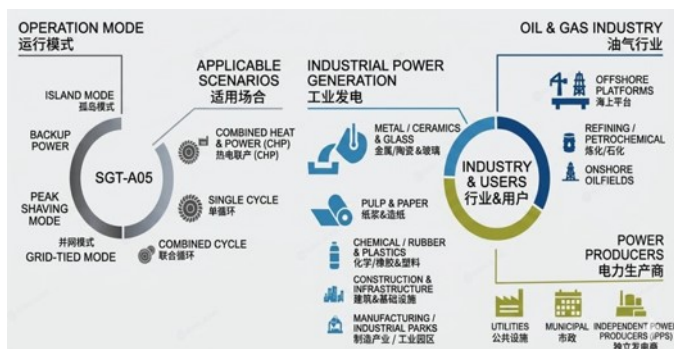
Baker Hughes (BKR US, NR) – aero-derivative leader

- The LM-series aero-derivative turbine platform (20–75MW) — machine-head supplier to Jereh's North America AIDC packages and the leading global aero-derivative OEM.
- LM9000 (73MW) and LM6000 (54MW) are the workhorses of aero-derivative AIDC power; current lead times 18–24 months. Jereh's AIDC success is simultaneously BKR's volume uplift — the two are structurally linked. LNG equipment is a parallel growth driver.
- Oil services, LNG, GT three drivers: softer upstream capex weighs on the non-GT businesses.

Jereh (002353 CH, NR) – EPC turnkey assembler

- Partnered with Baker Hughes: The only Chinese company executing large-scale gas turbine power packages in North America — US\$930m+ in 1Q2026 AIDC orders.
- Six North American AIDC contracts signed 4Q2025–1Q2026 totalling US\$930m+; package pricing repriced +20% to ~US\$12b/GW; Kawasaki partnership diversifies machine-head supply beyond Baker Hughes.
- Geopolitical risk: Chinese supplier in North America power infrastructure might face ongoing policy scrutiny. BKR LM-series allocation is the execution constraint. Not to miss: first contract delivery milestones (2026–2027); new AIDC contract announcements.

Jereh: Oil & Gas services + EPC for AIDC in North America



Source: Company, CCBIS

Baker Hughes: GE-inherited GT technology; cooperate with Jereh on AIDC projects

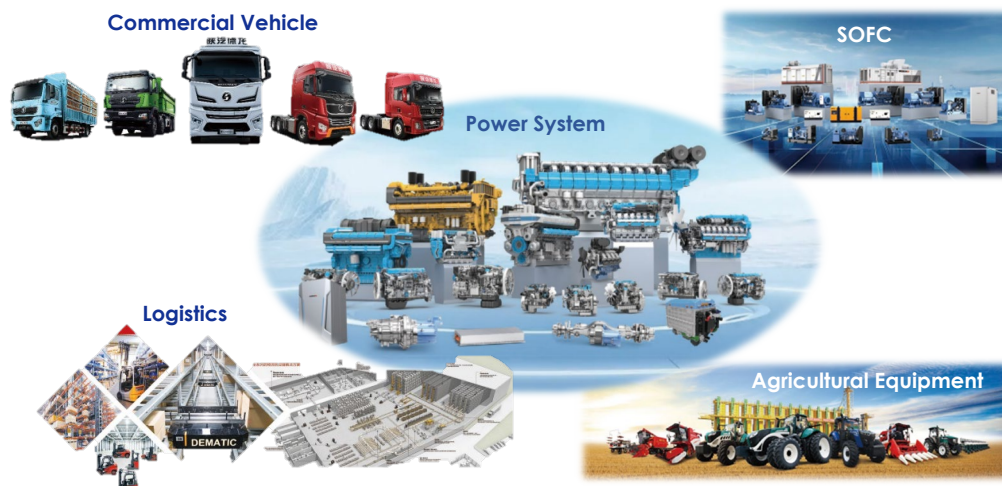


Source: Company, CCBIS

Weichai Power (2338 HK / 000338 CH, NR) – delivery moat and SOFC supplement

- Large-bore diesel gensets: same AI power demand theme as backup-power layer —displacing Cummins/CAT/MTU on a 2-month delivery window vs. competitors' 10+ months.
- SOFC medium-term angle: 20% stake in UK Ceres Power; manufacturing license signed Nov 2025; SOFC efficiency 55–65%, 90-day deployment.

Weichai: large-bore diesel genset backs up AIDC



Source: Company, CCBIS

Component Makers – Nested Bottleneck

Howmet Aerospace (HWM US, NR) – largest and hottest

- The global turbine blade casting oligopolist — Musk's named supply chain bottleneck; dual cycle (aviation recovery + IGT super-cycle) with 33% EBITDA margins and accelerating FCF.
- Primary supplier of GT blades to GEV, Siemens Energy, and MHI; Engine Products EBITDA margin 33.3% (+1,060bps since 2019); Q1 2026 FCF US\$359m (+168% YoY, record single quarter); FY2026 guidance raised twice to revenue of US\$9.58–9.73b; management confirms IGT growth 'starts to show in financials from 2026'. Analogous role to Yingliu in China but with 20x the scale.
- Debate: 48x 2027F consensus P/E is exceptional for an industrial company — priced for perfection; US\$1.8b CAM acquisition uncertain on integration; customer concentration concerns.

Wedge (000534 CH, NR) – a rising challenger

- China's #2 hot-section component play — penetrating into Siemens's AGT, SGT and MGT combustor and mid-stage blades, funded by a 90% gross margin pharma cash cow and anchored by a Saudi Arabia 10-year MRO contract.
- Sole Chinese batch supplier for Siemens SGT5-4000F combustors; additionally supplies DEC, Guanghan, AECC Taihang; Saudi Arabia MRO ~RMB200m/year recurring; Changjiang 1000/2000 aviation engine qualification entering production is a multi-year growth stream; pharma division (双奇药业) funding vehicle ~RMB300m/year at 90%+ GPM for R&D.
- Wedge vs. Yingliu: Wedge does combustors and mid-stage blades for small-mid GT; Yingliu does front-stage single-crystal blades for large GT — differentiated products, same tailwind, complementary rather than competing. 35x P/E vs. Yingliu's 48x reflects the certification gap. Not to miss: new OEM certifications.

Howmet's product offerings cover components for engine, fastenings, structures, wheels



Engine Products

Produce components enabling quieter, cleaner and more fuel-efficient aerospace engines and gas turbines.



Fastening Systems

Make aerospace and industrial fasteners to hold together aircraft, jet engines, commercial trucks, wind turbines, solar panels, and more.



Engineered Structures

Manufacture advanced, multi-material parts that make aircraft and vehicles lighter and more fuel-efficient.



Forged Wheels

Forge strong aluminum wheels that allow commercial trucks to run lighter and more fuel-efficiently.

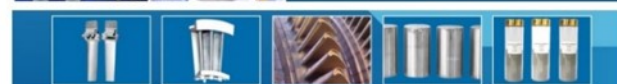
Source: Company, CCBIS

Wedge: Siemens' certified hot-section supplier for SGT, MGT, AGT

Hot-section Vane & Blade



Pharmaceutical (Bifid)



Source: Company, CCBIS

Himile (002595 CH, NR) – cold-section components leader

- Globally competitive cold-section manufacturer in GT casings and rings — all Big Three as customers, cheapest valuation in the sector at 17x P/E.
- GEV, Siemens Energy, and MHI all confirmed customers; cold-section ring/casing 20%+ global share achieved over two decades; 19% EPS CAGR; 40%+ GT segment gross margins. Tyre mold business (~40% of group) provides cashflow stability but obscures the turbine growth story and suppresses the multiple. Explicit GT revenue disclosure would be a re-rating trigger.
- 17x 2027E P/E with 19% growth— cheapest name in this universe. Management deliberately stays in cold section (lower risk) rather than hot section — disciplined focus or growth ceiling, depending on view.

Xizi Clean Energy (002534 CH, NR) – absolute leader in HRSG

- Domestic HRSG (Heat Recovery Steam Generator) monopoly (50%+ China share) with new export optionality as the natural bundling partner for Dongfang G50 overseas packages.
- Every G50 export creates a potential HRSG bundle order; first North American AIDC HRSG configurations discussed with Chinese OEMs in 2025–2026. Appropriate as a supplementary position when conviction on G50 export volume is high. Investment case is derivative of the G50 export story rather than standalone.

Himile: tire mold (cash cow) → CNC → precision casting for cold-section components

Tire Molds



Mechatronics

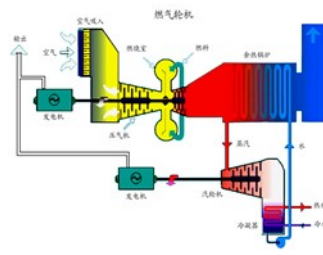


CNC



Xizi's HRSG 50% China share; nuclear components on certification progress

HRSG



Class 2/3 Nuclear Vessels



Heavy Industries



Compressors



Industrial Protective Coatings



Shanghai 9F
(SIE SGT 54000F)



Pakistan Balloki
with GE9HA.01



Source: Company, CCBIS

Source: Company, CCBIS

Gas turbine, aerospace, defense share the same precision manufacturing platform

If it flies, we're on it!

Howmet Aerospace Inc. is a leading global provider of advanced engineered solutions for the aerospace and transportation industries. The Company's primary businesses focus on jet engine components, aerospace fastening systems, and titanium structural parts necessary for mission-critical performance and efficiency in aerospace and defense applications, as well as forged wheels for commercial transportation. With nearly 1,150 granted and pending patents, the Company's differentiated technologies enable lighter, more fuel-efficient aircraft to operate with a lower carbon footprint. For more information, visit www.howmet.com.

Source: Company

RISKS

Demand Risks

- AI capex slowdown: If hyperscaler investment cycles moderate or LLM development plateaus, datacenter power demand growth slows. However, existing committed datacenter builds represent a floor.
- Energy transition policy reversal: A more hawkish regulatory environment on gas could tighten demand in Europe. Less likely in US/Asia but worth monitoring.
- Gas price spikes: High natural gas prices in North America could undermine the economic case for new gas-fired capacity. Henry Hub remains low but energy markets are volatile.
- Demand brought forward: If the 2026–2027 pipeline represents pull-forward demand from 2028–2030, earnings visibility could become murkier post-2027.

Supply Risks

- Faster-than-expected new entrant supply: If Chinese OEMs scale dramatically, or new players emerge, the supply shortage could normalize faster and compress margins.
- Component bottlenecks (especially blades): Ironically, the component supply chain (superalloy casting) could constrain OEM production even if there is demand. Yingliu's order backlog to 2030 is simultaneously reassuring and a warning.

Geopolitical Risks

- Geopolitical / technology restrictions: If the US imposes export controls on gas turbine technology (or pressures allies to do so), or restricts Chinese OEM sales to US customers, the export opportunity narrows. The MATCH Act framework (requiring allies to align export controls within 150 days) is a live risk to watch.
- IP litigation risk: As Chinese OEMs export to developed markets, IP challenges from GEV/Siemens/MHI are possible. G50 claims full indigenous IP but historical MHI collaboration may create gray areas.
- Execution risk on export ramp: Getting from "deposit paid" to "signed contract" to "delivered and running" is a long road. Chinese OEMs have thin international reference lists.
- Domestic policy risk: China's domestic gas power development could be curtailed if the government prioritizes other energy sources; though current policy is broadly supportive.

Yingliu | 603308 CH

The hottest section

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- **Niche in hot-section blades: Siemens LGT sole supplier in China**
- **Nested bottleneck in GT (hot section) + multiple drivers (GT, aerospace, nuclear)**
- **Initiate at Outperform; 48x P/E on 47% EPS CAGR**

Niche in hot-section blades. Yingliu is among the few global manufacturers of critical components and precision casting for gas turbines, aerospace, nuclear and oil & gas (i.e. high-temperature-alloys). Its products are exported to 40 countries and 100+ customers (54% overseas) including GE, Siemens Energy (sole Chinese supplier for heavy-duty GT blades), Baker Hughes, Ansaldo, Safran and Rolls-Royce, as well as domestic leaders AECC, CASIC, China Heavy Gas Turbine and Dongfang Electric. The niche hot-section manufacturing platform underpins five drivers: gas turbine, aerospace, nuclear power, commercial space, and UAV.

Nested Bottleneck + Multiple Drivers + Qualification Moat. We see Yingliu as the highest-conviction, highest-beta hot-section play on the gas-turbine theme. It sits at a nested bottleneck: (1) gas turbines are the shortage within the power supply chain, and (2) the niche hot-section blades and vanes are the shortage within gas turbines. Its superalloy casting know-how transfers across verticals — directly into aero-engine and turboshaft blades, and, via similar precision-casting know-how, into nuclear and low-altitude as optionality — unlocking multiple drivers for earnings leverage. Qualification is the second — but arguably the deeper — moat: built on multi-year supply-chain relationships across numerous OEM programs, it is genuinely hard to replicate.

Initiate at Outperform. Yingliu offers the highest earnings growth (47% CAGR) among its hot-section component manufacturer peers Howmet and Wedge. Current valuation of 48x 2027F P/E is justifiable given peer comparisons, the strong order book of itself as well as its OEM customers, and its niche positioning towards various high-growth verticals. We deduce our Target Price of RMB90.00 using 30x 2030F P/E and discounting back to 2027F, consistent with the long-term average of Howmet and itself. Initiate at Outperform.

Key risks include (1) slowdown in downstream markets including AI, aerospace, power etc., (2) capacity or qualification slippage, and (3) supply chain risks.

Forecast and valuation

Year to 31 Dec	2024	2025	2026F	2027F	2028F
Revenue (RMB m)	2,513	2,919	3,904	5,247	6,888
YoY (%)	4.2	16.1	33.8	34.4	31.3
Net profit (RMB m)	286	349	630	974	1,445
YoY (%)	(5.6)	21.7	80.7	54.7	48.3
Fully diluted EPS (RMB)	0.42	0.51	0.93	1.43	2.13
YoY (%)	(5.0)	21.7	80.7	54.7	48.3
P/E (x)	164.2	134.9	74.7	48.3	32.5
DPS (RMB)	0.07	0.16	0.28	0.43	0.64
Dividend yield (%)	0.1	0.2	0.4	0.6	0.9
P/B (x)	9.3	8.6	7.9	7.0	6.0
ROAE (%)	6.3	7.2	11.9	16.5	21.1
Net debt/equity (%)	60.9	33.8	43.0	48.2	46.9

Source: Bloomberg, CCBIS estimates

Outperform (Initiation)

Current price:

RMB69.26
(as at 22 Jun 2026)

Target:

RMB90.00
(initiation)

Trading data

52-week range	RMB19.50–84.66
Market capital (A-share)(US\$ m)	6,906
Shares outstanding (m)	679
Free float (%)	35
3M average daily T/O (m share)	12
3M average daily T/O (US\$ m)	178
Expected return (%) – 12 month	30

Source: Bloomberg, CCBIS estimates

Share price vs SH COMP



Source: Bloomberg

Stock performance

Performance over	1M	3M	12M
Absolute	(16)	(3)	170
Relative (%) to SH COMP	(15)	(4)	150

Source: Bloomberg

应流股份 | 603308 CH

最热的热端

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- **热端叶片利基: 西门子能源大型燃机唯一中国供货商**
- **燃机层叠瓶颈(热端叶片) + 多重增长动力(两机+核能)**
- **首予“优于大市”: 48 倍市盈率+47% EPS 复合增长率**

立足热端叶片利基。应流是全球少数能为燃气轮机、航空、核电及油气领域生产关键部件及精密铸件(即高温合金件)的厂商之一。其产品出口至 40 个国家、服务逾 100 家客户(海外占 54%),客户涵盖 GE、西门子能源(其重型燃机叶片的唯一中国供应商)、贝克休斯(Baker Hughes)、安萨尔多(Ansaldo)、赛峰(Safran)及罗罗(Rolls-Royce),以及中国航发(AECC)、中国航天科工(CASIC)、中国重燃及东方电气等国内龙头。这一稀缺的热端制造平台,支撑起五大增长动力:**燃气轮机、航空、核电、商业航天及无人机(UAV)**。

层叠瓶颈 + 多重动力 + 客户认证护城河。我们认为应流是燃机主题下确信度最高、弹性最高(high-beta)的热端首选标的。公司正处于一组**层叠的瓶颈**之中:(1) 燃气轮机是电力供应链中的短缺环节,(2) 而稀缺的热端叶片与导叶,又是燃机中的短缺环节。其**高温合金铸造**的技术积累可跨领域迁移——直接延伸至航空发动机及涡轴叶片,并通过相近的精密铸造技术,延伸至核电与低空(作为期权价值)——从而释放多重增长动力与盈利杠杆。**客户认证**是第二重、却可能更深的护城河:建立在与众多 OEM 项目多年的供应链关系之上,真正难以复制。

首予“优于大市”。在热端部件制造可比同业豪梅特(Howmet)与万泽股份当中,应流拥有**最高的盈利增速**(47% CAGR)。当前 48 倍 2027F 市盈率的估值合理——支撑因素包括可比同业对标、公司自身及其 OEM 客户双方强劲的在手订单,以及其面向多个高增长细分领域的利基定位。我们以 30 倍 2030F 市盈率并折现回 2027F,推算得出 **90.00 人民币的目标价**,与豪梅特及应流自身的长期平均水平一致。首予“优于大市”评级。

主要风险包括: (1) 下游市场(包括 AI、航空、电力等)放缓; (2) 产能或资质进度不及预期; (3) 供应链风险。

预测和估值

截至 12 月 31 日	2024	2025	2026F	2027F	2028F
收入(百万元 人民币)	2,513	2,919	3,904	5,247	6,888
同比(%)	4.2	16.1	33.8	34.4	31.3
净利润(百万元 人民币)	286	349	630	974	1,445
同比(%)	(5.6)	21.7	80.7	54.7	48.3
稀释每股收益(人民币)	0.42	0.51	0.93	1.43	2.13
同比(%)	(5.0)	21.7	80.7	54.7	48.3
市盈率(倍)	164.2	134.9	74.7	48.3	32.5
股息(人民币)	0.07	0.16	0.28	0.43	0.64
股息回报率(%)	0.1	0.2	0.4	0.6	0.9
市销率(倍)	9.3	8.6	7.9	7.0	6.0
股本回报率(%)	6.3	7.2	11.9	16.5	21.1
净负债率(%)	60.9	33.8	43.0	48.2	46.9

资料来源: 彭博、建银国际证券预测

优于大市 (首次覆盖)

现价:

69.26 人民币

(截至 2026 年 6 月 22 日)

目标价:

► 90.00 人民币

(首次覆盖)

交易数据

52 周股价高低	19.50–84.66 人民币
市值(百万美元)	6,906
由流通股数(百万股)	679
自由流通股份(%)	35
3 个月日均成交股数(百万股)	12
3 个月日均成交额(百万美元)	178
12 个月预期回报率(%)	30

资料来源: 彭博、建银国际证券预测

股价相对于上海综合指数及恒生指数



资料来源: 彭博

股价表现

表现区间	一个月	三个月	十二个月
绝对表现	(16)	(3)	170
相对于上海综合指数	(15)	(4)	150

资料来源: 彭博

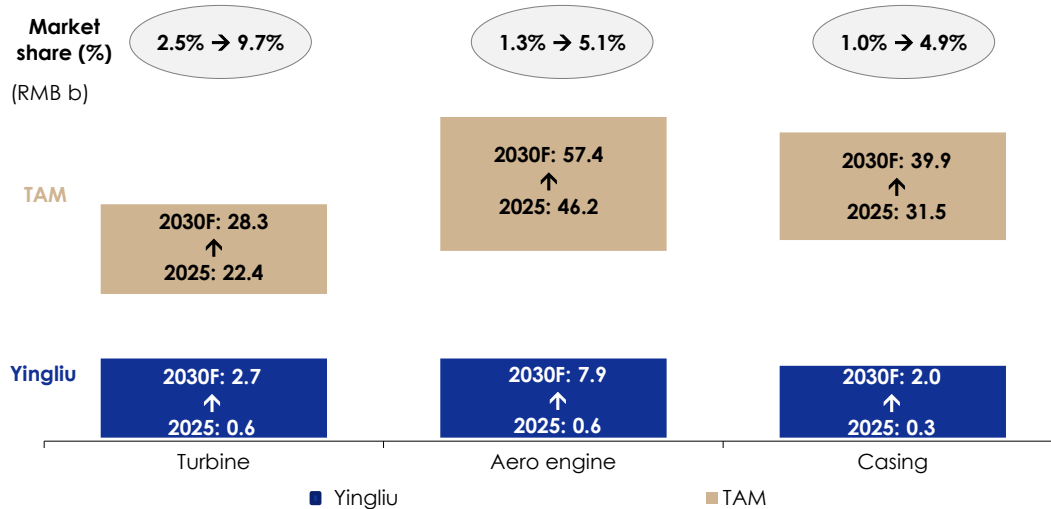
FOCUS CHARTS

TAM forecast for Yingliu's business lines

TAM (RMB b)	2025	2026F	2027F	2028F	2029F	2030F	CAGR (%)
GT TAM	112.1	117.2	121.9	129.2	136.6	141.3	4.7
Turbine BOM (%)	20	20	20	20	20	20	
Turbine	22.4	23.4	24.4	25.8	27.3	28.3	4.7
Aero engine	46.2	48.3	50.4	52.5	55.3	57.4	4.4
Casing	31.5	32.9	35.0	36.4	38.5	39.9	4.8
Market share (%)							
Turbine	2.5	3.8	5.2	6.7	8.2	9.7	
Aero engine	1.3	1.9	2.6	3.4	4.2	5.1	
Casing	1.0	1.7	2.5	3.3	4.1	4.9	
Revenue (RMB b)							
Turbine	0.6	0.9	1.3	1.7	2.2	2.7	37.4
Aero engine	0.6	0.9	1.3	1.8	2.3	2.9	37.3
Casing	0.3	0.6	0.9	1.2	1.6	2.0	44.1
Others	0.3	0.4	0.4	0.4	0.5	0.5	9.8
Two engine	1.8	2.7	3.8	5.1	6.6	8.1	35.3

Source: Company, CCBIS estimates

Upside opportunities for Yingliu's business lines



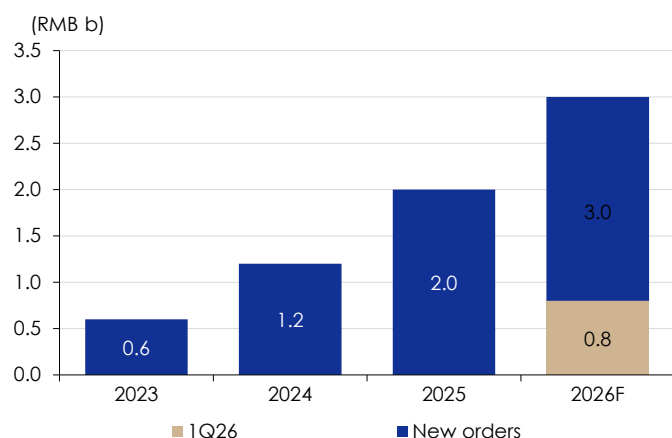
Source: Company, CCBIS estimates

Mapping between vertical, representative products, anchor customers, and underlying drivers

Vertical	Underlying drivers	Representative products	Key customers
GT hot-section	AIDC, energy transition, global supply shortage, exports, domestic replacement	Turbine blades (single-crystal/DS), casings	Siemens Energy, Ansaldo, Baker Hughes, Dongfang Electric, Shanghai Electric, China United Gas Turbine
Aerospace	Civil aviation, defense	LEAP casings, structural parts, C919 supply	GE Aerospace, Safran, Rolls-Royce, AECC Commercial
Commercial space	Space exploration, LEO constellation, defense	Rocket-engine castings	LandSpace, CASIC, SpaceX
Nuclear power	Energy Safety, Power, shortage, Decarbonization, SMR	Hualong-1 main-pump casing (sole), Class-1 nuclear castings, fuel-storage racks, fusion dewar	SNERDI, nuclear-plant owners
Low-altitude/UAV	Low-altitude Economy, Defense, Emergency	Turboshaft YLWZ-190/300/500, hybrid YLHD-300, whole-aircraft (Huayi 38.81%)	Low-altitude / UAV logistics

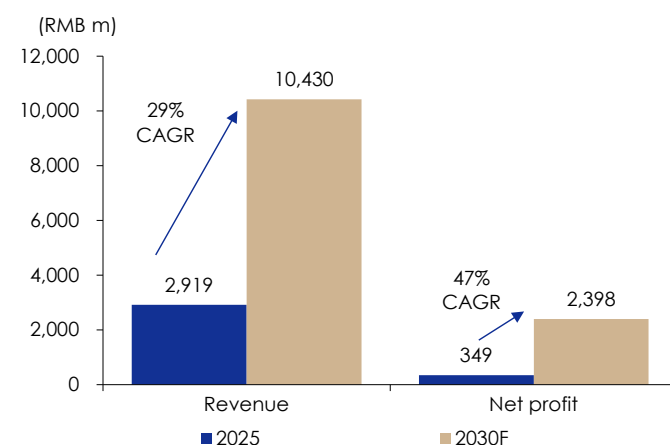
Source: Company, CCBIS estimates

Growth story underpinned by strong order flow



Source: Company, CCBIS estimates

Earnings growth driven by topline and operating leverage



Source: Company, CCBIS estimates

Target price RMB90.00 on 30x 2030F P/E; Outperform rating

	2025	2026F	2027F	2028F	2029F	2030F
EPS (RMB/share)	0.51	0.93	1.43	2.13	2.82	3.53
EPS - incl. CB (RMB/share)	0.00	0.86	1.34	1.98	2.63	3.29
Current P/E (x)	125.31	69.36	44.84	30.23	22.80	18.22
Target P/E (x)						30.00
Target price - 2030F (RMB)						118.52
Target price - 2027F (RMB)						90.00
Current price (RMB)						69.26
Upside						29.95
Rating						Outperform

Source: Bloomberg, CCBIS estimates

SCARCE SUPPLIER OF HOT-SECTION VANE AND BLADE

Business Segment: High-Temperature Alloy, Nuclear Casting, New Materials

Business segment and TAM

Segments	Products	FY25 revenue (RMB m)	GPM (%)	Verticals	TAM
High-temp alloy & precision steel castings	Hot-Section Vanes and Blades, Casing, Structural Casting	1,805	38.02	Gas Turbine (Power / AIDC) + Aviation + Aerospace	Global market: USD30b by 2030F, 10% CAGR; 30% in Turbine Blading
Nuclear & other med/large steel castings	Nuclear Class 1 Rupture Valve, Castings, RCP Casing, SMR Thermal Insulation, Shielding Materials	683	32.81	Nuclear Power (Fusion, SMR)	Domestic market: USD18b by 2030F, 6% CAGR, not including exports
New materials & equipment	Turboshaft Engine	293	32.72	Aero Engine, UAV	Global market USD28b
Other		540	16.35	—	
Total revenue		2,919	35.43		

Source: Company, CCBIS estimates

Moat: Qualification, Capacity, Vertical Integration

- **Strategy – GT, aerospace, nuclear:** the company concentrates on three high-end equipment domains — gas turbine, aerospace and nuclear — and pursues a “two extensions, two adjustments” playbook: extending up and down the industry chain (from castings to machined and coated finished parts, and towards whole-engine assembly) and along the value chain (from new-unit supply into long-cycle aftermarket). Domestically it targets national key programs, while in export markets it is extending its position in the global high-end supply chain.
- **Moat – superalloy casting + customer qualification:** the barrier is the combination of scarce casting capacity and long-cycle customer qualification. A single component can take two to three years to qualify with a top-tier OEM, after which the supplier is effectively designated in for the life of the program. Backed by hundreds of patents and a deep body of proprietary process know-how, Yingliu has developed more than 1,000 two-engine products and continues to expand the share that has reached mass production.
- **Platform reuse:** all five legs draw on the same high-temperature-alloy / single-crystal / HIP platform. The investment question is therefore one of capacity allocation and operating leverage across verticals, not five independent end-markets competing for separate capital — a distinction that drives the financial model.

Verticals / Products / Customers

Mapping between vertical, representative products, anchor customers, and underlying drivers

Vertical	Representative products	Key customers	Underlying driver
GT hot-section	Turbine blades (single-crystal/DS), casings	Siemens Energy, Ansaldo, Baker Hughes, Dongfang Electric, Shanghai Electric, China United Gas Turbine	AIDC, energy transition, supply shortage
Aero parts	LEAP casings, structural parts, C919 supply	GE Aerospace, Safran, Rolls-Royce, AECC Commercial	Civil aviation, defense
Commercial space	Rocket-engine castings	LandSpace, CASIC, SpaceX	Space exploration, LEO constellation, defense
Nuclear power	Hualong-1 main-pump casing (sole), Class-1 nuclear castings, fuel-storage racks, fusion dewar	SNERDI, nuclear-plant owners	Energy safety, power, shortage, decarbonization, SMR
Low-altitude/UAV	Turboshaft YLWZ-190/300/500, hybrid YLHD-300, whole-aircraft (Huayi 38.81%)	Low-altitude / UAV logistics	Low-altitude economy, defense, emergence

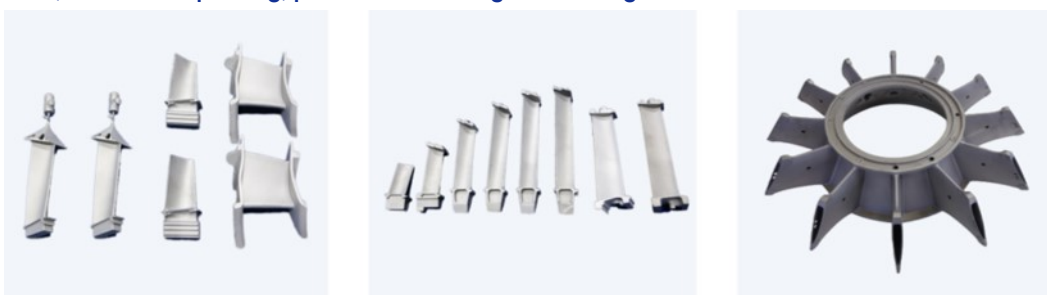
Source: Company, CCBIS estimates

Gas turbine products: vanes and blades at hot section; Key technologies: directional solidified, single-crystal superalloy



Source: Company

Aerospace products: hot-section blades, structural parts for aero engines; Key technologies: equiaxed, directional solidified, single-crystal superalloy, vacuum investment casting, master alloy and ceramic cores, hot isostatic pressing, precision machining and coating



Source: Company

Nuclear power products: Class 1 cast components, radiation-shielded metallic thermal insulation layers, fresh & spent fuel storage racks; Key certification: ASME (US) Nuclear Component Material Organization Certification; NNSA (China) Manufacturing Qualification Certificate for Civil Nuclear Safety Equipment



Source: Company

Yingliu's major customers are overseas market leaders



Source: Company

Development trajectory of Yingliu

Year	Key milestones
2000	Company founded, started high-end precision casting business
2008–2009	Acquired Nuclear Class 1 qualification, officially entered nuclear power industry
2014	Listed on SSE (603308.SH); transformed into nuclear, aero-engine and gas turbine sectors
2015	Formally entered aerospace and gas turbine component manufacturing
2016	Completed overseas M&A, expanded global layout and aero-engine whole-machine R&D
2018	Admitted into GE and Siemens global gas turbine supply chain
2019	Breakthrough in Hualong-1 Class 1 nuclear components; achieved world-leading nuclear metal insulation technology
2020	Realized mass production of equiaxed, DS and SX superalloy blades; built full hot-section component industrial chain
2021–2022	Became qualified supplier for GE Aerospace, Safran and Rolls-Royce
2023–2025	Obtained ASME MO nuclear certification; achieved heavy-duty gas turbine DS blade breakthrough; awarded national two-machine demonstration enterprise

Source: Company

YINGLIU'S INVESTMENT CASE

The investment case for Yingliu rests on two bottlenecks. The AI data-center power shortage makes **gas turbines the bottleneck of power supply**; and within the turbine, the **blades and vanes are the bottleneck within the gas-turbine supply** — the hardest part to cast, the slowest to add, the tightest globally. Yingliu sits at that innermost point. Our analysis below highlights how from a top-down view, demand will remain robust, while from a bottom-up perspective, Yingliu is well-positioned to capitalize on this demand.

Nested Bottlenecks In End Market And Component Market (Top-Down)

- **Power → turbines.** Global gas-turbine demand runs at ~40 GW/yr, rising to ~50 GW/yr by 2030 (base), with the three OEMs — GEV, Siemens Energy, MHI — effectively sold out to 2030. New-unit equipment TAM is ~US\$16b, and expected to rise to ~US\$20b a year.
- **Turbine → blades.** Hot-section blades and vanes are ~20% of unit value and the single tightest link — single-crystal / directional-solidification capacity takes 3–5 years to add and 2–3 years to qualify. That isolates a blade-casting TAM of ~US\$1.4–1.8b a year.
- **Yingliu's slice.** As the sole China supplier qualified on Siemens F/H blades (plus Ansaldo, Baker Hughes and domestic heavy-duty), Yingliu's addressable two-engine TAM will reach ~US\$4b by 2030F — a small fraction of the pool, so qualification and capacity, not demand, are the binding constraints.

Major players of hot section components manufacturing

Revenue	GT	Aero	Casing	Total	% of total
Howmet	0.94	3.26	0.20	4.40	40.6
PCC	1.10	2.30	0.60	4.00	36.9
Doncasters	0.35	0.45	0.09	0.89	8.2
Chromalloy / CPP / Others	0.55	0.50	0.30	1.35	12.4
Yingliu	0.08	0.08	0.05	0.21	1.9
Total	3.02	6.59	1.24	10.85	100.0

Source: Company, CCBIS estimates

TAM by product 2025 to 2030

Product	Unit	2025	2026	2027	2028	2029	2030
Gas turbine vane & blade	US\$ b	3.10	3.30	3.50	3.80	4.00	4.30
Aero engine parts	US\$ b	6.60	6.90	7.20	7.50	7.90	8.20
Casing & structure	US\$ b	4.50	4.70	5.00	5.20	5.50	5.70
TAM total	US\$ b	14.20	14.90	15.70	16.50	17.40	18.20

Source: Company, CCBIS estimates

Captured market — TAM x market share

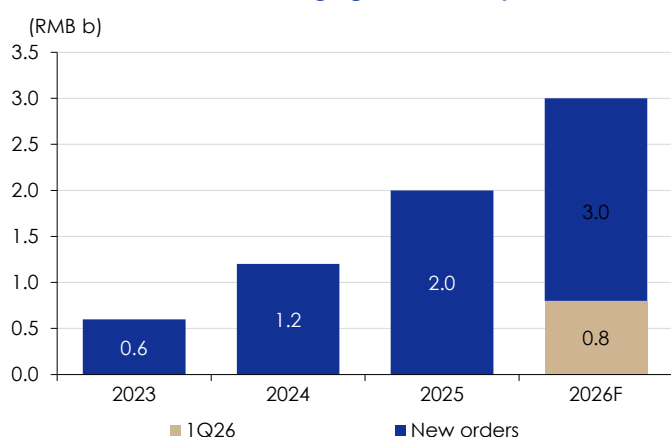
Market size	Unit	2025	2026	2027	2028	2029	2030
Market Share - GT Blade	%	2.5	3.8	5.2	6.7	8.2	9.7
Market Share - Aero	%	1.3	1.9	2.6	3.4	4.2	5.1
Market Share - Casing	%	1.0	1.7	2.5	3.3	4.1	4.9
Captured Market - GT Blade	RMB b	0.6	0.9	1.3	1.8	2.4	3.0
Captured Market - Aero	RMB b	0.6	0.9	1.3	1.8	2.4	3.0
Captured Market - Casing	RMB b	0.3	0.6	0.9	1.2	1.6	2.0
Major products total	RMB b	1.5	2.4	3.6	4.9	6.4	8.0
Accessories	RMB b	0.3	0.4	0.4	0.4	0.5	0.5
Total	RMB b	1.8	2.8	3.9	5.3	6.8	8.5

Source: Company, CCBIS estimates

Ramp-Up of Production Capacity (Bottom-Up)

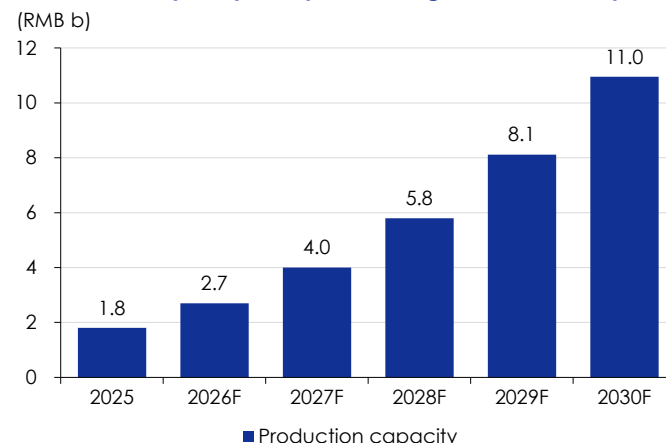
- **Demand-pull capacity +40-50% p.a.** Current production capacity is RMB1.8b in 2025 and we estimate this to expand by 50% p.a. in 2026-2028F and 40% p.a. in 2029-2030F. By 2028F and 2030F, production capacity is estimated at 6b and 10b respectively.
- **ASP step-up on in-house machining & TBC.** The convertible-bond will fund the project of in-house machining + thermal-barrier coating (TBC), lifting ASP up to ~2 times — though not on every product, as some customers do not need the finished part. This is one of the value levers, ahead of volume.
- **Order book surge.** New orders surged from RMB1.2b in 2024 to RMB2b in 2025 +67% (GT: ~60%). Management has set the new orders target to be RMB3b in 2026F, +50%. In 1Q26, RMB0.8b has been signed (run rate ~27%). At 1Q26, order backlog sits on RMB2.1b.
- **Oligopoly market.** Order backlog has been expanding into 2027F, with Yingliu already planning ahead for production in 2H27F. Unlike gas turbine OEM orders which extend into 2029-2030F, component makers generally do not overbook their production capacity for particular products nor conduct extensive capex expansion before order intakes. This preserves the scarce production capacity and retains pricing power.

New orders have been surging substantially



Source: Company, CCBIS estimates

Production capacity is expected to grow at 40-50% p.a.



Source: Company, CCBIS estimates

Various Catalysts for RMB10b Revenue in 2030F

Beyond gas turbines, Yingliu sits on the same single-crystal / superalloy platform across four areas that are **each** amongst 2026's most-bid investment themes. We carry them as upside rather than sizing them with the same precision — but the qualification, demand and re-rating are real and happening now.

- **Aero casings (aviation + defense) – RMB1.5b.** **Commercial aviation** is in a record-backlog supercycle — Boeing's backlog exceeds 6,100 aircraft and Airbus carries ~9,000, roughly a decade of production. The global engine MRO/spares market topped US\$136b in 2025 and is projected toward ~US\$193b by the end of the decade; **Defense** is healthy and supply of qualified hot-section casters is scarce. In China, COMAC's C919 carries hundreds of orders and is racing to lift output, with a priority towards domestic suppliers. Yingliu's exposure: aero-engine casings and blades for GE/CFM LEAP, Safran, Rolls-Royce and domestic AECC — the same platform as its GT work. Peers re-rating: Howmet (HWM) posted Q1'26 revenue +19% with Engine Products +29% and lifted FY26 guidance to ~US\$9.6–9.7b, with sharp share-price gains.
- **Nuclear – RMB1.6b.** Power generation's other strong leg: China is expected to approve 10–15 reactors a year from 2026–2030, with **Hualong One** at ~80% of new builds, and the 15th Five-Year Plan backing at-scale CAP-1400 production, with SPIC

earmarking RMB200b of 2026 investment. **Yingliu's exposure:** a leading supplier of Class-1 nuclear castings — notably the Hualong-One main coolant-pump (RCP) casing — with convertible bonds funding an advanced nuclear-shielding-materials expansion (FY25 nuclear revenue RMB0.68b). **Peers rallying:** nuclear-equipment names have run hard — China Nuclear Engineering (601611 CH, NR) jumped 33% in the first week of 2026 — alongside Dongfang Electric, Shanghai Electric and Harbin Electric, with operators CNNC (601985 CH, NR) / CGN (003816 CH / 1816 HK, NR) / and SPIC (000958 CH, NR).

- **Commercial space – hidden gem in LandSpace and SpaceX chain.** Arguably 2026's hottest theme — a LEO land-grab (**Starlink** past ~9,400 satellites vs China's **GuoWang** ~13,000 and **Qianfan** ~15,000 plans), national-security stakes and 15th-FYP priority; the A-share aerospace-equipment index led all sectors, +~37% YTD by mid-January 2026. LandSpace — maker of the Zhuque reusable rocket — won the nod for a ~RMB7.5b (US\$1.1b) STAR-Market IPO, set to be the first listed Chinese rocket maker, with SpaceX recently listed at ~US\$1.8t valuation. **Yingliu's exposure:** rocket-engine castings for LandSpace's Zhuque — a supplier into the very name driving the IPO wave (early-stage).
- **Low-altitude economy – intercepting into the aero engine space.** A 15th-FYP strategic cluster alongside AI and quantum, with official targets of ~RMB2t (US\$290b) by 2030 (and ~RMB3.5t by 2035); defense is a real pull, with PLA interest in turboshaft-powered medium-lifts, where engine reliability gates the whole category. **Yingliu's exposure:** unlike pure-casting peers, it builds its own powerplant — **turboshaft (YLWZ) and hybrid (YLHD) engines** via 60%-owned Yingliu Aerospace, plus drones and a whole-aircraft stake — RMB1b+ intent orders in 2025. Owning the engine is the scarce capability in the eVTOL/UAV chain.

Other growth legs are also red hot

Vertical	Downstream driver / position	TAM estimate	Potential revenue in 2030F (RMB b)
Aero parts	GE LEAP / Safran / Rolls-Royce; same-platform reuse with the gas-turbine line	Military / civil aircraft-engine casting market <u>TAM US\$10.5b</u> in 2025; engine component casting <u>+6.7%</u> ; aerospace superalloys <u>+7.7% CAGR</u> in 2025-2030F	1.5
Nuclear power	Sole Hualong-1 main-pump (RCP) casing, leading Class-1 castings; China nuclear build-out as driver	Nuclear power-plant equipment <u>TAM US\$22.5b</u> in 2025; Island equipment in China <u>+6% CAGR</u> on 110GW-by-2030 build	1.6
Commercial space	LandSpace Zhuque rocket-engine castings; high-profile but early-stage contribution	Propulsion market <u>TAM US\$7-9b</u> in 2025; 2025-2030F <u>CAGR +8% global / +9.5% China</u>	Optionality
Low-altitude / eVTOL	Turboshaft (YLWZ) + hybrid propulsion (YLHD) + whole-aircraft via Huayi (38.81%)	Turboshaft engine <u>TAM US\$5-6b</u> ; 2025-2030F <u>CAGR +7%</u>	Optionality

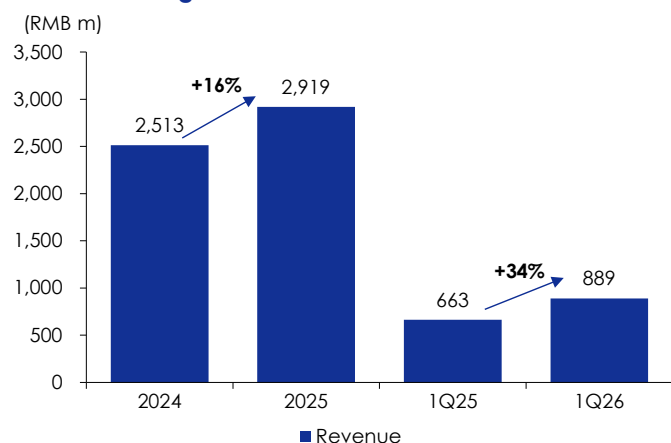
Source: Company, CCBIS

FINANCIAL FORECAST

Momentum into 2026: 1Q Sales +34%, Earnings +31%

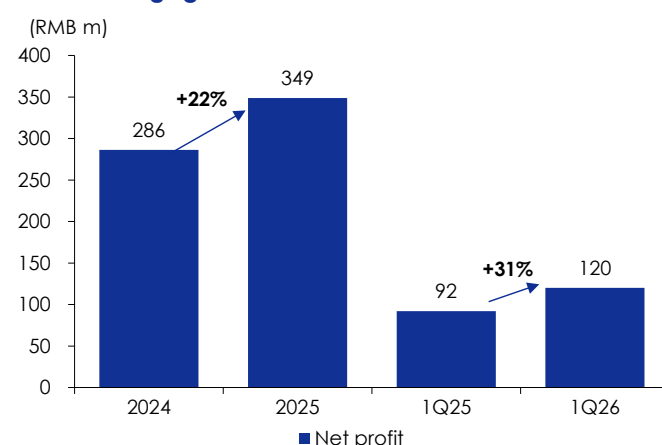
Revenue rose 16% to RMB2.9b in 2025, and growth has since accelerated — 1Q26 revenue was up 34% YoY at RMB889m, confirming the order book is converting. New orders exceeded RMB800m in 1Q26 against a full-year 2026F target of RMB3b (a ~27% first-quarter run-rate), and backlog stood at RMB2.1b at end-1Q, much of it gas-turbine and OEM-related. The next capacity step is scheduled for 2H26, when three ALD (directional-solidification / vacuum) furnaces are delivered and commissioned — the front-end casting capacity Yingliu needs to work down a backlog that already extends into 2027F.

1Q26 revenue growth accelerates to 34%



Source: Bloomberg, CCBIS estimates

1Q26 earnings growth accelerates to 31%



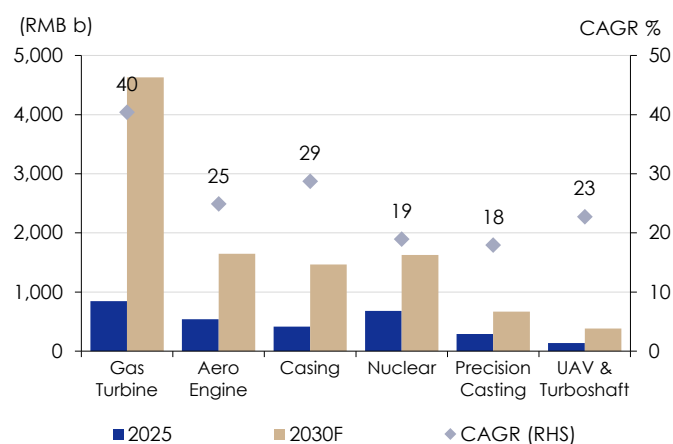
Source: Bloomberg, CCBIS estimates

Revenue Mix and Growth: GT +40% CAGR

By reported segment, 2025 revenue split into (1) **high-temperature alloy & precision castings** of RMB1.8b (+22.9%), (2) **nuclear & large metallic castings** RMB683m (+6.5%), (3) **new materials & equipment** RMB293m (+25.4%) and (4) **others** RMB138m (-18.5%). We disaggregate the high-temp-alloy segment into its three end-uses and forecast each through 2030F:

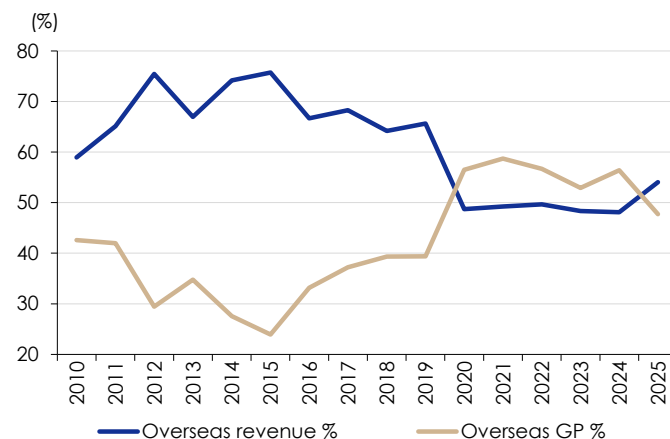
Gas Turbine is the engine of the forecast: a ~40% CAGR carries it to RMB4.6b by 2030F, while **Aero Blades** and **Casings** compound at 25% and 29%. **Nuclear** grows a steadier 19%, and the "other" bucket — **Aerospace** precision casting, **UAV** and turboshaft — at 20%. We flag that this mix is not fixed: it will flex with the relative pull of the gas-turbine versus aero-engine blade shortages on the demand side, against ramp constraints — furnace delivery cadence and skilled-labor availability — on the supply side.

GT as major growth driver



Source: Company, CCBIS estimates

Overseas business dominates



Source: Company, CCBIS estimates

Margin Expansion 36% → 40% on Pricing Power

We expect gross margin will widen from 35.8% in 2025 to 37% 38%/39% across 2026F/27F/28F and 40% in 2029–30F. The driver is structural rather than cyclical: as utilization rises toward full and the shortage of qualified hot-section supply persists behind high qualification thresholds, Yingliu gains pricing power. R&D points the same way — fine machining, thermal-barrier coating (TBC) and higher-stage hot-section parts (H-class GT 1st/2nd-stage vanes and blades, plus turboshaft) — each lifting ASP and enriching mix.

Earnings CAGR 47% in 2025-2030F

Operating leverage on top of the margin expansion will drive net profit up 80%/55%/48% in 2026F/27F/28F — a 47% CAGR through 2030F, well ahead of the 29% revenue CAGR and the core of the investment case.

Financial summary: Revenue +29% CAGR to RMB10b in 2030F; GPM +4.2ppt to 40%; NP +46% CAGR to RMB2.3b

(RMB b)	2023	2024	2025	2026F	2027F	2028F	2029F	2030F
Revenue	2,412	2,513	2,919	3,904	5,247	6,888	8,549	10,430
YoY (%)	9.7	4.2	16.1	33.8	34.4	31.3	24.1	22.0
GPM (%)	36.2	34.2	35.8	37.0	38.0	39.0	40.0	40.0
Selling (%)	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Admin (%)	8.0	7.9	8.2	8.2	8.0	7.8	7.8	7.8
R&D (%)	12.2	12.1	11.1	10.0	9.0	8.0	8.0	8.0
Net profit	303	286	349	630	974	1,445	1,889	2,311
YoY (%)	-24.5	-5.6	21.7	80.7	54.7	48.3	30.7	22.3
NPM (%)	12.6	11.4	11.9	16.1	18.6	21.0	22.1	22.2

Source: Company, CCBIS estimates

ROE Three-Folding to 20% by 2028F

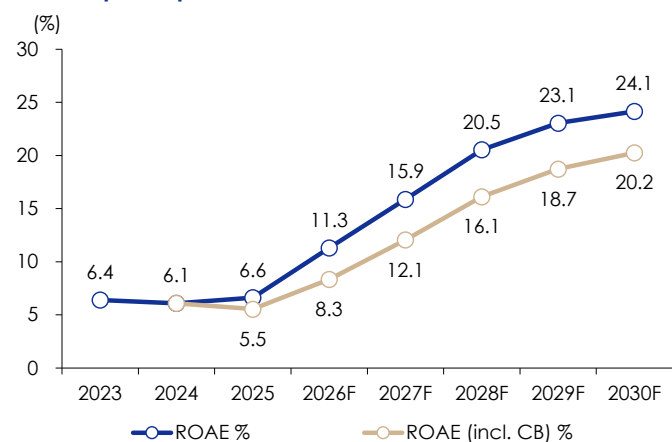
ROE is expected to climb from a depressed 6.6% in 2025 to 11.3%/15.9%/20.5% in 2026F/27F/28F, reaching 24.1% by 2030F as the capacity build matures into earnings — or 20.2% if the convertible bond fully converts and enlarges the equity base.

Convertible Bond Strengthen Balance Sheet

Net gearing will rise with the capacity build-out, from 94.7% in 2025 to a peak of 108.2% in 2027F, before easing to 66.1% by 2030F as volume scales and cash generation improves; treated as equity, the CB already brings 2025 net gearing down to a comfortable 42%. The

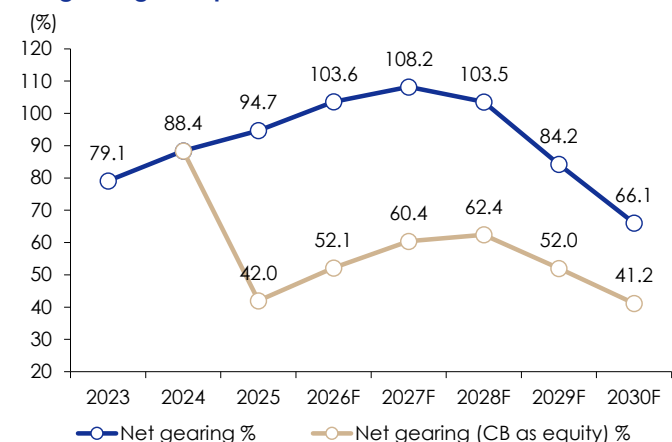
RMB1.5b CB, issued in September 2025, carries a conversion price of RMB30.47 against a current share price of ~RMB64.00 — deep in the money. It trades around 260, roughly 30% above its ~206 intrinsic value, signaling a high probability of conversion; conversion would de-lever the balance sheet and underpin the lower post-conversion gearing and ROE mentioned above.

ROE to spool up



Source: Company, CCBIS estimates

Net gearing to improve on exercise of CB



Source: Company, CCBIS estimates

PEER COMPARISON

We benchmark Yingliu with other upstream suppliers of gas turbines. However, we further identify Yingliu as one of the niche hot-section blade & casting specialists, distinguished from the upstream superalloy material producers – which are closer to commodities.

Hot-Section Blade Is a Sophisticated Product

- **Hot-blade casting is a high-moat, qualification-gated business.** The barrier is proprietary process know-how (cooling curves, casting sequences, impurity control — kept as trade secret, not patented), single-crystal / DS difficulty, and multi-year customer **validation** (2–3 years per part) that designates a supplier in for the life of the program. Switching costs are high and pricing power is real.
- **Superalloy / material producers (ingot, bar, powder) are closer to commodities** — lower switching costs, more volume/price cyclicality, thinner moat. We therefore exclude the superalloy names from the valuation comp and keep only hot-section specialists.

Closest peers: Howmet – global leader in both GT & aerospace, Wedge – domestic follower; Doncasters submitted S-1 in May 2026 to list in NYSE – nearest in terms of scale

Role	Company	公司	Stock code	Mkt cap (US\$ m)	Share price (LC)	P/E (x)			PEG (x)	Sales CAGR (%)	EPS CAGR (%)	GPM (%)	NPM (%)	Net gearing (%)	ROE (%)
Hot-Blade	Howmet Aerospace	豪梅特	HWM US	112,174	280.36	74.4	57.1	47.7	3.4	14	18	34	19	12.6	30.1
Hot-Blade	Yingliu	应流股份	603308 CH	6,906	69.26	135.8	74.9	48.3	1.5	34	52	36	12	97.2	8.3
Hot-Blade	Wedge	万泽股份	000534 CH	2,360	31.54	79.3	47.2	35.4	1.7	36	33	72	15	79.6	15.6
Superalloy	CISRI Gaona	钢研高纳	300034 CH	2,172	18.56	168.0	58.5	48.5	2.5	18	29	24	2	31.3	5.0
Superalloy	Longda	隆达股份	688231 CH	1,709	47.15	152.1	70.0	46.5	1.3	34	51	14	4	12.3	3.8
Superalloy	Toland Alloy	图南股份	300855 CH	1,985	34.18	87.6	34.2	25.9	1.3	36	30	28	14	(6.5)	11.5
Superalloy	Fushun Special Steel	抚顺特钢	600399 CH	1,309	4.52	(11.0)	48.4	9.5	0.0	11	27	1	(10)	47.5	(10.1)

Source: Bloomberg, Company, CCBIS estimates

A Privatization-Prone, Consolidating Space

- **Few listed players in a niche sector.** The decent business model — wide moat, qualification lock-in, long aftermarket/LTSA cash tails, critical-path bottleneck — make these assets prized by strategics and private equity, so most have been **taken private**: PCC by Berkshire (2016), Arconic by Apollo (2023), and Haynes by Acerinox (2024). The listed pure-play set is consequently thin.
- That trend is only now partly reversing — **Doncasters (DPC) filed for a NYSE listing in May 2026** — but the genuinely comparable listed names still number only a handful.

Howmet, Yingliu And Wedge Comparison

- **Howmet (HWM US, NR) — the leader and role model.** The largest GT hot-section component supplier globally, alongside private PCC the highest-quality, most-qualified name, with much larger scale and production capability than Yingliu and strength in **both** gas turbine and aero parts. Not cheap — Q1'26 revenue rose 19% and FY26 guidance was lifted to ~US\$9.58–9.73b — but high ROE (30%) on relatively lower earnings growth (18%) gives it the **highest PEG** (3.4x) of the three. Despite the scale difference, Howmet is the role model Yingliu is copying.
- **Yingliu (603308 CH, O) — China leader, tier-two top, highest growth.** Optically the most expensive on headline P/E (spot ~75x 2026F P/E), but it also has the **fastest EPS growth (52% CAGR)**, giving it the **lowest PEG (1.5x)**. Scale is ~1/20 of Howmet today,

closing toward ~1/10 by 2030F. It carries a decent roster of top-tier qualifications; technology reaches **3rd-stage blades on grade H and 1st-stage on grade F and below**; and it is extending down the chain into TBC and precision machining, with aerospace, nuclear parts and self-made aero engines — a deliberate replay of Howmet's playbook. ROE is the lowest of the three today but should climb toward **~20% by 2030F** as ASP step-up and CB conversion lift margins and de-lever.

- **Wedge (000534 CH, NR) — the little brother.** Also a premium valuation, but below Yingliu's, on slightly slower earnings growth (33%) and a higher ROE (16%), at smaller scale (half of Yingliu). Technology runs a couple of years behind Yingliu — and, more importantly, **customer qualifications are still few**, so it needs time to ramp before it earns Yingliu's multiple.

Yingliu's premium valuation is built on its fast earnings growth and market positioning

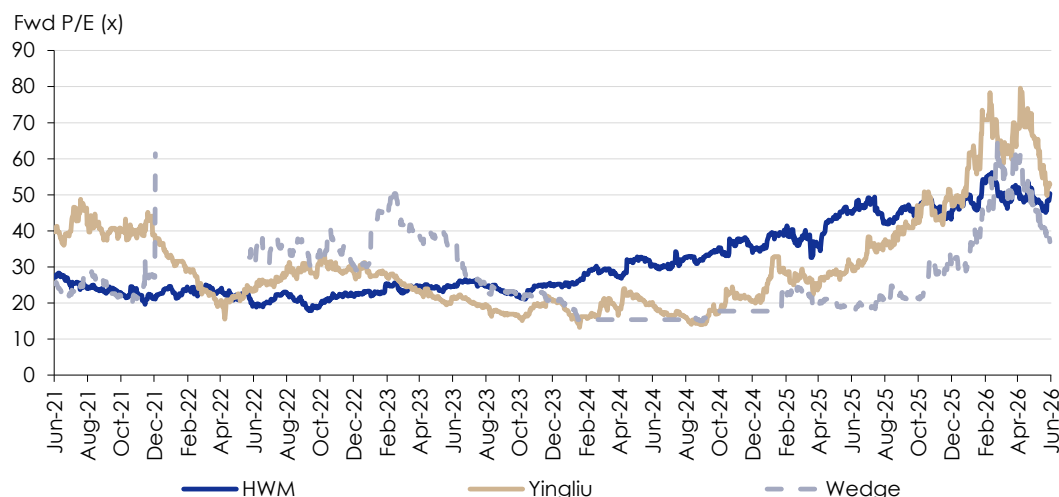
	2025 Revenue (RMB m)	2027F P/E (x)	EPS CAGR (%)	ROE (%)
Howmet Aerospace	56,196	47.7	18	30
Yingliu	2,919	48.3	52	8
Wedge	1,291	35.4	33	16

Source: Bloomberg, Company, CCBIS estimates

Why We Exclude Vertically-Integrated OEMs

Some OEMs (Siemens Energy, Doosan) have in-house blade capability, but they are not clean comparables. The OEM business is far larger and more diversified, which **drags the multiple**; the blade is a small slice of it; and even when they cast some blades in-house, OEMs remain net buyers and effectively **price-takers** to the specialist casters — reinforcing, not undermining, the blade-maker moat.

Forward P/E of hot-section players: Howmet's re-rating commenced in late 2023, Yingliu and Wedge followed up in 2025



Source: Bloomberg, CCBIS

RATING AND VALUATION

Valuation Method: 2030F P/E + Discount

30x 2030F P/E. Near-term multiples are distorted by the early, steep part of the growth curve and do not yet capture the booked blade + OEM order ramp. On 2026F the group screens optically expensive, and even 2027F multiples — converging into the 40–50x range — still under-reflect the growth runway:

Yingliu's high P/E has implicated high earnings growth

Company	2025	2026F	2027F
Howmet Aerospace	74.4	57.1	47.7
Yingliu	135.8	74.9	48.3
Wedge	79.3	47.2	35.4

Source: Bloomberg, Company, CCBIS estimates

Target Price RMB90.00; Initiate at Outperform

Why 30x. It is the **through-cycle average for both bellwethers** — Howmet's long-run average is ~30x (versus ~48x today) and Yingliu's own history also centers on ~30x (versus ~48x today). Applying it to 2030F assumes that by then the picture has **normalized**: growth decelerates from the current super-cycle pace, the gas-turbine supply shortage eases, and the competitive landscape stabilizes — by which point Yingliu's **scale and technology have closed much of the gap to the global top-three**, justifying a mature-leader multiple rather than a scarcity premium.

Target price RMB90.00, Outperform. We deduce our target price by multiplying 2030F EPS by 30x and discounting back to 2027F at 10% discount rate. Our target price for Yingliu is **RMB90.00**, implying ~40% upside from the current RMB64.34. Initiate at **Outperform**.

Target price RMB90.00 on 30x 2030F P/E; Outperform rating

	2025	2026F	2027F	2028F	2029F	2030F
EPS (RMB/share)	0.51	0.93	1.43	2.13	2.82	3.53
EPS - incl. CB (RMB/share)	0.00	0.86	1.34	1.98	2.63	3.29
Current P/E (x)	125.31	69.36	44.84	30.23	22.80	18.22
Target P/E (x)						30.00
Target price - 2030F (RMB)						118.52
Target price - 2027F (RMB)						90.00
Current price (RMB)						69.26
Upside						29.95
Rating						Outperform

Source: Bloomberg, CCBIS estimates

Key Risks

Slowdown in downstream Gas Turbine and Aerospace demand; Slippage in casting-capacity expansion or customer qualification (pushing out the order conversion); post-2030 order cliff (the 2000–05 US CCGT boom is the cautionary analogue); OEM price pressure or slower order spillover; balance-sheet leverage ahead of CB conversion; and timing/sentiment on the gas-turbine theme — the single biggest swing factor on a multiple this far forward.

Yingliu | 603308 CH – financial summary

Profit and loss

FYE 31 Dec (RMB m)	2024	2025	2026F	2027F	2028F
Operating income	2,513	2,919	3,904	5,247	6,888
Revenue	2,513	2,919	3,904	5,247	6,888
COGS	(1,695)	(1,938)	(2,499)	(3,306)	(4,271)
Gross profit	818	981	1,405	1,941	2,617
Operating expenses	(539)	(605)	(764)	(965)	(1,184)
Operating income	279	376	641	976	1,433
EBIT	248	335	606	937	1,390
EBITDA	544	667	971	1,339	1,832
Net financial income (exp)	(126)	(142)	(160)	(176)	(194)
JVs & associates	0	(0)	(0)	(0)	(0)
Other	95	102	125	137	151
Profit before tax	248	335	606	937	1,390
Tax	5	(23)	(41)	(63)	(94)
Total profit	254	313	565	874	1,297
Minority interest	33	36	65	100	149
Net profit attrib to shareholders	286	349	630	974	1,445
Core profit	277	321	599	940	1,407
Reported EPS (RMB)	0.42	0.51	0.93	1.43	2.13
Diluted EPS (RMB)	0.42	0.51	0.93	1.43	2.13
Core EPS (RMB)	0.42	0.51	0.93	1.43	2.13
Dividend (RMB m)	48	109	189	292	434
DPS (RMB)	0.07	0.16	0.28	0.43	0.64

Cash flow

FYE 31 Dec (RMB m)	2024	2025	2026F	2027F	2028F
Profit before tax	248	335	606	937	1,390
Amortization & depreciation	296	332	365	401	442
Net financial charge adj.	126	142	160	176	194
Tax paid	18	(30)	(73)	(106)	(152)
Other	(589)	(523)	(620)	(798)	(853)
Operating cash flow	99	256	438	611	1,020
Disposal of fixed assets	3	0	0	0	0
Capex	(592)	(625)	(688)	(757)	(832)
Investment	0	(262)	(507)	(659)	(714)
Net interest received (paid)	2	3	1	1	1
Other	(223)	15	1	1	1
Investment cash flow	(810)	(868)	(1,193)	(1,414)	(1,544)
Change in borrowings	574	1,152	1,639	2,130	2,308
Dividend paid	(229)	(196)	(204)	(265)	(337)
Other	338	(35)	0	0	0
Financing cash flow	684	921	1,435	1,865	1,970
Change in cash flow	(27)	309	680	1,063	1,447
Cash & equivalents, begin	182	151	480	1,160	2,223
Forex	(4)	20	0	0	0
Cash & equivalents, end	151	480	1,160	2,223	3,669

Source: Company data, CCBIS estimates

Balance sheet

FYE 31 Dec (RMB m)	2024	2025	2026F	2027F	2028F
Cash & equivalents	299	671	1,351	2,414	3,861
Receivables	1,198	1,111	1,438	1,863	2,324
Inventory	2,088	2,456	3,192	4,150	5,188
Other current assets	157	226	226	226	226
Total current assets	3,742	4,464	6,208	8,653	11,598
Property, plant & equipment	4,086	4,359	4,762	5,204	5,691
Intangible assets	29	29	29	29	29
JVs & associates	6	6	8	10	12
Other non-current assets	3,688	4,259	4,684	5,254	5,869
Total non-current assets	7,810	8,653	9,483	10,497	11,602
Total assets	11,552	13,117	15,691	19,151	23,200
Short-term borrowings	1,150	666	820	1,066	1,332
Trade & bill payable	1,330	1,517	1,962	2,541	3,168
Other current liabilities	1,004	1,066	1,311	1,704	2,130
Total current liabilities	3,483	3,249	4,093	5,310	6,630
Long-term borrowings	2,235	1,849	3,089	4,580	6,196
Other non-current liabilities	766	2,560	2,560	2,560	2,560
Total non-current liabilities	3,001	4,409	5,649	7,140	8,756
Total liabilities	6,485	7,658	9,742	12,451	15,386
Share capital	679	679	679	679	679
Reserves and retained profits	3,948	4,374	4,864	5,615	6,731
Shareholders' equity	4,627	5,054	5,543	6,294	7,410
Minority interest	441	405	405	405	405
Total equity	5,068	5,459	5,949	6,700	7,815
Total equity and liabilities	11,552	13,117	15,691	19,151	23,200

Ratios

FYE 31 Dec	2024	2025	2026F	2027F	2028F
Growth (%)					
Revenue	4.2	16.1	33.8	34.4	31.3
EBITDA	(2.2)	22.6	45.5	37.9	36.8
EBIT	(10.1)	34.8	70.4	52.2	46.8
Net profit	(5.6)	21.7	80.7	54.7	48.3
Core net profit	(1.1)	15.8	86.7	57.0	49.7
Profitability (%)					
Gross margin	32.6	33.6	36.0	37.0	38.0
EBITDA margin	21.7	22.9	24.9	25.5	26.6
EBIT margin	11.1	12.9	16.4	18.6	20.8
Net margin	11.4	11.9	16.1	18.6	21.0
Core net margin	11.0	11.0	15.3	17.9	20.4
Tax rate	(2.2)	6.7	6.7	6.7	6.7
Efficiency (days)					
Inventory turnover	425	428	413	405	399
Trades receivables	164	140	117	113	110
Trades payables	221	228	213	208	204
Returns & leverage (%)					
ROAA	2.6	2.8	4.4	5.6	6.8
ROAE	6.3	7.2	11.9	16.5	21.1
Net debt (cash)/equity	60.9	33.8	43.0	48.2	46.9
Payout ratio	16.6	31.2	30.0	30.0	30.0
Interest coverage	(2.2)	(2.6)	(4.0)	(5.5)	(7.4)
Liquidity (x)					
Current ratio	1.1	1.4	1.5	1.6	1.7
Quick ratio	0.5	0.6	0.7	0.8	1.0

Dongfang Electric | 600875 CH/1072 HK

The big brother of China gas turbines

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- **China power equipment leader: Six-power platform strengths at coal, nuclear, GT**
- **RMB160b orders and expanding margins drive earnings**
- **G50 orders and delivery as catalyst**

China's power-equipment leader. Dongfang Electric (DEC) is a central-SOE power-equipment champion and one of China's "Big Three" power groups. Its "six-power" platform spans coal, hydro, nuclear, wind, gas and solar, and extends into station services, hydrogen and storage; DEC has supplied roughly a quarter of China's installed generation capacity, with 30–70% share across coal / nuclear / hydro / gas. Crucially, through a decade of equipment oversupply (2015–25), DEC still compounded revenue / net profit at 8% / 24% CAGR — and overseas, just 6.5% of 2025 revenue, is the growth engine from here.

Orders and margins drive earnings. The AI-driven power shortage and 2030 sold-out lead times at Global Big Three gas turbine suppliers have pushed overseas buyers toward DEC's fully-domestic F-class G50. **Two earnings drivers** — (1) backlog conversion: ~RMB160b backlog at 1Q26 (~2x revenue) gives revenue visibility to 2028; (2) margin recovery: higher-priced coal orders, wind-turbine price hikes and overseas gas orders lift GPM from ~16% toward ~18%. **A re-rating argument** — station services run ~40% GPM and ~16% of profit, a high-margin recurring stream the market still prices as a cyclical. **Optionality on top** — coal capacity-payment revaluation plus a hydrogen call option.

Initiate at Outperform, G50 orders and delivery as catalyst. Our SOTP valuation assigns 15x 2027F P/E to coal / hydro, 25x to gas / nuclear and 30x to services— implying a target market cap of ~RMB165b, an A-share TP of RMB49.00 and an H-share TP of HK\$45.00. **Initiate at Outperform.** **Catalysts** include G50 new orders and commissioning (2H26+), the GPM inflection in 2026F, and the capacity-payment step-up.

Key risks slower domestic tendering, gas-turbine certification / delivery delays, price cuts by global incumbents, and slow wind-margin recovery.

Forecast and valuation

Year to 31 Dec	2024	2025	2026F	2027F	2028F
Revenue (RMB m)	69,695	78,615	86,799	96,767	107,064
YoY (%)	14.9	12.8	10.4	11.5	10.6
Net profit (RMB m)	2,922	3,831	4,925	6,078	7,402
YoY (%)	(17.7)	31.1	28.5	23.4	21.8
Fully diluted EPS (RMB)	0.94	1.11	1.42	1.76	2.14
YoY (%)	(17.7)	18.2	28.5	23.4	21.8
P/E (x) A-share	35.2	29.8	23.2	18.8	15.4
P/E (x) H-share	-	19.8	15.4	12.5	10.3
DPS (RMB)	0.40	0.53	0.78	0.97	1.18
Dividend yield (x) A-share	1.2	1.6	2.4	2.9	3.6
Dividend yield (x) H-share	-	2.3	3.4	4.3	5.2
P/S (x) A-share	1.5	1.5	1.3	1.2	1.1
P/S (x) H-share	1.0	1.0	0.9	0.8	0.7
ROAE (%)	7.7	9.1	10.5	12.2	13.8
Net debt/equity (%)	Net cash	Net cash	Net cash	Net cash	Net cash

Source: Bloomberg, CCBIS estimates

H-share: **Outperform** (Initiation)

A-share: **Outperform** (Initiation)

Current price:

RMB33.03

(as at 22 Jun 2026)

Target:

► **RMB49.00**

(initiation)

HK\$25.24

(as at 22 Jun 2026)

► **HK\$45.00**

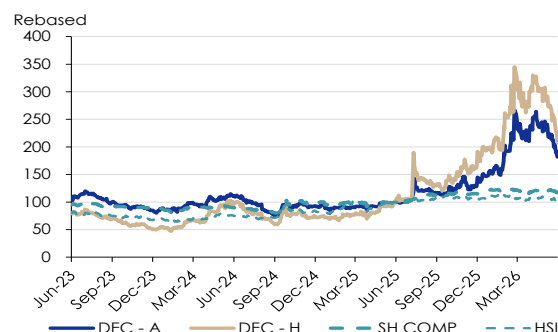
(initiation)

Trading data

	600875 CH	1072 HK
52-week range	RMB15.74–45.38	HK\$11.56–46.70
Market capital (m)	US\$14,882	US\$1,320
Shares outstanding (m)	3,050	408
Free float (%)	37	100
3M avg daily T/O (m share)	91	14
3M avg daily T/O (US\$ m)	447	67
Expected 12M return (%)	48	78

Source: Bloomberg, CCBIS estimates

Share price vs SH COMP & HSI



Source: Bloomberg

Stock performance

Performance over	1M	3M	12M
600875 CH			
Absolute	(22)	(26)	83
Relative (%) to SH COMP	(17)	(23)	66
1072 HK			
Absolute	(28)	(33)	105
Relative (%) to HSI	(21)	(28)	105

Source: Bloomberg

东方电气 | 600875 CH/1072 HK

中国燃气轮机带头大哥

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► 中国发电设备龙头: 六电并举

► 160 亿在手订单与毛利率改善驱动盈利

► G50 订单与交付为催化剂

中国发电设备龙头。东方电气(DEC)是央企发电设备龙头,中国“三大电气”集团之一。其“六电并举”平台覆盖火电、水电、核电、风电、气电及光伏,并延伸至电站服务、氢能与储能;东方电气累计提供了约四分之一的全国发电装机容量,在火电/核电/水电/气电的市占率达 30-70%。值得注意的是,在长达十年的设备产能过剩期(2015-25),东方电气的收入/归母净利仍以 8%/24%的 CAGR 复合增长——而仅占 2025 年收入 6.5% 的海外业务,将是未来的增长引擎。

订单与毛利率驱动盈利。AI 驱动的电力短缺,叠加全球三大燃机巨头交付排期已排满至 2030 年,促使海外买家转向东方电气全国产化的 F 级 G50。**两大盈利驱动**——(1)在手订单转化:1Q26 在手订单约 1,600 亿元人民币(≈2x 营收),为营收提供至 2028 年的能见度;(2)毛利率修复:煤电高价订单、风机涨价及海外燃机订单,推动毛利率由约 16%升至约 18%。一个重估逻辑——电站服务毛利率约 40%、占利润约 16%,是高毛利的经常性收入流,但市场仍按周期股给予估值。此外尚有**期权价值**——煤电容量电价重估,以及氢能期权。

首次覆盖给予“优于大市”,G50 订单与交付为催化剂。我们的 SOTP 估值对煤电/水电给予 15 倍 2027F 市盈率、对气电/核电给予 25 倍、对电站服务给予 30 倍——对应目标市值约 1,700 亿元人民币、A 股目标价 49.00 人民币、H 股目标价 45.00 港元。**首次覆盖给予“优于大市”。**催化剂包括:G50 新订单与商运投产(2H26 起)、2026F 毛利率拐点,以及容量电价上调。

主要风险:国内招标放缓、燃机认证/交付不及预期、海外巨头降价、风电盈利修复偏慢。

预测和估值

截至 12 月 31 日	2024	2025	2026F	2027F	2028F
收入(百万元 人民币)	69,695	78,615	86,799	96,767	107,064
同比(%)	14.9	12.8	10.4	11.5	10.6
净利润(百万元 人民币)	2,922	3,831	4,925	6,078	7,402
同比(%)	(17.7)	31.1	28.5	23.4	21.8
稀释每股收益(人民币)	0.94	1.11	1.42	1.76	2.14
同比(%)	(17.7)	18.2	28.5	23.4	21.8
市盈率(倍) A 股	35.2	29.8	23.2	18.8	15.4
市盈率(倍) H 股	-	19.8	15.4	12.5	10.3
股息(人民币)	0.40	0.53	0.78	0.97	1.18
股息回报率(倍) A 股	1.2	1.6	2.4	2.9	3.6
股息回报率(倍) H 股	-	2.3	3.4	4.3	5.2
市销率(倍) A 股	1.5	1.5	1.3	1.2	1.1
市销率(倍) H 股	1.0	1.0	0.9	0.8	0.7
股本回报率(%)	7.7	9.1	10.5	12.2	13.8
净负债率(%)	净现金	净现金	净现金	净现金	净现金

资料来源:彭博、建银国际证券预测

H 股: **优于大市** (首次覆盖)

A 股: **优于大市** (首次覆盖)

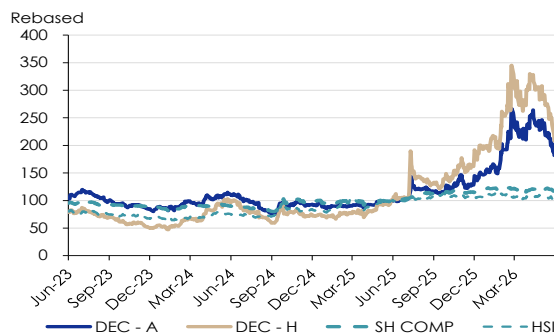
现价:	33.03 人民币	目标价:	49.00 人民币
(截至 2026 年 6 月 22 日)		(首次覆盖)	
25.24 港元		45.00 港元	
(截至 2026 年 6 月 22 日)		(首次覆盖)	

交易数据

	600875 CH	1072 HK
52 周股价高低	人民币 15.74-45.38	港元 11.56-46.70
市值(百万美元)	14,882	1,320
由流通股数(百万股)	3,050	408
自由流通股份(%)	37	100
3 个月日均成交股数(百万股)	91	14
3 个月日均成交额(百万美元)	447	67
12 个月预期回报率(%)	48	90

资料来源:彭博、建银国际证券预测

股价相对于上海综合指数及恒生指数



资料来源:彭博

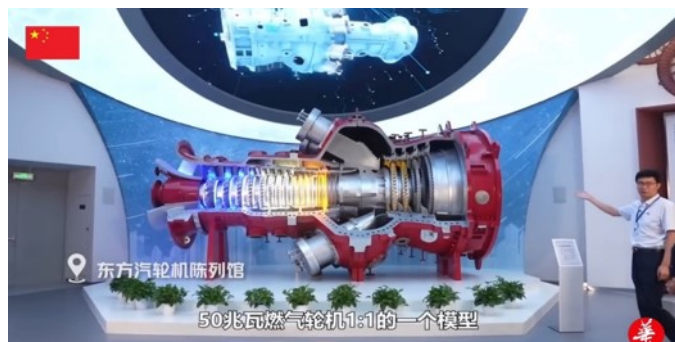
股价表现

表现区间	一个月	三个月	十二个月
600875 CH			
绝对表现	(22)	(26)	83
相对于上海综合指数	(17)	(23)	66
1072 HK			
绝对表现	(28)	(33)	105
相对于恒生指数	(21)	(28)	105

资料来源:彭博

FOCUS CHARTS

The 50MW F-class G50 is the first fully-domestic GT in China, receiving orders from Canada, Kazakhstan, etc.



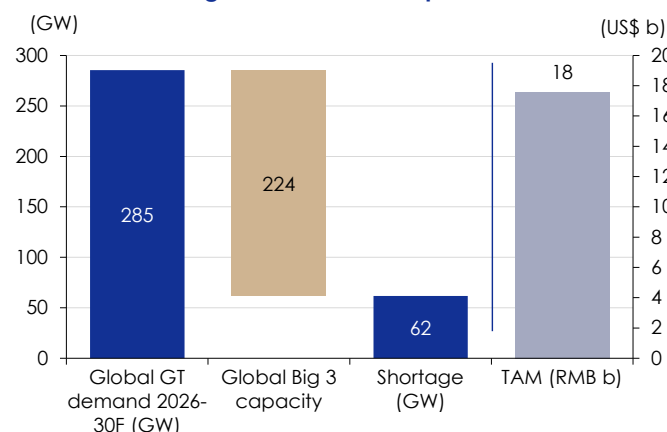
Source: Company

The 300MW F-class GT is a JV project by China United Gas Turbine and Big Three electric companies in China



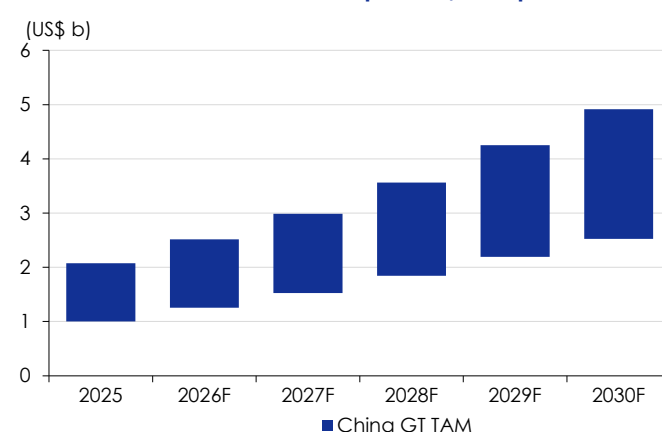
Source: Company

Global GT shortage in 2026-2030F: up to 62GW or RMB18b



Source: IEA, Company, CCBIS estimates

China GT TAM in 2026-2030F: up to US\$2-5b p.a.



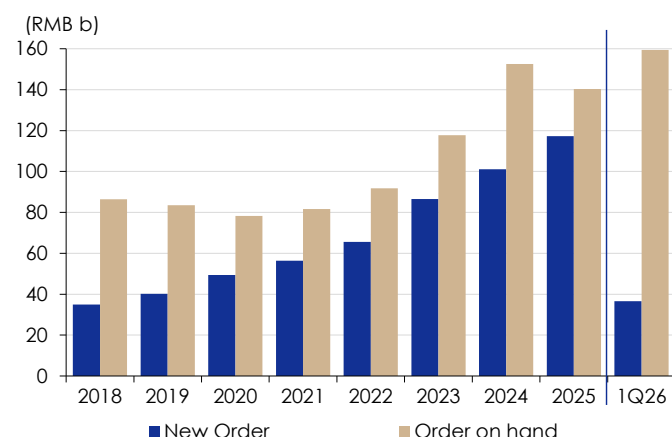
Source: IEA, Company, CCBIS estimates

SOTP Bridge — Stable Core + New Gas Units + Service Annuity

	2027F NP	2027F P/E (x)	Valuation	Rationale and assumptions
Coal-fired power	2,080	15	31,206	Anchored to global thermal OEM
Hydroelectric power	303	15	4,552	Utility-like, low margin, high visibility
Nuclear power	814	25	20,344	Premium on high margin / growth, discount to nuclear plays
Natural gas-fired power	489	25	12,221	Domestic replacement
Wind power	392	10	3,924	Low margin, immaterial
Power Plant Services	1,440	30	43,208	Recurring, high margin, benchmarked to global OEM
Engineering	195	15	2,926	Low margin EPC, finance
Emerging growth industries	542	20	10,837	Future growth potential
Gas-turbine + LTSA	781	30	27,094	25 units x 250m x 12% NPM + 5% LTSA
Total valuation			156,313	
Net cash			13,022	50% net cash not tied to finance
Net valuation			169,335	
	Sh px	TP	Upside	Rating
A share	33.03	49	48	Outperform
H share	25.24	45	78	Outperform

Source: Company, CCBIS estimates

Order backlog covers 2x revenue in 2025



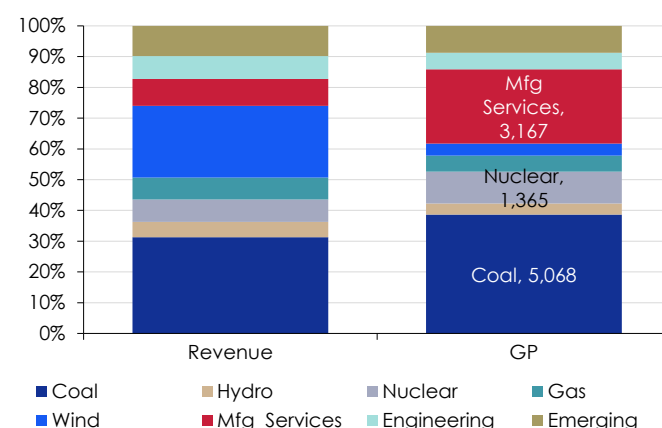
Source: Company, CCBIS

Decent fundamentals + Reasonable valuation + Best GT exposure among China Big Three

Company	EPS CAGR (%)	NPM (%)	ROE (%)	2027F P/E (x)	PEG (x)
Dongfeng Elec. H	23.0	4.9	9.3	12.5	1.1
Dngfeng Elec. A				18.8	1.3
Shanghai Elec. H	26.4	1.0	1.6	29.2	1.1
Shanghai Elec. A				65.6	3.2
Harbin Elec.	19.7	5.8	15.7	7.9	0.9

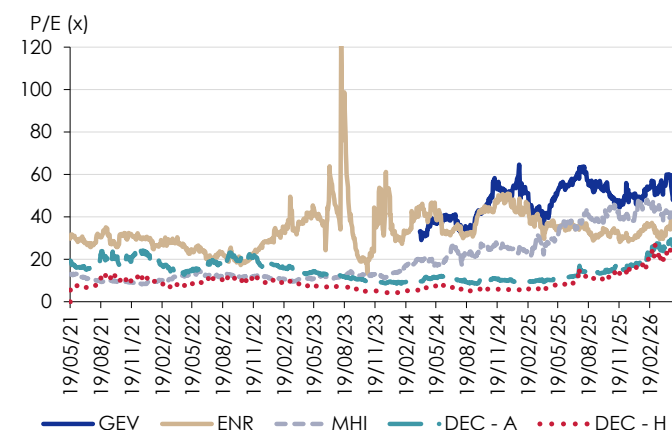
Source: Company, CCBIS estimates

Coal, Nuclear, Services drive earnings



Source: Company, CCBIS

A/H P/E band vs global big three



Source: Company, CCBIS

COMPANY OVERVIEW

Development Trajectory – From Sichuan To Global

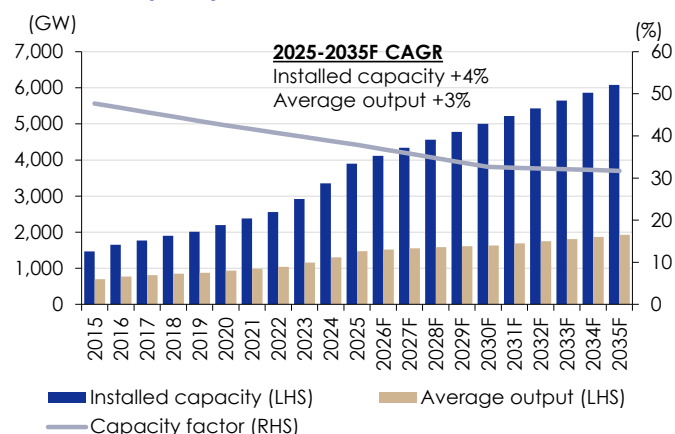
Development Trajectory

Year	Major events
1958	Deyang Hydro Power Equipment Works breaks ground — the origin of the DEC group
1966	Dongfang Turbine and Dongfang Boiler works established in Mianzhu / Zigong, Sichuan
1995	Dongfang Electric Machinery lists in both Shanghai and Hong Kong
2007	Full asset listing completed; "Dongfang Electric Corp." trades on A+H
2024	13MW offshore unit rolls off the line; backlog peaks at RMB153bn
2025	G50 wins first overseas orders (Kazakhstan / Iraq); Yarlung project ~RMB37.5b award

Source: Company

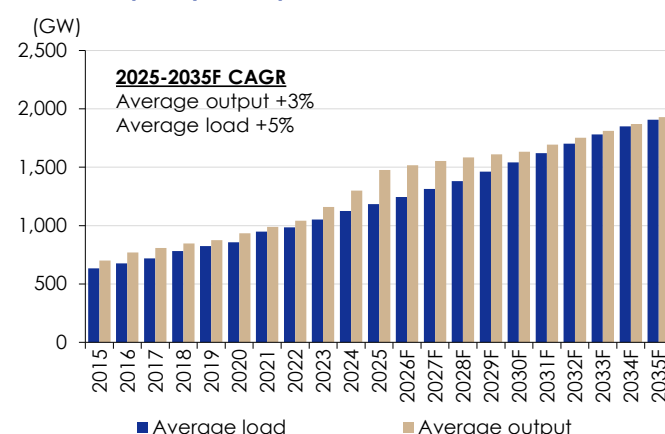
China Power Market

Installed Capacity +4% CAGR; CF lowers from 38% to 32%



Source: IEA, State Grid, CCBIS estimates

Excess Capacity for exports



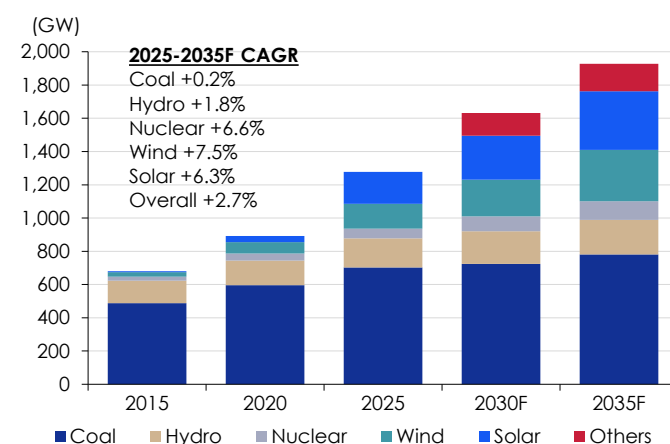
Source: IEA, State Grid, CCBIS estimates

Installed capacity and average output by power source in 2035F

Power Source	Installed Capacity (GW)	Utilization hour/year	CF	Average Output (GW)
Coal	1,850	3,700	0.42	781
Hydro	520	3,500	0.40	208
Nuclear	130	7,550	0.86	112
Wind	1,200	2,250	0.26	308
Solar	2,000	1,550	0.18	354
Others	380	3,800	0.43	165
Total	6,080	—	0.32	1,928

Source: IEA, State Grid, CCBIS

Average output driven by Wind and Solar



Source: IEA, State Grid, CCBIS

Business model — Six-power platform + Three order blocks

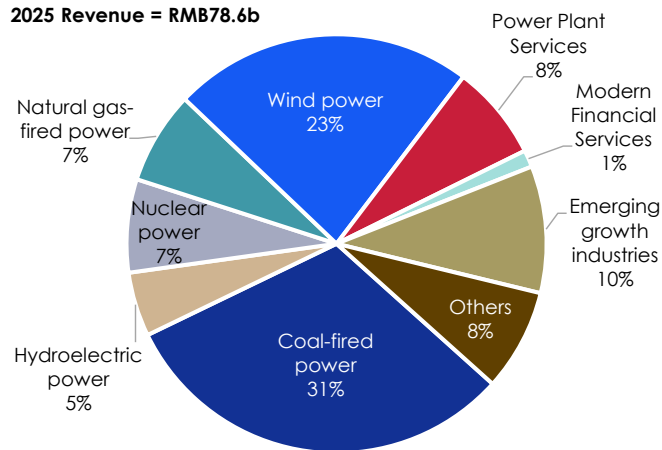
Six-power platform: coal, hydro, nuclear, wind, gas and solar — a breadth rarely seen in global power equipment; high share in coal / nuclear / hydro, with the mix shifting steadily toward clean energy.

Three order blocks: equipment manufacturing (~67%), manufacturing services (~22%; station O&M / retrofit / overhaul), and emerging industries (~11%; storage / hydrogen / solar gear).

Profit base: coal and nuclear contribute the most gross profit; station services (40%+ GPM) provide a recurring cash stream, gradually dampening cyclical and lifting growth quality.

Revenue: Coal > Wind > Emerging

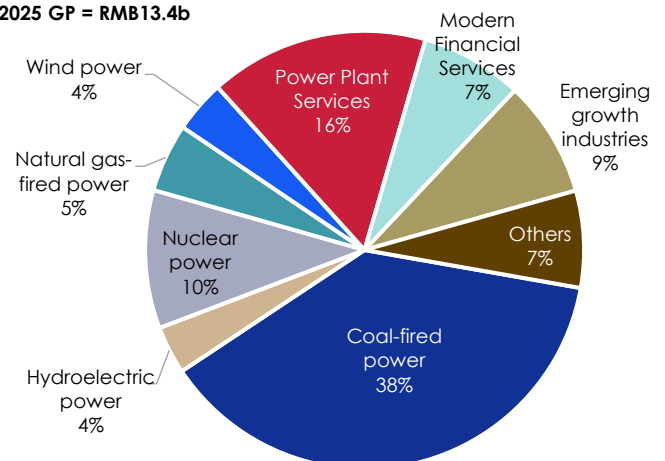
2025 Revenue = RMB78.6b



Source: Company, CCBIS estimates

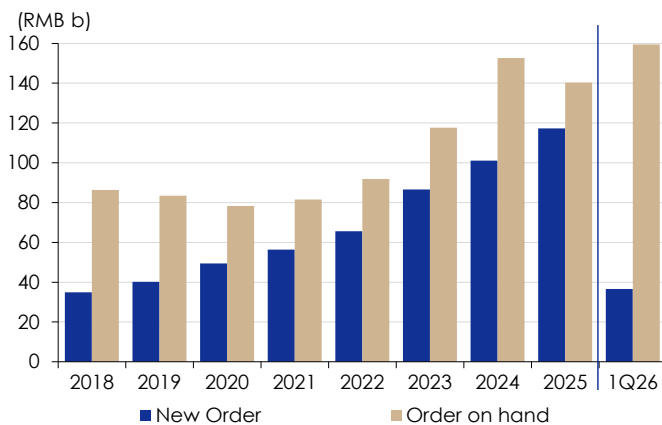
GP: Coal > Power Plant Services > Nuclear

2025 GP = RMB13.4b



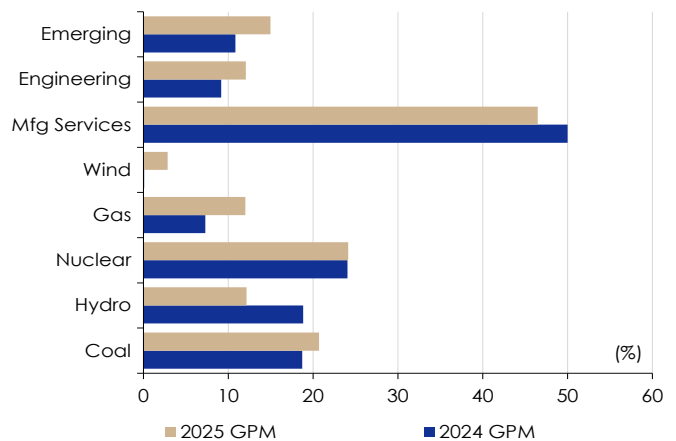
Source: Company, CCBIS estimates

Order on hands extends until 2028F



Source: Company, CCBIS estimates

GPM: Coal > Power Plant Services > Nuclear



Source: Company, CCBIS estimates

Product Portfolio

Six-Power Platform with each flagship products

Segment	Key products	Share / position	Angle
Gas	G50 (full-domestic F-class) / G15 / G80 / G200; M701 licensed range	70%+ domestic	Export lead, valuation catalyst
Coal	Ultra-supercritical main equipment + flexibility retrofit	30%+	Capacity payment + price hikes
Nuclear	Hualong One / Linglong One island + conventional island	50%+	High-margin cash cow
Hydro	Impulse / pumped-storage units (Yarlung 500MW)	40%+	Yarlung second curve
Wind	18MW / 26MW offshore turbines	Offshore leader	Volume; margin repair pending
Service	Installed-base O&M / retrofit / LTSA-style overhaul	—	Recurring, 40%+ GPM

Source: Company, CCBIS

Flagship Products include G50 GT (export breakthrough), Beihetan turbo-gen, Hualong nuclear Class 1, Offshore Wind turbine

National flagship equipment



Gas (G50 turbine — export breakthrough)

- G50 heavy-duty gas turbine (50MW class)
- Edge: proprietary IP; flagship projects in Canada (Synapse) & Kazakhstan



Hydro (strongest franchise — Baihetan flagship)

- 1,000MW Francis hydro turbine-generator (Baihetan)
- Edge: world's largest single-unit capacity; proprietary IP; all units



Nuclear (nuclear + conventional island — national flagship)

- Hualong One nuclear steam generator (Nuclear Class 1)
- Edge: China's first Nuclear Class 1 design license; GW-class serial production



Wind (onshore + offshore)

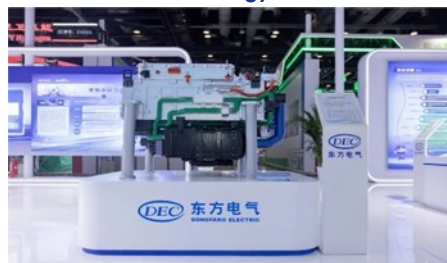
- Offshore wind turbine (10MW+)
- Edge: high capacity & reliability; leading domestic offshore-wind supplier

New-energy stars



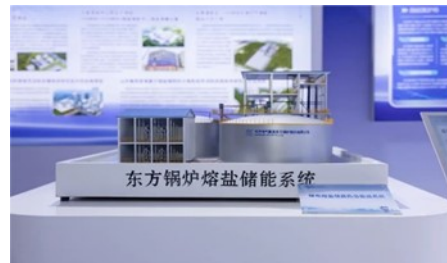
Hydrogen (full value chain)

- PEM electrolyser (500 Nm³/h)
- Edge: self-controlled tech; high current density; couples with wind/solar



Hydrogen fuel-cell system (40–270kW)

- Edge: automotive-grade; 1,200+ vehicles in service; up to 550,000 km each



Storage / CSP (solar-thermal)

- Molten-salt storage & heat-exchange island (CSP)
- Edge: core equipment for the Hami CSP project; integrated storage + generation

Core components / systems



Large castings & forgings (nuclear / hydro / gas)

Edge: Nuclear Class 1; GW-class rotors; rare globally

Source: Company, CCBIS

Customers

Domestic: the big five gensco's (Huaneng / Datang / Huadian / CHN Energy / SPIC), CNNC / CGN, and provincial energy platforms.

Overseas (the new gas-turbine angle): Kazakhstan and Iraq unit orders, plus a Canadian data-center customer with a 10-unit intent plus deposit — entry into developed-market DC power supply from emerging markets.

Key Customers include Big Five Gensco in China, Oil & Gas giants, Hydro / Nuclear Power companies, Hyperscalers, Overseas data centers



Source: Company, CCBIS

INVESTMENT CASE — ONE THEME AND TWO DRIVERS

Core Investment Theme: Gas-Turbine Breakthrough (AI Infrastructure × Global Benchmarking)

- **Demand:** the AI data-center power shortage has driven GEV / Siemens Energy / Mitsubishi new-unit lead times out to 2029–2030F, forcing global buyers toward mid-size / domestic units.
- **DEC's hook:** the fully-domestic F-class G50 (100% in-house, >12,000 commercial hours cumulative) — new units plus 20-year LTSA profit carry no imported hot-section premium, so DEC keeps the high value-add.
- **Valuation logic (core):** gas turbines are not the single earnings engine over the next 2–3 years (~8% of 2027F earnings), but they are the re-rating catalyst — the market benchmarks DEC against global peer PE, lifting the whole valuation.

Driver 1: Order Surge + Visibility to 2028

- **All-in backlog ~RMB160b (1Q26)**, 70% domestic / 30% overseas (6.5% overseas revenue in 2025); year-end basis RMB140b, clean energy ~70%; equivalent to ~1.8–2.8x annual revenue.
- **Structural shift:** new effective orders rose from ~RMB55b (2018) to RMB117b (2025, +15.9%), with 2023 the inflection (+32%); from “coal-cycle stock” to “clean-energy growth stock”.
- **Multiple legs relay:** coal (capacity payment), nuclear (≥10 approvals/year), hydro (Yarlung + pumped storage) and gas (exports) rotate, extending order visibility to 2028.

Driver 2: Margin Recovery

- **Coal:** low-price orders fully delivered; higher-priced orders now flowing (main equipment RMB0.8b → RMB1.1b/ GW); 2025 coal GPM up to ~20.7% (+4.3pct).
- **Wind:** recovery from near-0% gross profit is the biggest lever; with 2026 turbine price hikes, GPM could rebound to 8–10%.
- **Gas + mix:** rising localization dilutes the imported hot section; mix shift toward nuclear / gas / services plus operating leverage push blended GPM into an uptrend.

A Re-rating Argument: Re-rating of the Profit Structure via Station Services

- **An overlooked gem:** services are ~8% of revenue but 16%+ of gross profit, with 37% GPM, and grows with the installed base.
- **Benchmark GEV:** ~50% of GEV's power-equipment orders come from services; the market still treats DEC as a pure cyclical, so the service annuity is under-priced — it deserves a higher multiple.

An Option to the Future: Coal Capacity Payment + Hydrogen

- **Coal capacity payment:** RMB330/kW-yr nationwide, 30 – 50% compensated in 2025 and ≥50% from 2026 — coal shifts from “selling power” to “selling power + standby capacity,” revaluing units and underpinning a coal order / profit inflection.
- **Hydrogen / storage option:** emerging industries are ~11% of orders; electrolyzers / storage are early-stage — a free option not in the core valuation but offering long-term upside.

FINANCIAL FORECAST

Six-Power Segments: Revenue / GPM / Backlog (2025)

Clean energy is the earnings driver; Renewable is the revenue driver; Services is the value driver

Segment	2025 Revenue (RMB b)	2018-2025 CAGR (%)	2025 GP (RMB b)	2025 GPM (%)	Orders on hand (RMB b)	Order / Book	Watching points
Energy Equipment							
Clean energy							
Coal	24.5	6.1	5.1	20.7	30.0	1.2	Capacity payment, supply guarantee
Nuclear	5.7	9.7	1.4	24.1	20.0	3.5	High visibility, high margin
Gas	5.6	9.2	0.7	12.0	15.0	2.7	Gas turbine export
Renewable energy							
Wind	18.2	26.6	0.5	2.8	20.0	1.1	Dragger / margin-repair lever
Hydro	3.9	17.4	0.5	12.1	19.0	4.9	Yarlung / PSH high order book
Manufacturing Services							
Modern manufacturing services	6.8	16.0	3.2	46.5	16.1	2.4	LTSA recurring
Engineering & supply chain	5.8	5.9	0.7	12.1	7.7	1.3	EPC, financial
Emerging	7.7	14.2	1.2	15.0	11.4	1.5	Storage, hydrogen, etc
Total	78.6	11.3	13.4	17.0	140.3	1.8	

Source: Company, CCBIS estimates

Earnings Forecast

Earnings forecast

(RMB m, %)	2023	2024	2025	2026F	2027F	2028F	2025-2028F
Revenue	60,677	69,695	78,615	86,799	96,767	107,064	11
Coal-fired power	13,897	16,371	24,492	25,961	27,259	28,349	5
Hydroelectric power	2,311	2,955	3,903	4,879	6,098	7,623	25
Nuclear power	2,903	4,877	5,660	7,075	9,197	11,497	27
Natural gas-fired power	3,789	7,110	5,628	7,034	9,145	11,431	27
Wind power	10,840	12,970	18,224	19,135	20,092	21,097	5
Power plant services	3,788	6,135	5,781	6,706	7,578	8,335	13
Modern financial services	1,126	1,102	1,033	1,053	1,074	1,096	2
Project contracting	3,966	5,220	5,000	5,250	5,513	5,788	5
Supply chain mgmt & services	7,922	721	800	840	882	926	5
Emerging growth industries	8,843	11,150	7,709	8,865	9,929	10,922	12
GPM (%)	18.8	15.5	17.0	16.9	17.5	18.2	+1ppt
Selling (%)	2.6	1.2	1.1	1.1	1.1	1.1	Flat
Admin (%)	5.6	5.1	4.8	4.8	4.8	4.8	Flat
R&D (%)	4.5	4.3	4.1	4.1	4.1	4.1	Flat
Net profit	3,550	2,922	3,831	4,925	6,078	7,402	23
NPM (%)	5.9	4.2	4.9	5.7	6.3	6.9	+2ppt
ROE (%)	7.1	5.2	7.6	10.5	12.2	13.8	+5.7ppt
Gearing (%)	(40.5)	(61.5)	(37.0)	(44.0)	(50.4)	(56.5)	Net cash

Source: Company, CCBIS estimates

PEER COMPARISON

Peer comparison – Global Big Three vs China Big Three vs Korea champion

Company	Business mix	Sales / NP (latest FY)	GPM / NPM	EPS CAGR	P/E (fwd)	PEG	One-liner
DEC (600875 CH / 1072 HK)	Six-power: coal 31%, wind 23%, services 8%, nuclear 7%, gas 7%, hydro+solar bal.	RMB78.6b / RMB3.8b	17% / 5%	25	19/12	1.3/1.1	The only fully-domestic F-class (G50) integrated OEM; cheapest credible way to own the China gas-turbine export story, with coal/nuclear/services as ballast.
SHE (601727 CH / 2727 HK)	Coal, wind, nuclear, T&D, storage, industrial; no hydro	RMB127b / RMB1.2b	18% / 1%	33	66/29	3.2/1.1	Biggest and broadest, but margins razor-thin and earnings erratic; trades on asset/revenue scale not earnings — its rich multiple flatters DEC's by comparison.
Harbin (1133 HK)	Coal 43%, hydro 11%, nuclear, gas (small HTGT only)	RMB46b / RMB2.7b	15% / 6%	14	8	0.9	Cheapest of the Big Three, but no G50 — only a ~16MW unit still in test, exports uncertified; "cheap but incomplete," a value trap unless the gas gap closes.
GEV (GEV US)	Gas power, electrification, wind, nuclear (SMR); ~50% services	RMB259b / RMB16b	20% / 6%	25	47	2.6	The global benchmark and price-setter; slots sold out to 2028, huge services RPO (US\$87b). DEC trades at a fraction of its multiple — the re-rating yardstick.
SIE (ENR GR)	Gas services, grid tech, transformation, wind (Gamesa)	RMB309b / RMB11b	17% / 4%	33	29	1.4	H-class tech leader; record €138bn backlog, wind drag finally narrowing. Re-rated hard off the turnaround — shows what fixing a margin story is worth.
MHI (7011 JP)	Energy (GTCC), aero/defense, logistics, plants	RMB214b / RMB14b	22% / 7%	29	33	1.2	Heavy-duty / J-class powerhouse doubling turbine capacity; conglomerate structure dilutes the gas pure-play, but the defense+energy combo earns a premium.
Doosan (034020 KS)	Gas turbines, nuclear/SMR, thermal EPC, wind, castings	RMB77b / RMB381m	16% / 0%	62	104	1.9	Asia's other domestic gas-turbine challenger (proprietary large GT since 2019) + an SMR supply-chain story (Rolls-Royce, NuScale, X-energy); rerating on orders, not earnings — DEC's closest "national champion" analogue.

Source: Bloomberg, Company, CCBIS estimates

Gas-Turbine-Only Reference (non-power peers)

- **CSSC / China Power (600482 CH, NR):** heavy-duty / marine-derived turbines (703 Institute / Guanghan); ~80-unit overseas order in hand; ship-led, different valuation regime — used as a gas-turbine integration reference; discounted valuation 20x 2027F P/E due to export restrictions on Chinese shipbuilding companies; gas turbine business held by Guanghan not yet injected into listco.
- **AECC Aviation Power (600893 CH, NR):** aero-derivative (Taihang 110), defence-led; carries a defence multiple — used as an aeroderivative reference, not blended into power-equipment valuation; premium valuation 124x 2027F P/E due to aerospace hype in 2026; fast-growing topline but heavy capex cycle leading to thin margin.

VALUATION & RATING (SOTP)

SOTP valuation of Dongfang Electric

	2027F NP	2027F P/E (x)	Valuation	Rationale and assumptions
Coal-fired power	2,080	15	31,206	Anchored to global thermal OEM
Hydroelectric power	303	15	4,552	Utility-like, low margin, high visibility
Nuclear power	814	25	20,344	Premium on high margin / growth, discount to nuclear plays
Natural gas-fired power	489	25	12,221	Domestic replacement
Wind power	392	10	3,924	Low margin, immaterial
Power Plant Services	1,440	30	43,208	Recurring, high margin, benchmarked to global OEM
Engineering	195	15	2,926	Low margin EPC, finance
Emerging growth industries	542	20	10,837	Future growth potential
Total Valuation	6,256		129,219	

Gas-turbine optionality

	2027F	
No. of GT exports	25	Kazakhstan 3, Iraq 2, Canada 20 units
ASP (RMB m)	250	200m for now
Revenue	6,250	
NP	781	
P/E (x)		30
Valuation		23,438
Total valuation (incl. LTSA)		27,094

Total valuation	156,313	
Net cash	13,022	50% net cash not tied to finance
Total valuation	169,335	

Target price - A (RMB) 49.00 Rating: Outperform

H/A discount (%) 20

Target price - H (HK\$) 45.00 Rating: Outperform

Current price - A (RMB) 33.03

Upside (%) 48

Current price - H (HK\$) 25.24

Upside (%) 78

Source: Company, CCBIS estimates

Key Risks

- Slower domestic tendering / installations; gas-turbine certification, delivery and capacity ramp below expectations; price cuts by global incumbents compressing the export premium; slow wind-margin recovery; delayed recognition of large projects such as Yarlung; non-recurring investment gains (1Q2026 fair-value gain +RMB0.325b).

Dongfang Electric | 600875 CH/ 1072 HK – financial summary

Profit and loss

FYE 31 Dec (RMB m)	2024	2025	2026F	2027F	2028F
Operating income	68,593	77,583	85,766	95,734	106,031
Other	1,102	1,033	1,033	1,033	1,033
Revenue	69,695	78,615	86,799	96,767	107,064
COGS	(59,255)	(65,678)	(72,529)	(80,281)	(88,139)
Gross profit	10,441	12,937	14,270	16,486	18,925
Operating expenses	(7,354)	(7,891)	(8,712)	(9,713)	(10,746)
Operating income	3,086	5,046	5,558	6,774	8,179
EBIT	3,884	4,785	5,966	7,368	8,979
EBITDA	4,801	5,956	7,372	8,915	10,679
Net financial income (exp.)	(45)	(118)	47	47	47
JVs & associates	187	169	186	204	225
Other	656	(312)	176	343	528
Profit before tax	3,884	4,785	5,966	7,368	8,979
Tax	(597)	(819)	(867)	(1,075)	(1,313)
Total profit	3,288	3,966	5,099	6,294	7,665
Minority interest	(365)	(135)	(174)	(216)	(264)
Net profit attributable to shareholders	2,922	3,831	4,925	6,078	7,402
Core profit	1,980	3,192	4,925	6,078	7,402
Reported EPS (RMB)	0.94	1.11	1.42	1.76	2.14
Diluted EPS (RMB)	0.94	1.11	1.42	1.76	2.14
Core EPS (RMB)	0.94	1.11	1.42	1.76	2.14
Dividend (RMB m)	1,256	1,833	2,709	3,343	4,071
DPS (RMB)	0.40	0.53	0.78	0.97	1.18

Cash flow

FYE 31 Dec (RMB m)	2024	2025	2026F	2027F	2028F
Profit before tax	3,884	4,785	5,966	7,368	8,979
Amortization & depreciation	917	1,171	1,406	1,546	1,701
Net financial charge adj	45	118	(47)	(47)	(47)
Tax paid	(2,753)	(3,136)	(1,213)	(1,462)	(1,742)
Other	7,967	(923)	5,030	5,390	5,782
Operating cash flow	10,059	2,014	11,142	12,796	14,672
Disposal of fixed assets	1	1	0	0	0
Capex	(2,999)	(4,284)	(4,927)	(5,420)	(5,962)
Investment	(35,651)	(31,456)	(691)	(760)	(836)
Net interest received (paid)	120	131	954	1,049	1,154
Other	35,755	25,021	8	8	9
Investment cash flow	(2,774)	(10,587)	(4,656)	(5,122)	(5,634)
Change in borrowings	2,196	4,484	298	313	328
Equity issues	110	5,236	0	0	0
Dividend paid	(1,779)	(2,238)	(1,961)	(2,878)	(3,560)
Other	563	(2,380)	0	0	0
Financing cash flow	1,089	5,102	(1,663)	(2,565)	(3,232)
Change in cash flow	8,375	(3,471)	4,823	5,108	5,806
Cash & equivalents, begin	14,160	22,495	18,973	23,796	28,904
Forex	(39)	(51)	0	0	0
Cash & equivalents, end	22,495	18,973	23,796	28,904	34,711

Source: Company data, CCBIS estimates

Balance sheet

FYE 31 Dec (RMB m)	2024	2025	2026F	2027F	2028F
Cash & equivalents	26,856	22,680	27,503	32,611	38,417
Receivables	22,120	27,018	29,681	32,611	35,834
Inventory	21,685	26,171	28,788	31,667	34,834
Other current assets	23,118	25,815	25,998	26,200	26,422
Total current assets	93,780	101,684	111,971	123,089	135,507
Property, plant & equipment	5,641	6,693	10,441	14,563	19,097
Intangible assets	15	51	51	51	51
JVs & associates	2,924	3,623	3,986	4,384	4,823
Other non-current assets	39,649	50,623	49,042	47,453	45,855
Total non-current assets	48,230	60,991	63,519	66,451	69,826
Total assets	142,009	162,674	175,490	189,540	205,333
Short-term borrowings	75	141	148	156	164
Trade & bill payable	84,491	94,261	103,687	114,055	125,461
Other current liabilities	4,347	7,921	7,966	8,014	8,063
Total current liabilities	88,913	102,323	111,802	122,225	133,688
Long-term borrowings	2,167	4,542	4,788	5,046	5,317
Other non-current liabilities	7,787	7,641	7,641	7,641	7,641
Total non-current liabilities	9,954	12,182	12,428	12,686	12,957
Total liabilities	98,867	114,506	124,230	134,911	146,645
Share capital	3,118	3,458	3,458	3,458	3,458
Reserves and retained profits	35,679	41,776	44,868	48,237	52,296
Shareholders' equity	38,796	45,235	48,326	51,696	55,754
Minority interest	4,346	2,934	2,934	2,934	2,934
Total equity	43,142	48,168	51,260	54,629	58,688
Total equity and liabilities	142,009	162,674	175,490	189,540	205,333

Ratios

FYE 31 Dec	2024	2025	2026F	2027F	2028F
Growth (%)					
Revenue	14.9	12.8	10.4	11.5	10.6
EBITDA	(1.7)	24.1	23.8	20.9	19.8
EBIT	0.0	0.0	10.1	21.9	20.7
Net profit	N/A	N/A	28.5	23.4	21.8
Core net profit	N/A	N/A	54.3	23.4	21.8
Profitability (%)					
Gross margin	15.0	16.5	16.4	17.0	17.7
EBITDA margin	6.9	7.6	8.5	9.2	10.0
EBIT margin	4.4	6.4	6.4	7.0	7.6
Net margin	4.2	4.9	5.7	6.3	6.9
Core net margin	2.8	4.1	5.7	6.3	6.9
Tax rate	15.4	17.1	14.5	14.6	14.6
Efficiency (days)					
Inventory turnover	123	133	138	137	138
Trades receivables	114	112	118	116	115
Trades payables	246	252	249	247	247
Returns & leverage (%)					
ROAA	2.2	2.5	2.9	3.3	3.7
ROAE	7.7	9.1	10.5	12.2	13.8
Net debt (cash)/equity	(57.1)	(37.4)	(44.0)	(50.2)	(56.1)
Payout ratio	43.0	47.8	55.0	55.0	55.0
Interest coverage	(69.3)	(42.9)	119.2	145.3	175.4
Liquidity (x)					
Current ratio	1.1	1.0	1.0	1.0	1.0
Quick ratio	0.8	0.7	0.7	0.7	0.8

Rating definitions:**Outperform (O) – expected return > 10% over the next twelve months****Neutral (N) – expected return between -10% and 10% over the next twelve months****Underperform (U) – expected return < -10% over the next twelve months****Analyst certification:**

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